Molecular Simulation of Gas Adsorption in Metal–Organic Frameworks

From Thermodynamics to Grand Canonical Monte Carlo and RASPA

A comprehensive training handbook

Detailed physical explanations, derivations, figures, equations, solved examples, independent exercises, and manual-matched RASPA2 laboratories.

Thermodynamics	GCMC & Sampling	RASPA Practice

Prepared from the supplied TMU molecular-simulation teaching material and expanded into a self-contained guide

Dr Mehrdad Asgari

Expanded and visually refined edition – October 2026

How to use this handbook

This handbook is written for a student who may know chemical engineering, chemistry, materials science, or machine learning but has not yet run a molecular simulation. It starts with the physical meaning of adsorption, builds the thermodynamics and statistical mechanics needed for grand-canonical Monte Carlo, and then turns those ideas into practical simulation workflows for metal–organic frameworks (MOFs). The emphasis is not merely on obtaining a number from software. The emphasis is on understanding what that number means, what assumptions produced it, what can make it wrong, and how to validate it.

The supplied teaching slides are used as the conceptual backbone. Session 1 develops equilibrium, chemical potential, Langmuir/Henry behaviour, temperature dependence, and heats of adsorption. Session 2 develops statistical thermodynamics, ensembles, Metropolis sampling, GCMC, Widom insertion, and a first RASPAworkflow. The present document retains that progression but adds the topics that students need for independent research: force fields, partial charges, Ewald electrostatics, mixture adsorption, IAST/RAST, uncertainty, convergence, reproducibility, RASPA2 and RASPA3 workflows, high-throughput screening, diffusion, process-relevant descriptors, and project design.

A recommended learning path

First pass: Chapters 1–6, 9–13, 20–23. This is enough to understand what a GCMC adsorption calculation is doing.
Second pass: Chapters 14–19 and 24–27. This builds research-level judgement about force fields, charges, mixture adsorption, and validation.
Hands-on pass: Chapters 28–31 and the practice worksheets. Run simple calculations immediately. A small, well-understood Widom or low-pressure GCMC calculation is more educational than reading another 100 pages without touching the software.

Scope and important cautions

Classical molecular simulation is extremely powerful, but it is a model of a model. A crystal structure may be imperfect, the framework may be treated as rigid, the force field may neglect polarization, partial charges may be approximate, and a simulation may represent a perfect infinite crystal while an experiment contains defects, binders, moisture, grain boundaries, or inaccessible pores. Good simulation practice therefore separates *sampling error* from *model error*, and separates both from *experimental mismatch*. Throughout this handbook, numerical values in worked examples are illustrative unless explicitly identified as literature or software defaults.

RASPA evolves. The supplied seminar uses the familiar RASPA2-style `simulation.input` syntax as a teaching aid. RASPA3 uses JSON input/output and provides a Python interface; the exact keyword set must always be checked against the installed version and official documentation. This handbook teaches the physical mapping from theory to input variables so that students can adapt when syntax changes.

Contents

1 Why simulate adsorption in MOFs? . . . 1

1.1 MOFs as molecularly structured adsorbents . . . 1

1.2 What simulation can and cannot answer . . . 1

1.3 Why GCMC is central for adsorption . . . 1

1.4 A research mindset . . . 1

2 Adsorption experiments and the meaning of an isotherm . . . 2

2.1 What is measured in a volumetric experiment . . . 2

2.2 Common uptake bases . . . 2

2.3 Shape contains physics . . . 2

3 Thermodynamic equilibrium and chemical potential . . . 4

3.1 Why chemical potential is the natural variable . . . 4

3.2 Ideal-gas chemical potential . . . 4

3.3 Rigid-host approximation . . . 4

4 Henry law, Langmuir, dual-site Langmuir, and model fitting . . . 5

4.1 Henry regime . . . 5

4.2 Langmuir model . . . 5

4.3 Dual-site Langmuir . . . 5

4.4 Fitting strategy . . . 6

4.5 Other useful forms . . . 6

5 Temperature dependence and heats of adsorption . . . 7

5.1 Temperature dependence of affinity . . . 7

5.2 Differential and isosteric heats . . . 7

5.3 Loading dependence . . . 8

5.4 Working capacity and TSA . . . 8

6 From classical thermodynamics to statistical mechanics . . . 8

6.1 Microstates and macrostates . . . 8

6.2 Entropy as counting . . . 10

6.3 Fluctuations . . . 10

7 Ensembles: NVE, NVT, NPT, and grand canonical μVT . . . 10

7.1 Microcanonical ensemble . . . 10

7.2 Canonical ensemble . . . 11

7.3 Isothermal-isobaric ensemble . . . 11

7.4 Grand canonical ensemble . . . 11

8 Monte Carlo, Markov chains, and the Metropolis rule . . . 12

8.1 Markov-chain idea . . . 12

8.2 Metropolis acceptance . . . 12

8.3 Detailed balance . . . 12
8.4 Equilibration and production . . . 12
9 Grand Canonical Monte Carlo: the core algorithm . . . 13
9.1 State space . . . 13
9.2 Insertion and deletion acceptance . . . 13
9.3 Building an isotherm . . . 14
9.4 Move selection . . . 14
10 From pressure to chemical potential: fugacity and equations of state . . . 14
10.1 Fugacity . . . 15
10.2 Pure-component workflow . . . 15
10.3 Mixture fugacity . . . 15
11 Widom test-particle insertion and the Henry coefficient . . . 15
11.1 The insertion average . . . 15
11.2 Energy–entropy competition . . . 15
11.3 When Widom becomes difficult . . . 16
11.4 Practical educational task . . . 16
12 Isosteric heat from simulation . . . 17
12.1 Fluctuation expression . . . 17
12.2 Clausius–Clapeyron route . . . 17
12.3 Interpretation . . . 17
13 Force fields: what energy function is GCMC sampling? . . . 17
13.1 Energy decomposition . . . 17
13.2 Transferability . . . 18
13.3 Cross interactions . . . 18
14 Lennard–Jones interactions, cutoffs, and long-range dispersion . . . 19
14.1 The 12–6 form . . . 19
14.2 Cutoffs . . . 20
14.3 Shifting and tail corrections . . . 20
15 Electrostatics, partial charges, and Ewald summation . . . 20
15.1 Coulomb interactions . . . 21
15.2 Partial-charge methods . . . 21
15.3 Ewald idea . . . 21
15.4 Neutrality and framework chemistry . . . 21
16 Guest models: CO2, CH4, N2, H2, and water . . . 22
16.1 CO2 . . . 22
16.2 Methane and nitrogen . . . 22
16.3 Hydrogen . . . 22
16.4 Water . . . 23
17 Framework preparation: CIFs, cleaning, atom typing, and supercells . . . 23
17.1 Inspect the CIF . . . 23
17.2 Activation model . . . 23
17.3 Atom typing and charges . . . 23
17.4 Unit-cell replication . . . 23
18 Periodic boundary conditions and finite-size effects . . . 24

18.1 Periodic images . . . 24
18.2 Minimum image and cutoff geometry . . . 24
18.3 Finite-size effects beyond cutoffs . . . 24
19 Rigid versus flexible frameworks . . . 24
19.1 Why rigidity is attractive . . . 24
19.2 When rigidity fails . . . 25
19.3 Flexible simulation options . . . 25
20 Pure-component adsorption isotherms: a complete workflow . . . 25
20.1 Define the scientific state space . . . 25
20.2 Preflight the structure and model . . . 25
20.3 Equilibrate each pressure . . . 25
20.4 Production and uncertainty . . . 25
20.5 Post-process consistently . . . 26
21 Mixture adsorption and competitive GCMC . . . 26
21.1 Multicomponent grand canonical ensemble . . . 27
21.2 Adsorption selectivity . . . 27
21.3 Why pure-gas ratios can fail . . . 27
21.4 Sampling challenges . . . 27
22 IAST and RAST: mixture prediction from pure-component isotherms . . . 28
22.1 Spreading pressure . . . 28
22.2 What IAST needs . . . 28
22.3 When IAST works well . . . 28
22.4 Why RAST may be needed . . . 28
22.5 A sensible hierarchy . . . 28
23 Absolute, excess, and accessible-volume conventions . . . 29
23.1 Absolute loading . . . 29
23.2 Excess loading . . . 29
23.3 Why excess can decrease . . . 29
24 Density maps, adsorption sites, and microscopic interpretation . . . 30
24.1 Three-dimensional density . . . 30
24.2 Site occupancy . . . 30
24.3 Radial distribution functions . . . 30
24.4 Energy decomposition . . . 30
25 Statistical convergence and uncertainty estimation . . . 30
25.1 Stationarity first . . . 30
25.2 Autocorrelation . . . 30
25.3 Block averaging . . . 32
25.4 Replicate runs . . . 32
25.5 Convergence criteria . . . 32
26 Validation against experiment . . . 32
26.1 Like-for-like comparison . . . 32
26.2 Shape before score . . . 33
26.3 Avoid retuning on the test set . . . 34
26.4 What good agreement means . . . 34
27 Working capacity, selectivity, and process-relevant molecular descriptors . . . 34

27.1 Working capacity . . . 34
27.2 Selectivity versus recovery . . . 35
27.3 Volumetric metrics . . . 35
27.4 Heat and regenerability . . . 35
28 Molecular dynamics and diffusion: what GCMC does not provide . . . 35
28.1 MC time is not physical time . . . 35
28.2 Molecular dynamics . . . 36
28.3 Transport versus equilibrium . . . 36
28.4 Coupled workflows . . . 36
29 Advanced Monte Carlo moves and difficult adsorption problems . . . 37
29.1 Reinsertion and rotation . . . 37
29.2 Configurational-bias Monte Carlo . . . 37
29.3 Continuous fractional component methods . . . 37
29.4 Identity-change moves . . . 37
30 Open metal sites, polarization, and other hard chemistry . . . 37
30.1 Why open metal sites are hard . . . 37
30.2 Possible strategies . . . 38
30.3 Validation targets . . . 38
31 High-throughput molecular simulation and data management . . . 38
31.1 Standardise inputs . . . 39
31.2 Fail loudly . . . 39
31.3 Checkpoint and resume . . . 39
31.4 Quality-control metrics . . . 39
31.5 From simulation to ML . . . 39
32 RASPA2: the classic workflow and input anatomy . . . 40
32.1 Core files . . . 40
32.2 Conceptual input . . . 40
32.3 Keyword mapping . . . 40
32.4 Outputs to inspect . . . 40
33 RASPA3: modern JSON inputs, outputs, and Python integration . . . 41
33.1 What changed conceptually . . . 41
33.2 A minimal JSON-style pattern . . . 41
33.3 Why JSON helps . . . 42
33.4 Python integration . . . 42
34 Installing and testing RASPA on a workstation or HPC system . . . 42
34.1 Choose a version deliberately . . . 42
34.2 Environment isolation . . . 42
34.3 Smoke test . . . 43
34.4 HPC considerations . . . 43
35 A first hands-on CO2 GCMC project . . . 43
35.1 Goal . . . 43
35.2 Task A: preflight . . . 43
35.3 Task B: low-cost affinity . . . 43
35.4 Task C: isotherm . . . 43
35.5 Task D: interpretation . . . 43

36 A CO2/CH4 mixture project and comparison with IAST . . . 44
36.1 Pure-component stage . . . 44
36.2 IAST stage . . . 44
36.3 Mixture-GCMC stage . . . 44
36.4 Interpret discrepancies . . . 44
37 Linking molecular simulation to process simulation . . . 44
37.1 Isotherm models for process simulators . . . 45
37.2 Thermal effects . . . 45
37.3 Mass transfer . . . 45
37.4 Uncertainty propagation . . . 45
38 Common failure modes and a troubleshooting logic . . . 45
38.1 Zero uptake everywhere . . . 45
38.2 Astronomical energies or immediate crashes . . . 45
38.3 Unphysical high uptake . . . 45
38.4 Poor insertion acceptance at high loading . . . 46
38.5 Nonmonotonic pure isotherm . . . 46
38.6 Simulation–experiment mismatch . . . 46
39 Reproducibility, provenance, and research-quality reporting . . . 46
39.1 Minimum metadata . . . 46
39.2 Immutable raw outputs . . . 46
39.3 Version control . . . 46
39.4 Reproducible figures . . . 46
39.5 Negative and failed runs . . . 47
40 Designing a research project around molecular simulation . . . 47
40.1 Project type 1: descriptor physics . . . 47
40.2 Project type 2: force-field fidelity map . . . 47
40.3 Project type 3: mixture-model validity . . . 47
40.4 Project type 4: molecular-to-process ranking . . . 47
40.5 Project type 5: active simulation . . . 47
A1 Mathematical primer for molecular simulation . . . 48
A1.1 Exponentials and Boltzmann factors . . . 48
A1.2 Logarithms and free energies . . . 48
A1.3 Probability, averages, variance, covariance . . . 48
A1.4 Derivatives and response functions . . . 48
A1.5 Dimensional analysis . . . 48
A2 Derivation sketch: GCMC insertion and deletion . . . 48
A3 Units and conversion tables . . . 49
A4 RASPA2 quick-reference checklist . . . 49
A5 RASPA3 quick-reference checklist . . . 49
A6 Glossary . . . 50
A7 Twenty-five practice worksheets . . . 51
A8 Suggested answers and discussion points for worksheets . . . 76
A9 Twenty-five worked training examples . . . 77

I Expanded Explanations and Guided Examples 84
A9.1 Expanded tutorial 1: Why simulation? . . . 85
A9.2 Expanded tutorial 2: Experimental isotherms . . . 87
A9.3 Expanded tutorial 3: Chemical potential . . . 90
A9.4 Expanded tutorial 4: Henry regime . . . 92
A9.5 Expanded tutorial 5: Langmuir model . . . 95
A9.6 Expanded tutorial 6: Dual-site Langmuir . . . 97
A9.7 Expanded tutorial 7: Isotherm fitting . . . 100
A9.8 Expanded tutorial 8: Temperature dependence . . . 102
A9.9 Expanded tutorial 9: Heat of adsorption . . . 105
A9.10 Expanded tutorial 10: Microstates and probabilities . . . 107
A9.11 Expanded tutorial 11: Ensembles . . . 110
A9.12 Expanded tutorial 12: Metropolis MC . . . 112
A9.13 Expanded tutorial 13: Detailed balance . . . 115
A9.14 Expanded tutorial 14: GCMC insertion/deletion . . . 117
A9.15 Expanded tutorial 15: Fugacity . . . 120
A9.16 Expanded tutorial 16: Widom insertion . . . 122
A9.17 Expanded tutorial 17: Isosteric heat in GCMC . . . 125
A9.18 Expanded tutorial 18: Force fields . . . 127
A9.19 Expanded tutorial 19: Lennard-Jones . . . 130
A9.20 Expanded tutorial 20: Cutoffs and supercells . . . 132
A9.21 Expanded tutorial 21: Electrostatics . . . 135
A9.22 Expanded tutorial 22: Partial charges . . . 137
A9.23 Expanded tutorial 23: Guest models . . . 140
A9.24 Expanded tutorial 24: CIF preparation . . . 142
A9.25 Expanded tutorial 25: Periodic boundaries . . . 145
A9.26 Expanded tutorial 26: Rigid versus flexible MOFs . . . 147
A9.27 Expanded tutorial 27: Pure isotherm workflow . . . 150
A9.28 Expanded tutorial 28: Mixture GCMC . . . 152
A9.29 Expanded tutorial 29: IAST . . . 155
A9.30 Expanded tutorial 30: RAST . . . 157
A9.31 Expanded tutorial 31: Absolute versus excess . . . 160
A9.32 Expanded tutorial 32: Density maps . . . 162
A9.33 Expanded tutorial 33: RDFs . . . 165
A9.34 Expanded tutorial 34: Convergence . . . 167
A9.35 Expanded tutorial 35: Block averaging . . . 170
A9.36 Expanded tutorial 36: Validation . . . 172
A9.37 Expanded tutorial 37: Working capacity . . . 175
A9.38 Expanded tutorial 38: Volumetric metrics . . . 177
A9.39 Expanded tutorial 39: MD and diffusion . . . 180
A9.40 Expanded tutorial 40: Configurational bias . . . 182
A9.41 Expanded tutorial 41: Open metal sites . . . 185
A9.42 Expanded tutorial 42: High-throughput GCMC . . . 187
A9.43 Expanded tutorial 43: RASPA2 files . . . 190
A9.44 Expanded tutorial 44: RASPA2 units . . . 192
A9.45 Expanded tutorial 45: RASPA2 move keywords . . . 195
A9.46 Expanded tutorial 46: RASPA2 fugacity . . . 197

A9.47 Expanded tutorial 47: RASPA2 excess adsorption . . . 200
A9.48 Expanded tutorial 48: RASPA2 restarts . . . 202
A9.49 Expanded tutorial 49: HPC arrays . . . 205

II Visual Reasoning Atlas 208

A10 Reading adsorption physics from pictures . . . 209
A10.1 Chemical potential is the bridge between pressure and GCMC . . . 209
A10.2 Metropolis acceptance is a probability, not an optimizer . . . 209
A10.3 Why Widom insertion is dominated by rare favourable events . . . 210
A10.4 Move acceptance changes with loading . . . 210
A10.5 Pressure versus fugacity . . . 211
A10.6 Periodic boundaries, cutoffs and supercell size . . . 211
A10.7 Langmuir regimes are physically different regimes . . . 212
A10.8 Mixture selectivity is state dependent . . . 214
A10.9 Use IAST as a model, direct GCMC as a test . . . 214
A10.10 Convergence means reproducibility across time and seeds . . . 215
A10.11 Heat of adsorption can reveal site filling . . . 215
A10.12 A RASPA run is a chain of physical decisions . . . 215

III Statistical-Mechanics Lessons for Reliable Monte Carlo 217

A11 Practical lessons adapted from the standard molecular-simulation framework . . . 218
A11.1 The ensemble is part of the model . . . 218
A11.2 Sampling a distribution is different from minimizing energy . . . 218
A11.3 Detailed balance is a route to the desired stationary distribution . . . 218
A11.4 Importance sampling spends effort where probability is high . . . 218
A11.5 Correlation reduces the number of independent samples . . . 218
A11.6 Finite size and periodicity are modelling choices . . . 219
A11.7 Widom insertion is a free-energy method . . . 219
A11.8 Validation should separate sampling from model error . . . 219

IV RASPA2 Practical Workbook Matched to the Supplied Manual 220

A12 How to use the laboratories . . . 221
A12.1 Five file roles . . . 221
A12.2 Pressure and fugacity . . . 221
A12.3 Initialization and production . . . 221

A13 Laboratory 1: a fixed-N methane box . . . 221
A13.1 Input template . . . 221
A13.2 Physical interpretation . . . 221
A13.3 Step-by-step checks . . . 222
A13.4 Worked interpretation example . . . 222
A13.5 Independent extensions . . . 222

A14 Laboratory 2: N2 GCMC in MIL-47 . . . 222
A14.1 Input template . . . 222
A14.2 Physical interpretation . . . 223
A14.3 Step-by-step checks . . . 223
A14.4 Worked interpretation example . . . 223
A14.5 Independent extensions . . . 223

A15 Laboratory 3: scout the pressure range . . . 224
A15.1 Input template . . . 224
A15.2 Physical interpretation . . . 224
A15.3 Step-by-step checks . . . 224
A15.4 Worked interpretation example . . . 224
A15.5 Independent extensions . . . 224
A16 Laboratory 4: Henry coefficient with Widom insertion . . . 225
A16.1 Input template . . . 225
A16.2 Physical interpretation . . . 225
A16.3 Step-by-step checks . . . 225
A16.4 Worked interpretation example . . . 226
A16.5 Independent extensions . . . 226
A17 Laboratory 5: infinite-dilution adsorption enthalpy . . . 226
A17.1 Input template . . . 226
A17.2 Physical interpretation . . . 226
A17.3 Step-by-step checks . . . 227
A17.4 Worked interpretation example . . . 227
A17.5 Independent extensions . . . 227
A18 Laboratory 6: methane in MFI . . . 227
A18.1 Input template . . . 227
A18.2 Physical interpretation . . . 228
A18.3 Step-by-step checks . . . 228
A18.4 Worked interpretation example . . . 228
A18.5 Independent extensions . . . 228
A19 Laboratory 7: CO2 in Cu-BTC with CIF charges . . . 229
A19.1 Input template . . . 229
A19.2 Physical interpretation . . . 229
A19.3 Step-by-step checks . . . 229
A19.4 Worked interpretation example . . . 230
A19.5 Independent extensions . . . 230
A20 Laboratory 8: a binary CO2/CH4 mixture . . . 230
A20.1 Input template . . . 230
A20.2 Physical interpretation . . . 231
A20.3 Step-by-step checks . . . 231
A20.4 Worked interpretation example . . . 231
A20.5 Independent extensions . . . 231
A21 Laboratory 9: crash recovery . . . 231
A21.1 Input template . . . 231
A21.2 Physical interpretation . . . 232
A21.3 Step-by-step checks . . . 232
A21.4 Worked interpretation example . . . 232
A21.5 Independent extensions . . . 232
A22 Laboratory 10: a SLURM isotherm array . . . 233
A22.1 Input template . . . 233
A22.2 Physical interpretation . . . 233

A22.3 Step-by-step checks . . . 233
A22.4 Worked interpretation example . . . 233
A22.5 Independent extensions . . . 234
A23 Laboratory 11: output parsing and QC . . . 234
A23.1 Input template . . . 234
A23.2 Physical interpretation . . . 234
A23.3 Step-by-step checks . . . 234
A23.4 Worked interpretation example . . . 235
A23.5 Independent extensions . . . 235
A24 Laboratory 12: complete student mini-project . . . 235
A24.1 Input template . . . 235
A24.2 Physical interpretation . . . 235
A24.3 Step-by-step checks . . . 235
A24.4 Worked interpretation example . . . 236
A24.5 Independent extensions . . . 236
V Extended Solved Examples and Independent Exercises 237
A25 Sixty solved examples . . . 238
A25.1 Solved example 1: Pressure units . . . 238
A25.2 Solved example 2: Half saturation . . . 238
A25.3 Solved example 3: Henry slope . . . 238
A25.4 Solved example 4: Working capacity . . . 239
A25.5 Solved example 5: Volumetric loading . . . 239
A25.6 Solved example 6: Selectivity . . . 240
A25.7 Solved example 7: Fugacity . . . 240
A25.8 Solved example 8: Chemical potential . . . 240
A25.9 Solved example 9: Metropolis . . . 241
A25.10 Solved example 10: Autocorrelation . . . 241
A25.11 Solved example 11: Excess loading . . . 241
A25.12 Solved example 12: Supercell . . . 242
A25.13 Solved example 13: Charge replication . . . 242
A25.14 Solved example 14: Job count . . . 242
A25.15 Solved example 15: Core hours . . . 243
A25.16 Solved example 16: Dual-site Henry . . . 243
A25.17 Solved example 17: LJ minimum . . . 243
A25.18 Solved example 18: Temperature trend . . . 244
A25.19 Solved example 19: Widom ratio . . . 244
A25.20 Solved example 20: Molecules per unit cell . . . 245
A25.21 Solved example 21: Selectivity variant 21 . . . 245
A25.22 Solved example 22: Pressure conversion variant 22 . . . 245
A25.23 Solved example 23: Effective-sample variant 23 . . . 246
A25.24 Solved example 24: Fugacity variant 24 . . . 246
A25.25 Solved example 25: Working-capacity variant 25 . . . 246
A25.26 Solved example 26: Selectivity variant 26 . . . 247
A25.27 Solved example 27: Pressure conversion variant 27 . . . 247
A25.28 Solved example 28: Effective-sample variant 28 . . . 248

A25.29 Solved example 29: Fugacity variant 29 . . . 248
A25.30 Solved example 30: Working-capacity variant 30 . . . 248
A25.31 Solved example 31: Selectivity variant 31 . . . 249
A25.32 Solved example 32: Pressure conversion variant 32 . . . 249
A25.33 Solved example 33: Effective-sample variant 33 . . . 249
A25.34 Solved example 34: Fugacity variant 34 . . . 250
A25.35 Solved example 35: Working-capacity variant 35 . . . 250
A25.36 Solved example 36: Selectivity variant 36 . . . 251
A25.37 Solved example 37: Pressure conversion variant 37 . . . 251
A25.38 Solved example 38: Effective-sample variant 38 . . . 251
A25.39 Solved example 39: Fugacity variant 39 . . . 252
A25.40 Solved example 40: Working-capacity variant 40 . . . 252
A25.41 Solved example 41: Selectivity variant 41 . . . 252
A25.42 Solved example 42: Pressure conversion variant 42 . . . 253
A25.43 Solved example 43: Effective-sample variant 43 . . . 253
A25.44 Solved example 44: Fugacity variant 44 . . . 254
A25.45 Solved example 45: Working-capacity variant 45 . . . 254
A25.46 Solved example 46: Selectivity variant 46 . . . 254
A25.47 Solved example 47: Pressure conversion variant 47 . . . 255
A25.48 Solved example 48: Effective-sample variant 48 . . . 255
A25.49 Solved example 49: Fugacity variant 49 . . . 255
A25.50 Solved example 50: Working-capacity variant 50 . . . 256
A25.51 Solved example 51: Selectivity variant 51 . . . 256
A25.52 Solved example 52: Pressure conversion variant 52 . . . 257
A25.53 Solved example 53: Effective-sample variant 53 . . . 257
A25.54 Solved example 54: Fugacity variant 54 . . . 257
A25.55 Solved example 55: Working-capacity variant 55 . . . 258
A25.56 Solved example 56: Selectivity variant 56 . . . 258
A25.57 Solved example 57: Pressure conversion variant 57 . . . 258
A25.58 Solved example 58: Effective-sample variant 58 . . . 259
A25.59 Solved example 59: Fugacity variant 59 . . . 259
A25.60 Solved example 60: Working-capacity variant 60 . . . 259

A26 One hundred and twenty unsolved exercises . . . 260

A26.1 Thermodynamics . . . 260
A26.2 Isotherm fitting . . . 261
A26.3 Monte Carlo . . . 261
A26.4 GCMC and Widom . . . 262
A26.5 Force fields . . . 262
A26.6 Structures and PBC . . . 263
A26.7 Mixtures and IAST . . . 264
A26.8 Uncertainty and validation . . . 264
A26.9 RASPA2 . . . 265
A26.10 HPC and reproducibility . . . 265

A27 Hints for the independent exercises . . . 266

A28 Selected further reading . . . 266

Sources used in this expanded edition

This edition keeps the two supplied TMU lecture decks as the conceptual backbone: adsorption equilibrium, chemical potential, Henry and Langmuir behaviour, temperature dependence and heats of adsorption in Session 1; statistical thermodynamics, ensembles, Metropolis sampling, grand-canonical Monte Carlo, Widom insertion, and practical RASPA work in Session 2.

The supplied **RASPA 2.0.45 manual** is used as the authoritative source for the RASPA2 file structure, input syntax, units, simulation keywords, move definitions, restart behaviour, output conventions, and tutorial-style examples. The practical examples in this edition are intentionally written in RASPA2 style to match that manual. In particular, pressure inputs are treated in pascal; `simulation.input`, CIF, force-field, pseudo-atom, and molecule-definition files are distinguished explicitly; `SwapProbability` is connected to insertion/deletion exchange; `WidomProbability` is connected to Henry and infinite-dilution properties; and `HeliumVoidFraction` is treated as part of the absolute/excess adsorption convention.

The statistical-mechanical discussion is also informed by the standard presentation in D. Frenkel and B. Smit, *Understanding Molecular Simulation: From Algorithms to Applications*, 2nd ed. Only concepts directly relevant to the scope of these lectures are used: ensembles, probability weights, importance sampling, detailed balance, Metropolis Monte Carlo, grand-canonical particle moves, Widom insertion, configurational bias, periodic boundaries, and statistical error analysis. The discussion here is independently written and adapted specifically to MOF adsorption.

Version-specific software caution

A RASPA2 input that is correct for the supplied manual should not be assumed to be valid syntax for another major version. Preserve the physical model, then check the installed version's documentation before running production calculations.

1 Why simulate adsorption in MOFs?

This chapter establishes the big picture. Molecular simulation is a bridge between an atomic structure and experimentally meaningful adsorption observables. The central question is not "can software calculate an isotherm?" but "which microscopic assumptions are sufficient to explain, predict, or compare adsorption behaviour?"

1.1 MOFs as molecularly structured adsorbents

Metal–organic frameworks are crystalline porous solids assembled from inorganic nodes and organic linkers. Their attraction for adsorption science is the combination of high porosity with chemically specific internal surfaces. Changing a linker, metal, functional group, defect population, or pore geometry can change the energetic landscape experienced by a guest molecule. Thus two frameworks with similar pore volume may have very different low-pressure affinity, while two frameworks with similar Henry coefficients may differ strongly in saturation capacity.

A simulation starts from an atomistic representation of the framework. It then supplies an interaction model for guest–framework and guest–guest interactions and a statistical-mechanical rule for sampling configurations. The outputs can include adsorption isotherms, mixture loadings, heats of adsorption, density maps, and in some cases transport properties.

1.2 What simulation can and cannot answer

A rigid-framework GCMC calculation is naturally suited to equilibrium uptake in a specified crystalline model. It is not, by itself, a prediction of pellet-scale kinetics, pressure drop, cyclic steady state, mechanical shaping, or stability under water. Those require additional models. This distinction matters because molecular screening is often only the first level of a multiscale workflow.

For research, think of three layers: **molecular evidence** (adsorption thermodynamics and transport), **material evidence** (stability, defects, morphology, shaping), and **process evidence** (cycle, membrane, or reactor performance). A strong paper is explicit about which layer is being addressed.

1.3 Why GCMC is central for adsorption

In an adsorption experiment the pore is in diffusive contact with a gas reservoir. The reservoir fixes temperature and chemical potential; molecules enter and leave the pore. The number of adsorbed molecules is therefore not fixed. This is precisely the physical situation represented by the grand-canonical ensemble, in which T, V, and μ are fixed while N fluctuates. GCMC is the Monte Carlo algorithm most commonly used to sample this ensemble for adsorption in rigid nanoporous materials.

1.4 A research mindset

Do not begin by asking which button to press in RASPA. Begin by writing the scientific question. Is the target a low-pressure Henry coefficient, a full high-pressure isotherm, a binary selectivity, an isosteric heat, or a working capacity? The target determines the ensemble, pressure range, guest model, framework treatment, and convergence test. It also determines what experimental data are appropriate for validation.

First principle

A simulation is only useful when the observable, thermodynamic ensemble, molecular model, and validation target are aligned.

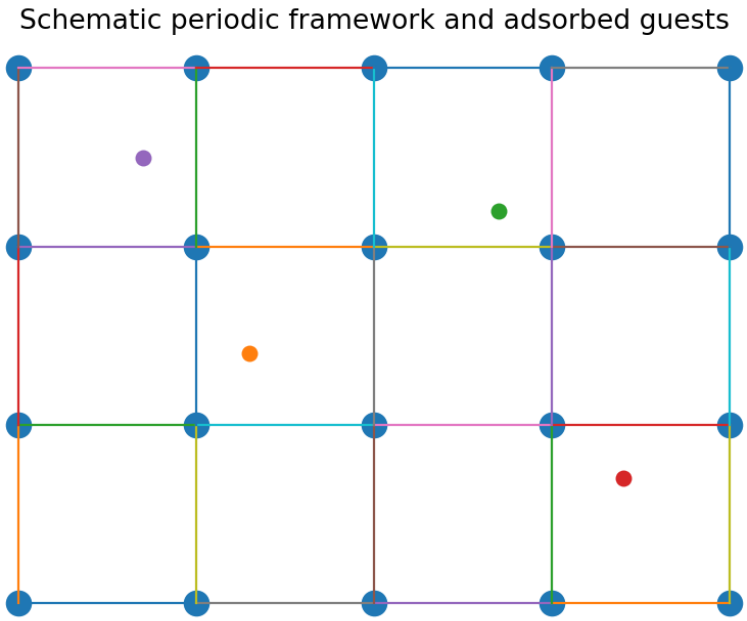


Figure 1.1: Schematic of a periodic framework with adsorbed guest molecules. The simulation cell is a mathematical representation of an effectively extended crystal, not an isolated pore.

2 Adsorption experiments and the meaning of an isotherm

Before learning Monte Carlo, it is essential to understand what an adsorption isotherm represents experimentally. The supplied teaching material begins from a volumetric experiment because it makes the reservoir picture intuitive.

2.1 What is measured in a volumetric experiment

A gas reservoir and sample cell are maintained at a known temperature. A dose of gas is introduced, the pressure is allowed to equilibrate, and a gas-phase mass balance is used to infer how much gas is no longer in the free volume. Repeating the dose–equilibrate–measure sequence generates uptake as a function of equilibrium pressure at fixed temperature. That curve is an adsorption isotherm.

The key conceptual point is that the pressure is measured in the external gas phase. Inside a rigid microporous solid, a scalar mechanical "pressure of the adsorbed phase" is usually not the useful control variable. The control variable for molecular exchange is chemical potential.

2.2 Common uptake bases

Experimental and simulation data may be reported as mmol per gram, mol per kilogram, molecules per unit cell, mass percent, standard-volume per gram, or volume of gas at STP per volume of adsorbent. A comparison is meaningless until the basis is harmonised. For high-pressure gases one must also distinguish *absolute* adsorption from *excess* adsorption. RASPA and experimental instruments may report different conventions depending on configuration and post-processing.

2.3 Shape contains physics

A steep low-pressure rise indicates strong affinity or a concentrated population of strong sites. A gradual rise can indicate weaker adsorption or large accessible volume. A saturation plateau indicates that the

finite pore volume or site capacity is being filled. Steps, inflections, or hysteresis can arise from structural transitions, cooperative adsorption, capillary effects, pore blocking, or experimental artefacts. Classical rigid-framework GCMC reproduces some mechanisms but not all.

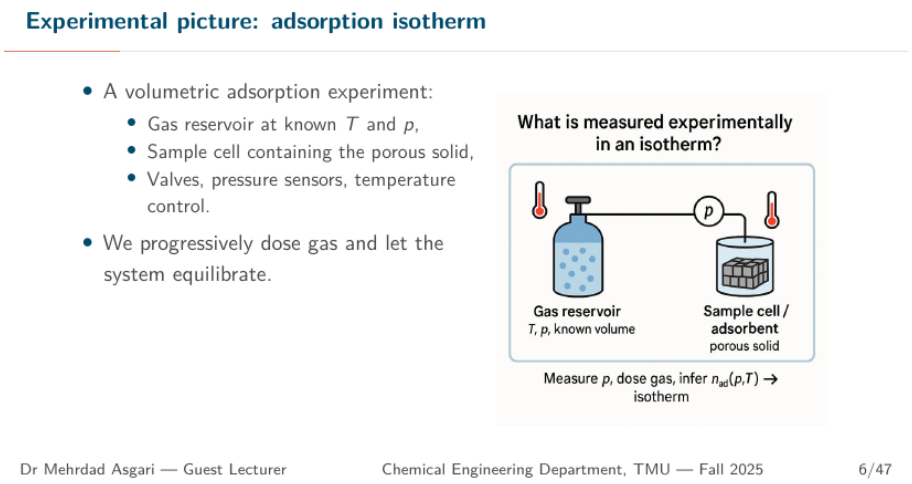


Figure 2.1: Original teaching slide: the experimental reservoir/sample-cell picture used to introduce adsorption isotherms.

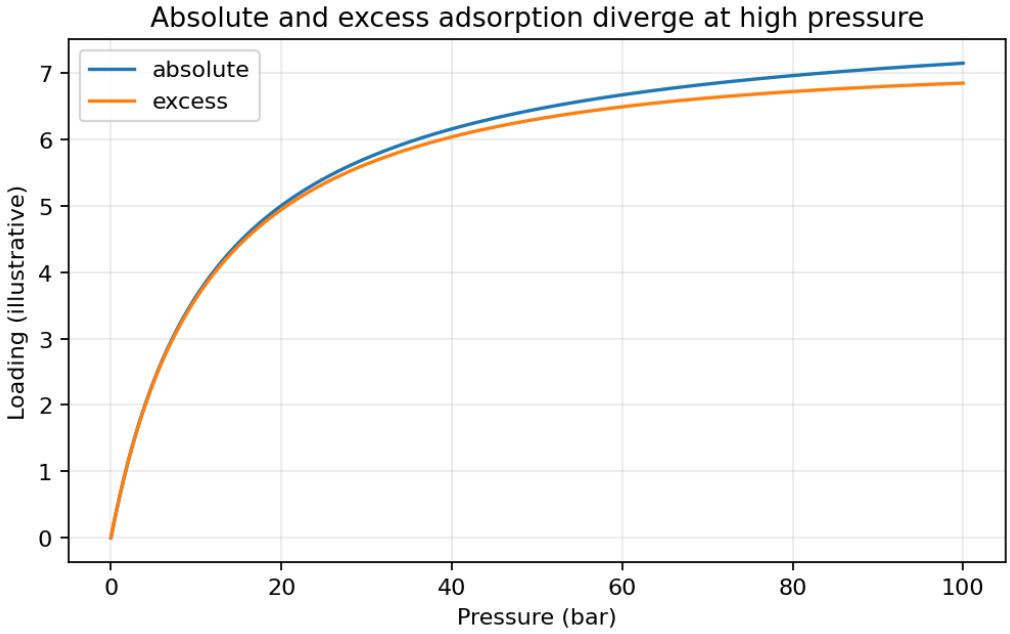


Figure 2.2: Illustrative difference between absolute and excess adsorption at elevated pressure.

Unit discipline

Always record temperature, pressure unit, uptake basis, adsorbent mass basis (dry/activated), gas purity, and whether the loading is absolute or excess. Most apparent simulation–experiment disagreements are impossible to diagnose if these metadata are missing.

3 Thermodynamic equilibrium and chemical potential

The first deep idea is that adsorption is a problem of diffusive equilibrium. The gas reservoir and the adsorbed phase exchange molecules until their chemical potentials are equal.

3.1 Why chemical potential is the natural variable

For a bulk gas, pressure is a convenient mechanical variable because the gas volume can change. A rigid crystal, however, has an essentially fixed cell volume. Its mechanical response is more naturally described by a stress tensor than by one internal scalar pressure. For particle exchange, the intensive variable that equalises at equilibrium is chemical potential, μ.

At equilibrium between gas and pore,

$$T_{\mathrm{gas}} = T_{\mathrm{ads}}, \qquad \mu_{\mathrm{gas}}(T,p) = \mu_{\mathrm{ads}}(T,V,N).$$

Raising pressure at fixed temperature raises the gas chemical potential and therefore changes the equilibrium loading.

3.2 Ideal-gas chemical potential

For an ideal gas, the pressure dependence may be written

$$\mu(T,p) = \mu^{\circ}(T) + RT\ln\left(\frac{p}{p^{\circ}}\right).$$

At low pressure this is often sufficient. At higher pressure the real gas should be described using fugacity f:

$$\mu(T,p) = \mu^{\circ}(T) + RT\ln\left(\frac{f}{f^{\circ}}\right), \qquad f = \phi p,$$

where ϕ is the fugacity coefficient. GCMC fundamentally requires chemical potential or activity; pressure is only a convenient external specification that software converts through an equation of state or supplied fugacity coefficient.

3.3 Rigid-host approximation

Many MOF adsorption calculations treat framework coordinates as fixed. This removes host vibrational and structural degrees of freedom and makes V fixed. The approximation is often practical for screening, but can fail for breathing frameworks, gate-opening materials, soft linker rotations, or adsorption-induced structural transitions. When flexibility matters, the ensemble and force field become more demanding.

Bridge to GCMC

Once we accept a rigid volume, fixed temperature, and exchange of particles with a reservoir, the grand-canonical ensemble is not an arbitrary computational choice. It is the statistical-mechanical representation of the experiment.

4 Henry law, Langmuir, dual-site Langmuir, and model fitting

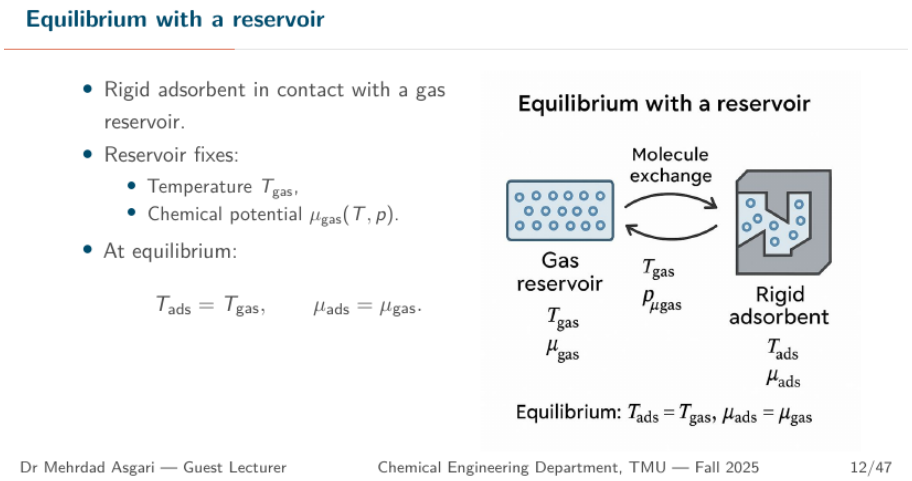


Figure 3.1: Original teaching slide: equilibrium between a gas reservoir and a rigid adsorbent.

Simple isotherm models are useful because they compress a curve into interpretable parameters. They are also dangerous when fitted mechanically. This chapter explains what the parameters mean and how to recognise when a model is identifiable.

4.1 Henry regime

At sufficiently low pressure the adsorbed molecules are sparse and guest–guest interactions are negligible. Uptake is linear in pressure or fugacity,

$$n \approx K_H p \quad \text{or} \quad n \approx K_H f.$$

The Henry coefficient is therefore the zero-loading slope. It is strongly sensitive to the strongest accessible binding environments and is often a more revealing low-pressure descriptor than saturation capacity.

4.2 Langmuir model

The Langmuir model assumes a finite set of equivalent sites, one molecule per site, and no lateral interactions. With saturation capacity q_{sat} and affinity parameter b,

$$q(p) = q_{\mathrm{sat}} \frac{bp}{1 + bp}.$$

At low pressure $q \simeq q_{\mathrm{sat}} bp$, so $K_H = q_{\mathrm{sat}} b$ on the chosen basis. At high pressure $q \to q_{\mathrm{sat}}$. The model is most useful as a compact empirical representation when its assumptions are not taken too literally.

4.3 Dual-site Langmuir

Heterogeneous MOFs often have at least two classes of adsorption environments. The dual-site Langmuir model writes

$$q(p) = q_{s,1} \frac{b_1 p}{1 + b_1 p} + q_{s,2} \frac{b_2 p}{1 + b_2 p}.$$

This can represent a small population of strong sites plus a larger population of weaker pore-filling sites. However, four parameters can be poorly identifiable if the pressure range is narrow. Different parameter combinations may fit the same data nearly equally well.

4.4 Fitting strategy

Fit in physically meaningful units and inspect residuals, parameter correlations, and confidence intervals. Use weighting that reflects experimental uncertainty rather than automatically giving high-pressure points dominance. Check whether the fitted saturation capacity is plausible relative to pore volume. Compare models by predictive behaviour, not only R^2. For multi-temperature fits, imposing physically consistent temperature dependence on $b(T)$ can reduce degeneracy.

4.5 Other useful forms

Sips/Langmuir–Freundlich, Toth, virial expansions, and spline representations are common. Flexible frameworks or cooperative adsorption may require forms that can represent steps. For process simulation, smoothness, extrapolation behaviour, and thermodynamic consistency matter as much as fit error.

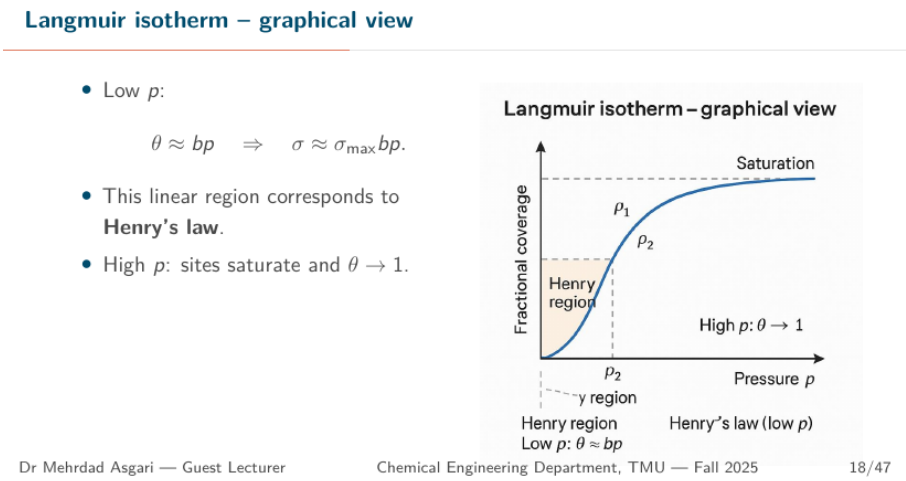


Figure 4.1: Original teaching slide: Langmuir saturation and the Henry limit.

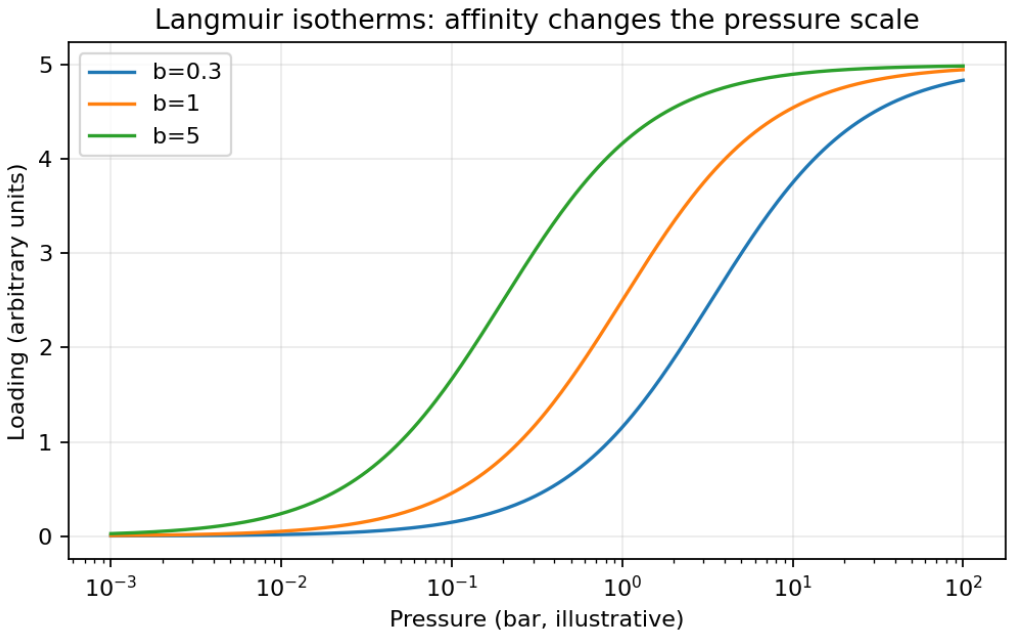


Figure 4.2: How the affinity parameter changes the pressure scale of a Langmuir isotherm.

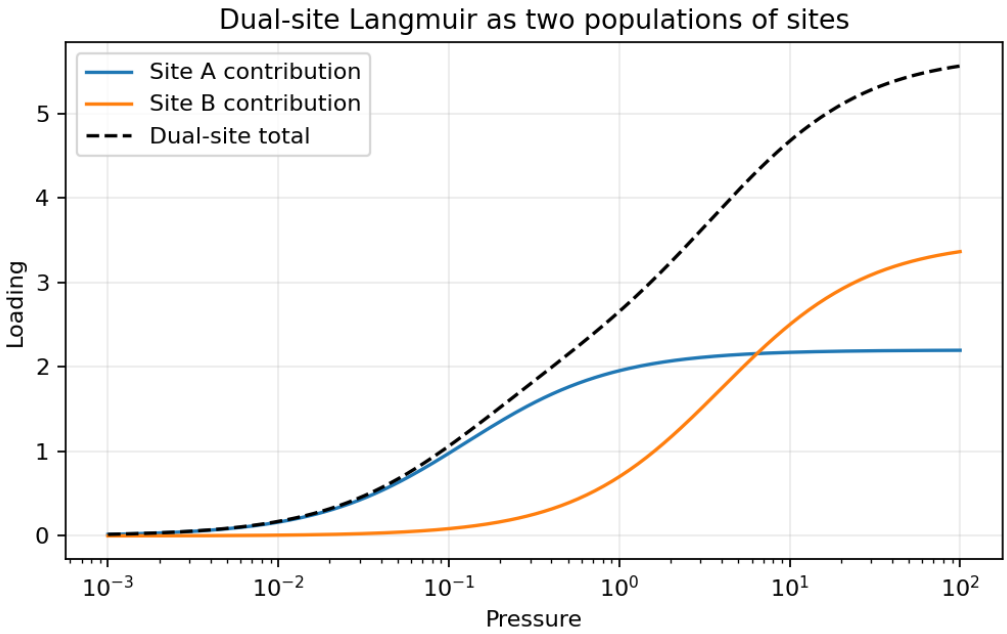


Figure 4.3: Dual-site Langmuir as the sum of two site populations.

Identifiability test

If you have only two pressure points, you cannot independently identify even a two-parameter Langmuir model in a defensible way, and certainly not a four-parameter dual-site model. More parameters do not create more information.

5 Temperature dependence and heats of adsorption

Adsorption is usually exothermic. Temperature therefore changes both equilibrium affinity and process working capacity. The heat of adsorption is also a sensitive diagnostic of binding-site chemistry.

5.1 Temperature dependence of affinity

In the Henry regime a van't Hoff-like relation is often written

$$\frac{d \ln K_H}{d(1/T)} \approx \frac{q_{\mathrm{ads}}}{R},$$

with sign conventions varying across communities. The safest practice is to state explicitly whether the reported q is positive for exothermic adsorption. Stronger binding generally produces a stronger temperature dependence.

5.2 Differential and isosteric heats

The phrase "heat of adsorption" is not unique unless boundary conditions are stated. Calorimetric differential heats and isosteric heats inferred from equilibrium isotherms are related but not identical definitions. The supplied teaching material emphasises this because students often compare values from different methods without checking the thermodynamic path.

The isosteric heat is commonly estimated from isotherms at multiple temperatures using the Clausius–Clapeyron relation at fixed loading,

$$q_{st} \approx -R \left(\frac{\partial \ln p}{\partial (1/T)} \right)_n ,$$

again with a declared sign convention.

5.3 Loading dependence

At low coverage the strongest sites fill first, so q_{st} can be high. As loading increases, weaker regions are populated and the heat may decrease. Guest–guest attractions can instead cause a rise in some systems. A flat heat does not prove energetic homogeneity; different mechanisms can compensate.

5.4 Working capacity and TSA

Temperature swing adsorption exploits the change in equilibrium loading between a cold adsorption temperature and a warmer regeneration temperature. The usable capacity is a difference between state points, not simply the maximum uptake at one condition. Stronger binding can increase capture but make regeneration harder.

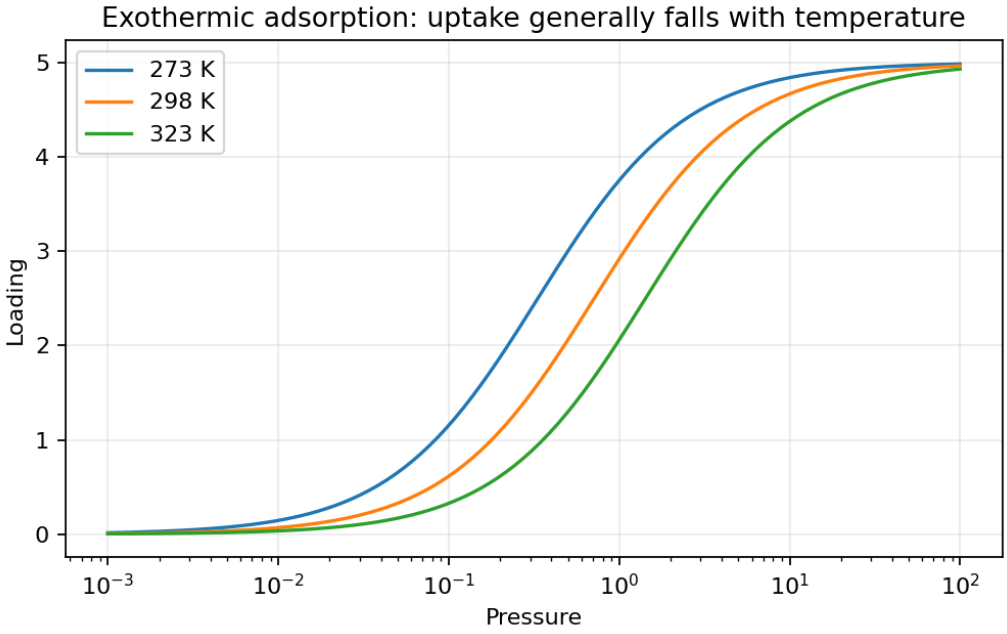


Figure 5.1: Illustrative multi-temperature isotherms for exothermic adsorption.

6 From classical thermodynamics to statistical mechanics

Statistical mechanics explains how macroscopic adsorption observables arise from a huge number of microscopic configurations. It also tells us which probability distribution a Monte Carlo algorithm must sample.

6.1 Microstates and macrostates

A macrostate is described by a small number of variables such as T, V, N, and energy. A microstate specifies the detailed positions, orientations, and (where relevant) momenta of all particles. Many microstates

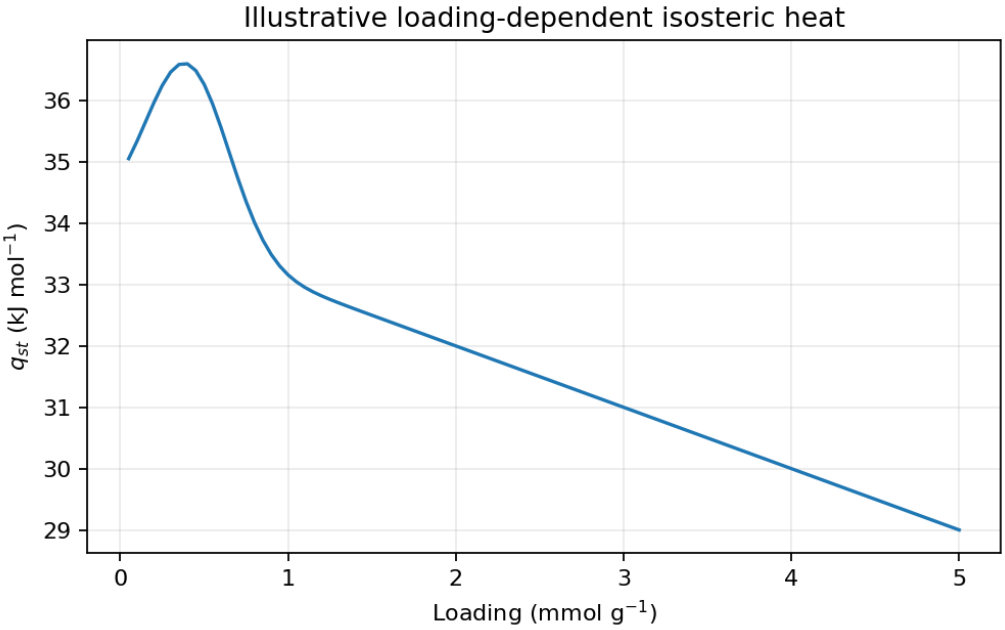
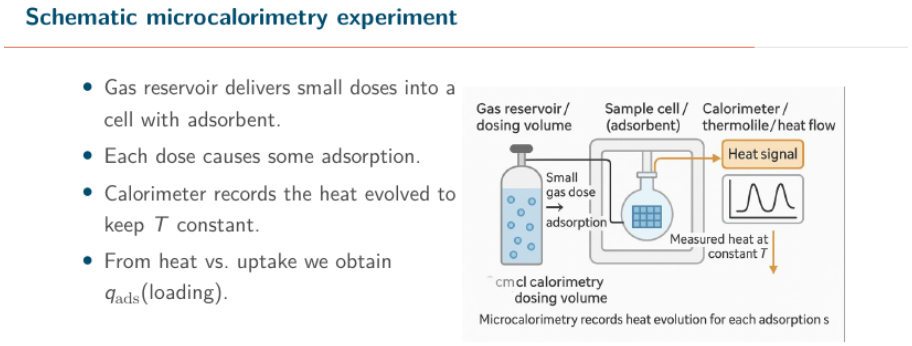


Figure 5.2: Illustrative loading dependence of the isosteric heat.



- Gas reservoir delivers small doses into a cell with adsorbent.
- Each dose causes some adsorption.
- Calorimeter records the heat evolved to keep T constant.
- From heat vs. uptake we obtain q_{ads}(loading).

Figure 5.3: Original teaching slide: conceptual microcalorimetry experiment.

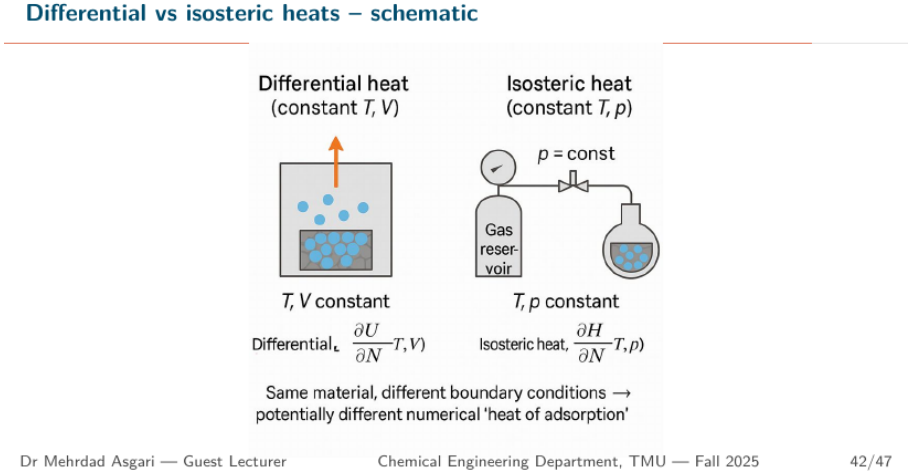


Figure 5.4: Original teaching slide: why differential and isosteric heats must not be conflated.

correspond to the same macrostate. Thermodynamic observables are averages over their probability distribution.

6.2 Entropy as counting

For an isolated system, Boltzmann's relation connects entropy and multiplicity,

$$S = k_{\mathrm{B}} \ln \Omega.$$

When two subsystems exchange energy, the most probable division maximises total entropy. Differentiating the entropy condition gives equality of temperature. Allowing particle exchange similarly leads to equality of chemical potential. This is the conceptual bridge from equilibrium thermodynamics to adsorption ensembles.

6.3 Fluctuations

Small systems fluctuate strongly. A simulation cell may contain tens or hundreds of adsorbates, not Avogadro's number. Those fluctuations are not numerical noise to be removed: in the correct ensemble they carry thermodynamic information. For example, number fluctuations in GCMC are linked to adsorption response and appear in fluctuation expressions for thermodynamic properties.

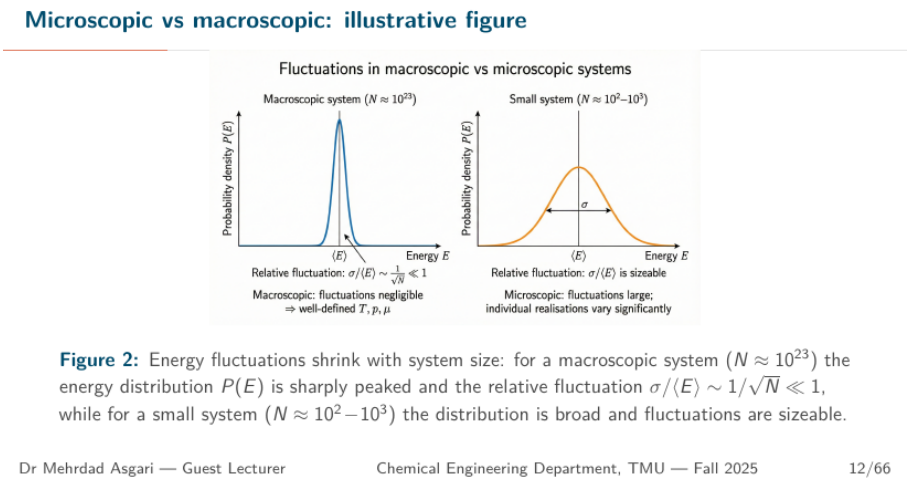


Figure 2: Energy fluctuations shrink with system size: for a macroscopic system ($N \approx 10^{23}$) the energy distribution $P(E)$ is sharply peaked and the relative fluctuation $\sigma/\langle E \rangle \sim 1/\sqrt{N} \ll 1$, while for a small system ($N \approx 10^2 - 10^3$) the distribution is broad and fluctuations are sizeable.

Figure 6.1: Original teaching slide: relative fluctuations decrease as system size grows.

Why students should care

Monte Carlo is not random guessing. It is a controlled method for drawing samples from a mathematically specified equilibrium distribution.

7 Ensembles: NVE, NVT, NPT, and grand canonical μVT

An ensemble is a statement about what the system can exchange with its surroundings. Choosing the ensemble is part of the physics, not a software preference.

7.1 Microcanonical ensemble

The microcanonical ensemble fixes N, V, and E. It represents an isolated system. It is fundamental conceptually but is not the natural choice for adsorption in contact with an external reservoir.

7.2 Canonical ensemble

The canonical ensemble fixes N, V, and T. Microstate probabilities are proportional to the Boltzmann factor,

$$P(\alpha) = \frac{e^{-\beta E_\alpha}}{Z}, \qquad \beta = (k_{\mathrm{B}}T)^{-1}.$$

The partition function Z normalises the distribution and connects to Helmholtz free energy $F = -k_{\mathrm{B}}T \ln Z$. NVT Monte Carlo is useful for bulk fluids and for equilibration at fixed loading, but by itself it does not produce an adsorption isotherm because N cannot respond to pressure.

7.3 Isothermal-isobaric ensemble

NPT fixes N, p, and T while volume fluctuates. It is useful for bulk-phase equilibria, density prediction, and flexible-cell studies. It is not the standard ensemble for rigid-pore adsorption.

7.4 Grand canonical ensemble

The grand-canonical ensemble fixes μ, V, and T while N fluctuates. Its probability weight for a configuration with particle number N and energy U contains

$$\exp[-\beta(U - \mu N)].$$

The grand partition function sums over all particle numbers. The natural thermodynamic potential is the grand potential $\Omega = -pV$ for a homogeneous bulk fluid; in confined systems the ensemble remains valid even though interpreting an internal scalar pressure is subtle.

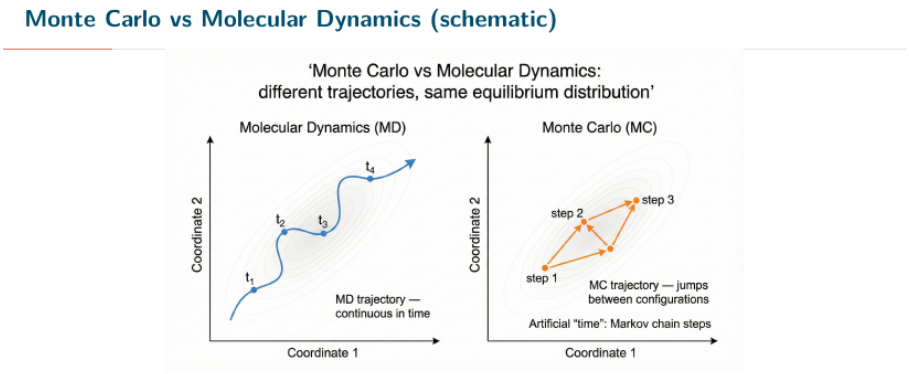


Figure 7.1: Original teaching slide: molecular dynamics and Monte Carlo follow different paths but can target the same equilibrium distribution.

Ensemble matching

Rigid MOF + thermal bath + gas reservoir $\Rightarrow$ fixed T and V, imposed μ, fluctuating N $\Rightarrow$ grand-canonical ensemble.

8 Monte Carlo, Markov chains, and the Metropolis rule

Monte Carlo replaces the impossible task of enumerating all microscopic configurations with a stochastic walk that samples important configurations in the correct proportion.

8.1 Markov-chain idea

A Markov chain proposes a new configuration from the current configuration. If the proposal and acceptance rules satisfy the required balance conditions and the chain is ergodic, long-run samples follow the target distribution. The states can be molecular coordinates, orientations, identities, and, in GCMC, particle number.

8.2 Metropolis acceptance

For a symmetric trial move in NVT, the familiar Metropolis rule is

$$A(o \to n) = \min\{1, \exp[-\beta(U_n - U_o)]\}.$$

Energy-lowering moves are accepted. Energy-increasing moves may still be accepted because entropy matters: rejecting every uphill move would trap the simulation in local minima rather than sample equilibrium.

8.3 Detailed balance

Detailed balance requires that equilibrium probability flow between two states is balanced pairwise. Metropolis acceptance is one construction that satisfies this. Detailed balance is sufficient but not always strictly necessary for stationarity; for beginners it is the safest conceptual framework.

8.4 Equilibration and production

The initial configuration is usually atypical. A first block of cycles is therefore discarded as equilibration. Only after observables appear stationary should production averaging begin. Production length must be long enough relative to the autocorrelation time to provide an adequate number of effectively independent samples.

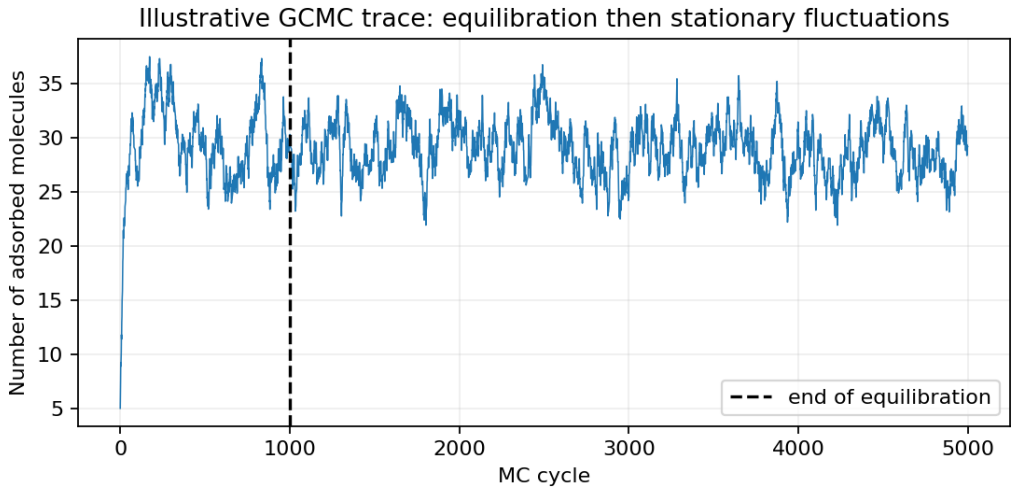


Figure 8.1: Illustrative time series showing an equilibration period followed by stationary number fluctuations.

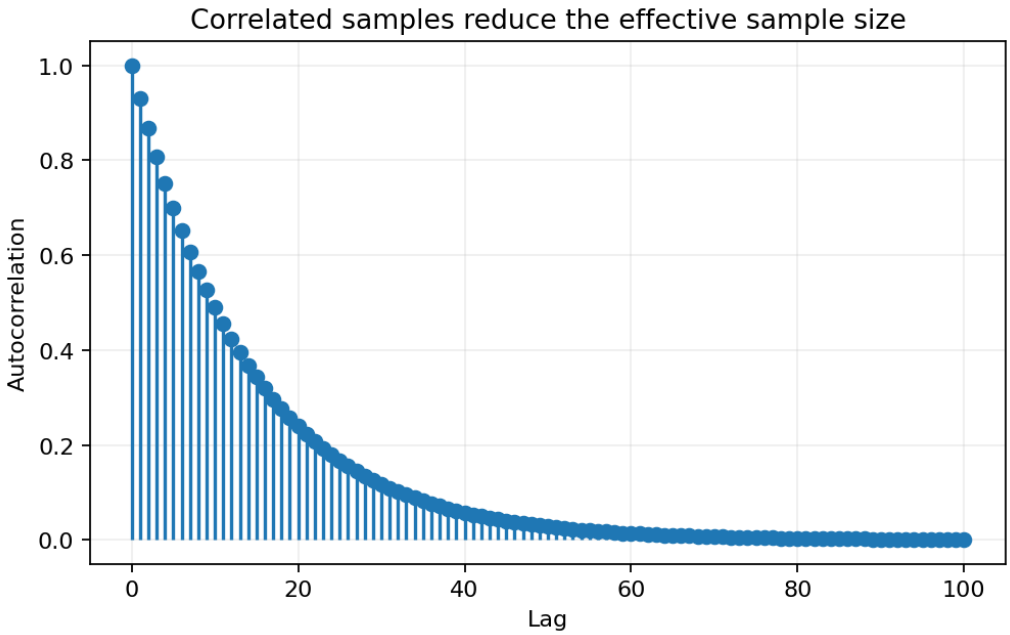


Figure 8.2: Autocorrelation persists over multiple Monte Carlo steps.

Common mistake

A million Monte Carlo steps do not imply a million independent samples. Statistical uncertainty depends on correlation, not just the raw step count.

9 Grand Canonical Monte Carlo: the core algorithm

GCMC adds particle insertion and deletion moves to ordinary configurational moves. Those moves allow the adsorbed population to respond to chemical potential and therefore generate an equilibrium isotherm.

9.1 State space

A GCMC state contains both coordinates and particle number: $(N, \mathbf{r}^N)$. Translation and rotation moves explore configurations at fixed N; insertion and deletion moves connect sectors with different particle number. For molecular adsorbates, internal/configurational-bias moves may also be needed.

9.2 Insertion and deletion acceptance

For a simple monatomic classical fluid with uniform insertion proposals, one common form is

$$A_{\text{ins}} = \min\left[1, \frac{V}{(N+1)\Lambda^3}\exp(\beta\mu)\exp(-\beta\Delta U)\right],$$

$$A_{\text{del}} = \min\left[1, \frac{N\Lambda^3}{V}\exp(-\beta\mu)\exp(-\beta\Delta U_{\text{del}})\right].$$

For rigid polyatomic molecules, orientations, molecular partition factors, Rosenbluth weights, and implementation conventions modify the expression. Software handles these details, but the physical structure remains: an entropic insertion factor competes with energetic cost and reservoir activity.

9.3 Building an isotherm

At one temperature and reservoir fugacity, equilibrate and then average the number of adsorbed molecules,

$$\langle N \rangle \approx \frac{1}{M} \sum_{k=1}^{M} N_k.$$

Convert the average to the desired loading basis. Repeat at many fugacities or pressures. Use a pressure grid dense enough to resolve the low-pressure rise, inflection region, and saturation behaviour.

9.4 Move selection

Translation, rotation, reinsertion, insertion, and deletion probabilities should be chosen to explore the relevant degrees of freedom efficiently. Absolute probabilities need not sum in the same way across all codes; some packages use relative move weights. Monitor acceptance statistics. Very low insertion acceptance at high loading is expected but may motivate configurational-bias or continuous-fractional-component methods for difficult systems.

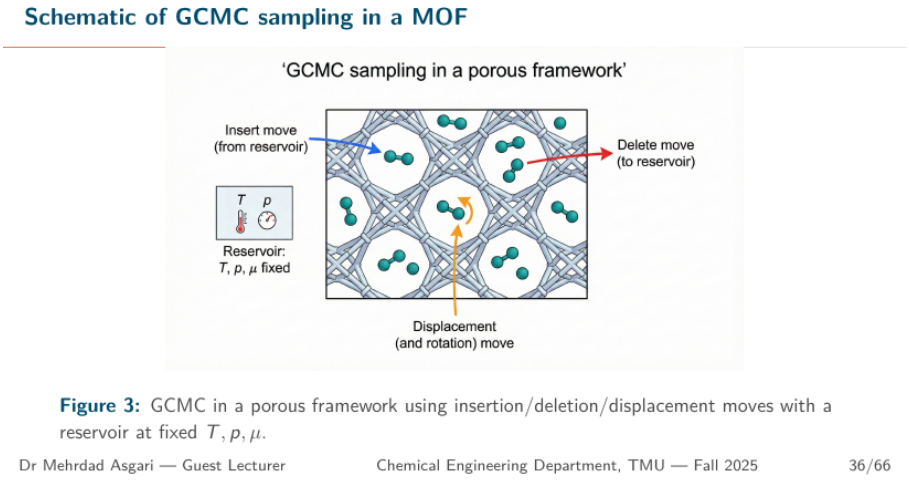


Figure 9.1: Original teaching slide: insertion, deletion, and displacement moves in a pore.

Think physically

Insertion answers: "Can the reservoir place another molecule here at this chemical potential?" Deletion asks the reverse. Equilibrium emerges when these competing exchanges are sampled in the correct statistical proportion.

10 From pressure to chemical potential: fugacity and equations of state

Simulation packages often ask for pressure, but the grand-canonical ensemble is controlled by chemical potential or activity. At elevated pressure, replacing fugacity by pressure can become a significant modelling error.

10.1 Fugacity

For a real gas, $f = \phi p$ defines the fugacity coefficient ϕ. The chemical potential relation becomes

$$\mu(T, p) = \mu^{\circ}(T) + RT \ln(f/f^{\circ}).$$

At low pressure $\phi \to 1$. At tens of bar, especially for strongly non-ideal or mixed gases, an equation of state should be used.

10.2 Pure-component workflow

For each pressure point, calculate or obtain $\phi(T, p)$ from an equation of state or let the simulation software do so when that capability is validated. Pass pressure and the appropriate fugacity convention consistently. Never mix a pressure-based input with a separately pre-corrected chemical potential unless you understand the code's expectation.

10.3 Mixture fugacity

For mixture component i, the reservoir fugacity is $f_i = \phi_i y_i p$. The fugacity coefficient depends on mixture composition and total pressure. At low pressure $f_i \approx y_i p$. At higher pressure, a cubic equation of state such as Peng–Robinson is often used in engineering workflows. The molecular simulation then samples adsorption at the specified component activities.

Debugging principle

If two codes disagree at high pressure, compare the actual component fugacities used before blaming the force field.

11 Widom test-particle insertion and the Henry coefficient

Widom insertion is one of the cheapest and most instructive calculations a beginner can run. It probes the free-energy cost of inserting a ghost molecule without changing the system composition.

11.1 The insertion average

A ghost particle is placed at randomly sampled positions and orientations. For each trial, the interaction energy change ΔU is evaluated but the particle is not retained. The excess chemical potential is related to an exponential average,

$$\mu^{\mathrm{ex}} = -k_{\mathrm{B}} T \ln \left\langle e^{-\beta \Delta U} \right\rangle .$$

At infinite dilution in an empty or effectively guest-free framework, the same insertion idea gives the Henry coefficient after applying the appropriate volume and unit prefactors.

11.2 Energy–entropy competition

A tiny region with extremely favourable energy can dominate the Boltzmann weight, but only if random insertions have a reasonable chance of finding it. Thus Henry affinity reflects both energetic depth and accessible configurational volume. This is why a narrow strong pocket and a broad moderately attractive pore can have comparable Henry constants.

11.3 When Widom becomes difficult

In dense fluids or very tight pores, successful random insertions become rare and the exponential average is dominated by rare events. The estimator then converges slowly. Biasing, staged insertion, or other free-energy methods may be necessary.

11.4 Practical educational task

Run a Widom calculation for the same gas in two structurally different MOFs. Compare K_H, the distribution of insertion energies, and a density or site map if available. Then explain the difference using pore size, chemical functionality, and strong binding environments rather than merely reporting the number.

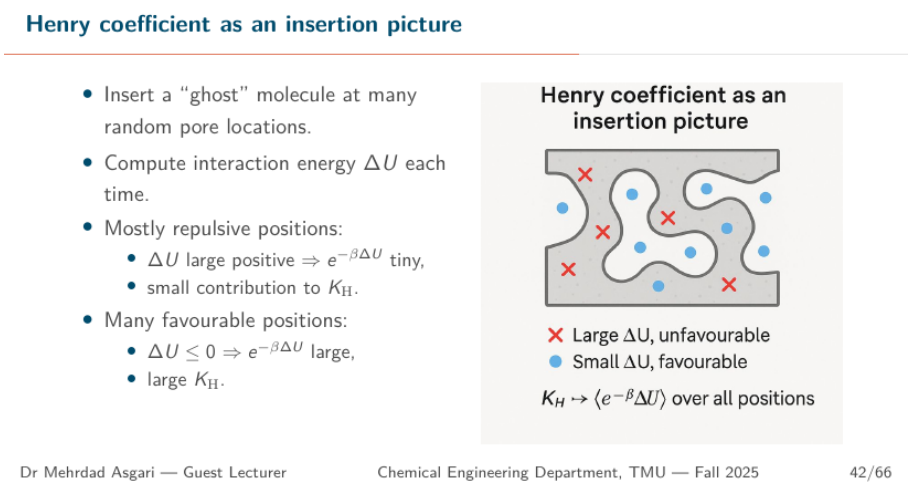


Figure 11.1: Original teaching slide: the ghost-particle insertion picture.

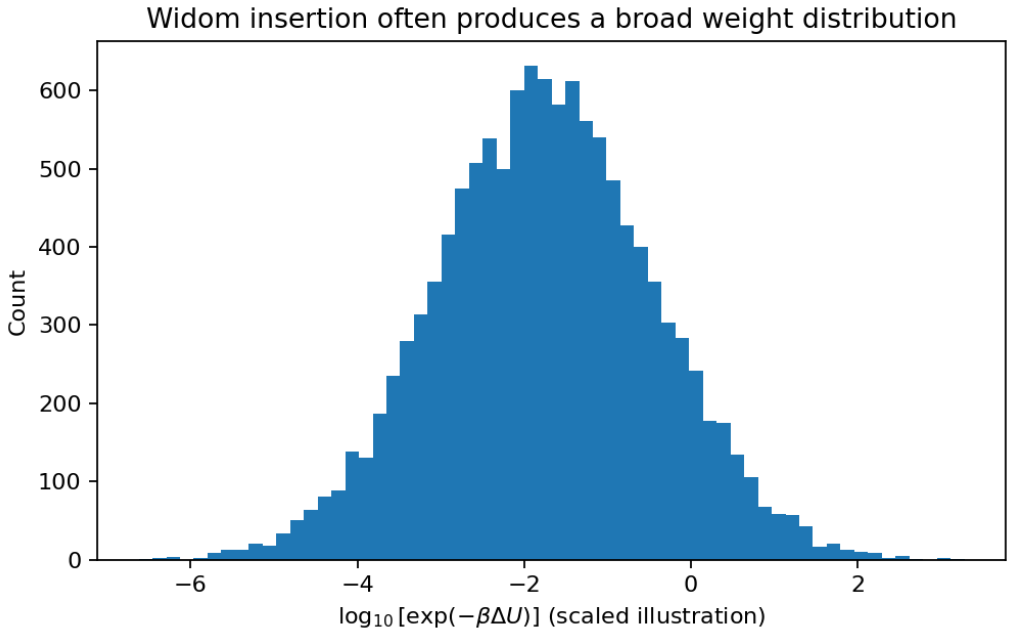


Figure 11.2: Illustrative broad distribution of Widom insertion weights.

Why start here

Widom insertion is computationally cheap, connects directly to statistical mechanics, and teaches students how force fields and pore chemistry translate into affinity.

12 Isosteric heat from simulation

The isosteric heat provides an energetic fingerprint of adsorption. GCMC can estimate it from fluctuations at a single state point or it can be inferred from multi-temperature isotherms.

12.1 Fluctuation expression

A commonly used grand-canonical estimator is related to the covariance of energy and particle number. One teaching form is

$$q_{st} \approx \frac{\langle UN \rangle - \langle U \rangle \langle N \rangle}{\langle N^2 \rangle - \langle N \rangle^2} - k_{\mathrm{B}} T,$$

with precise sign and energy conventions depending on implementation. Always check whether framework energy, intramolecular energy, and ideal-gas contributions are included consistently.

12.2 Clausius–Clapeyron route

Alternatively, simulate isotherms at several temperatures. At fixed loading, fit $\ln p$ against $1/T$. This route is intuitive and directly comparable to experimental isosteric analysis, but interpolation between isotherms and statistical uncertainty can be substantial.

12.3 Interpretation

A high low-loading heat usually points to strong binding sites. Declining heat with loading often indicates site heterogeneity. A plateau suggests more uniform environments, while increasing heat can signal cooperative guest–guest interactions. Compare trends, not just one reported average.

Sign convention

Some communities quote exothermic heats as positive magnitudes, others retain a negative enthalpy of adsorption. State the convention explicitly every time.

13 Force fields: what energy function is GCMC sampling?

GCMC does not know chemistry. It samples whatever potential-energy model you provide. Understanding the force field is therefore at least as important as understanding the Monte Carlo algorithm.

13.1 Energy decomposition

A typical classical adsorption energy contains guest–framework van der Waals interactions, electrostatics, guest–guest interactions, and, for flexible molecules or frameworks, bonded terms. A schematic decomposition is

$$U = U^{gf}_{\mathrm{vdW}} + U^{gf}_{\mathrm{elec}} + U^{gg}_{\mathrm{vdW}} + U^{gg}_{\mathrm{elec}} + U_{\mathrm{bonded}}.$$

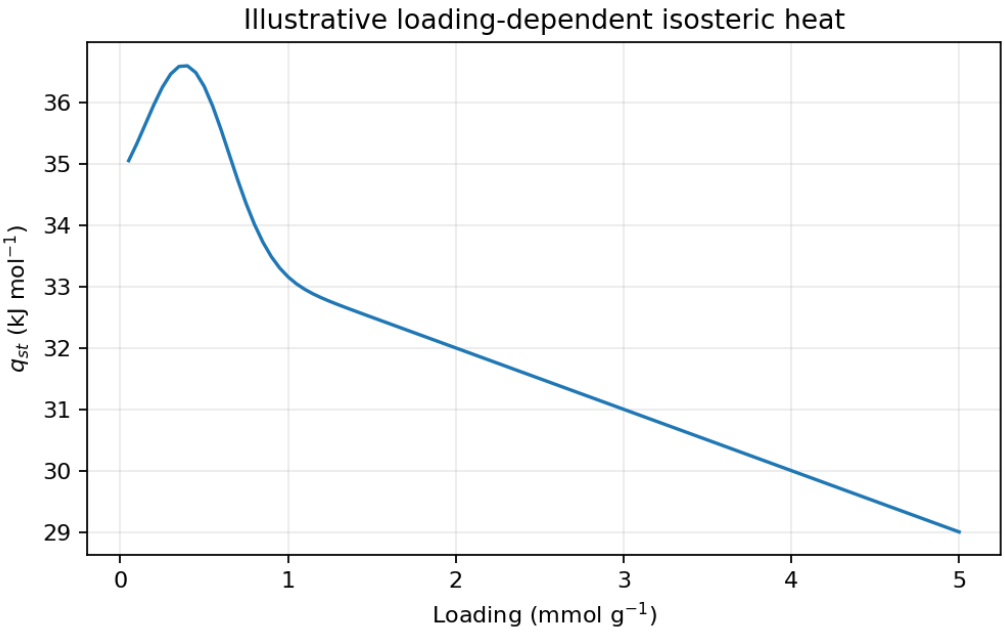


Figure 12.1: A schematic q_{st} profile showing strong-site filling followed by weaker-site occupation.

For a rigid framework the framework internal energy is constant and does not affect relative GCMC acceptance directly, but the framework coordinates define the external field.

13.2 Transferability

Generic force fields such as UFF or DREIDING are convenient for screening but were not fitted specifically to every MOF–adsorbate interaction. Their errors can be systematic for open metal sites, strongly polar groups, hydrogen bonding, or flexible coordination environments. A high-throughput study should treat force-field choice as a model assumption and, ideally, test sensitivity on representative materials.

13.3 Cross interactions

When atom-type parameters are defined separately, unlike-pair interactions are commonly produced by mixing rules such as Lorentz–Berthelot,

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}, \qquad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}.$$

Alternative rules or explicitly fitted cross terms can be more accurate. The important point is that mixing rules are part of the model and must be recorded.

Model vs sampling

Longer simulations reduce statistical error. They do not fix an inaccurate potential.

14 Lennard–Jones interactions, cutoffs, and long-range dispersion

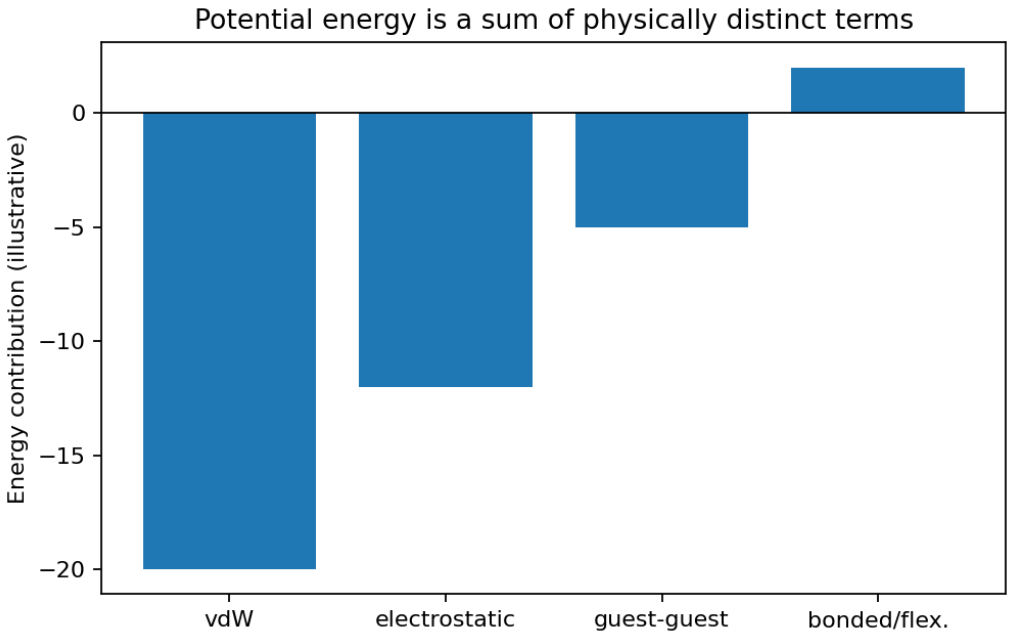


Figure 13.1: Conceptual decomposition of the energy model used in an adsorption simulation.

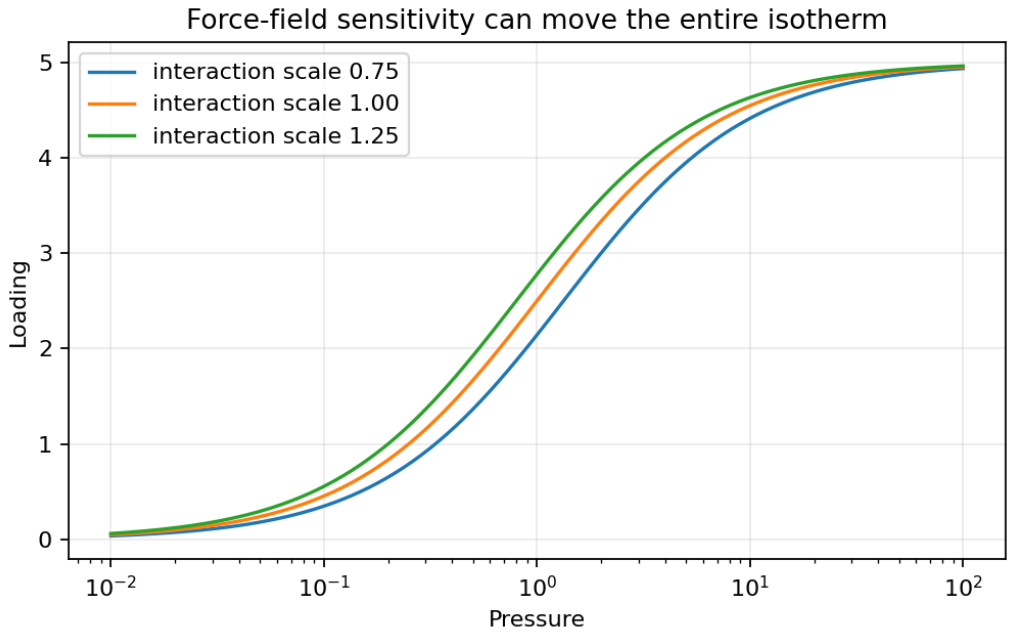


Figure 13.2: Changing interaction strength shifts the adsorption isotherm; sampling cannot correct a biased force field.

The 12–6 Lennard–Jones potential is the workhorse model for short-range repulsion and dispersion in classical adsorption simulations.

14.1 The 12–6 form

The pair potential is

$$U_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right].$$

The r^{-12} term produces steep repulsion; the r^{-6} term models dispersion. The minimum occurs at $r = 2^{1/6}\sigma$ with depth $-\epsilon$. Although convenient, the repulsive exponent is empirical rather than fundamental.

14.2 Cutoffs

Pair interactions are usually truncated beyond a cutoff r_c for efficiency. If the simulation cell is too small relative to the cutoff, periodic images can be double-counted incorrectly or minimum-image assumptions can fail. Replicating the MOF unit cell until each cell dimension comfortably exceeds twice the cutoff is a common practical criterion for simple minimum-image treatments.

14.3 Shifting and tail corrections

A raw truncation creates a discontinuity at r_c. Energy-shifted or force-shifted forms remove some discontinuity. Tail corrections approximate the neglected homogeneous long-range contribution, but their validity in strongly inhomogeneous micropores is not automatic. Use the treatment implemented and validated by the chosen force field/software rather than mixing conventions silently.

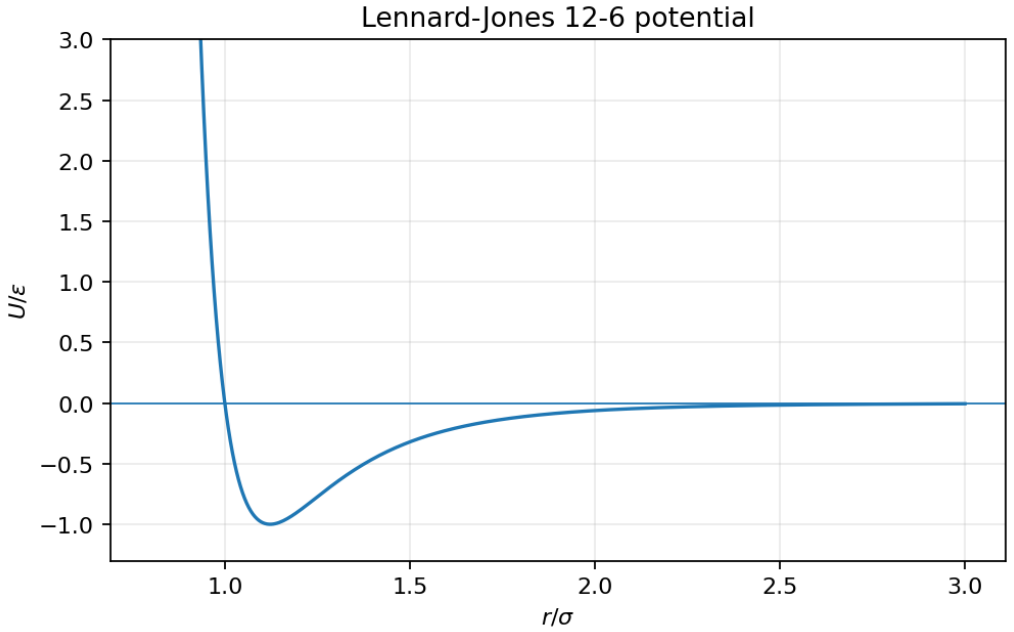


Figure 14.1: Lennard–Jones potential in reduced units.

Reproducibility metadata

Record the potential form, ϵ and σ source, mixing rule, cutoff, shifting/tail correction convention, and unit-cell replication.

15 Electrostatics, partial charges, and Ewald summation

Electrostatics often control low-pressure CO2 adsorption in polar MOFs. Charge assignment and long-range summation therefore deserve explicit validation.

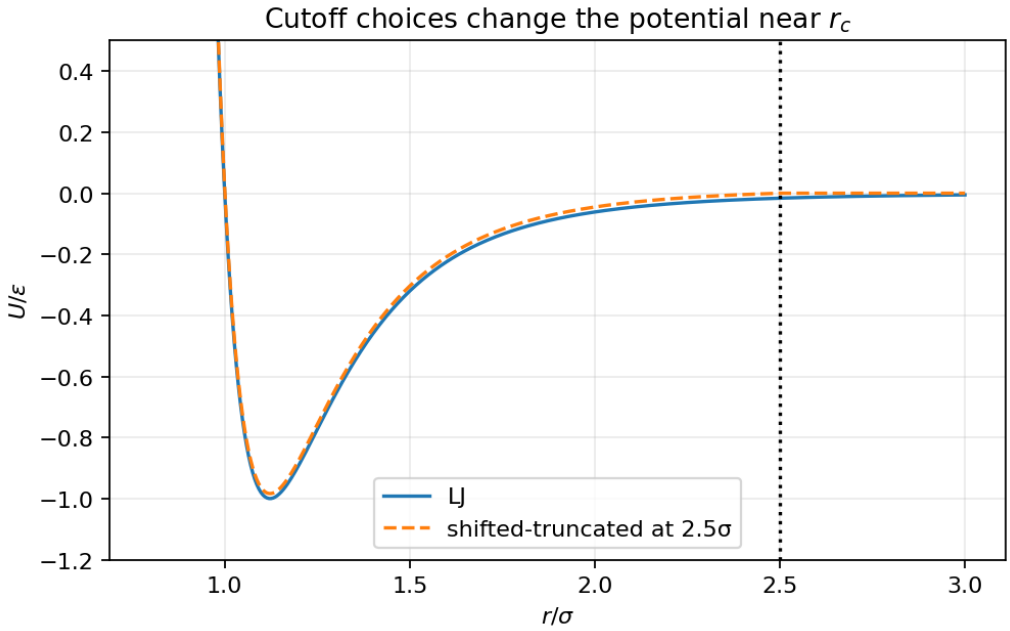


Figure 14.2: Illustration of how truncation/shift conventions alter the potential near the cutoff.

15.1 Coulomb interactions

For point charges,

$$U_{ij}^{\mathrm{elec}} = \frac{1}{4\pi\epsilon_0}\frac{q_i q_j}{r_{ij}}.$$

Unlike dispersion, Coulomb interactions decay slowly. A simple finite cutoff can lead to serious artefacts, especially in periodic ionic or highly polar structures.

15.2 Partial-charge methods

MOF charges may come from quantum-derived schemes such as REPEAT or DDEC, from charge equilibration methods such as QEq/EQeq, or from force-field-specific assignments. Different methods can produce materially different electrostatic fields. The charge method should therefore be treated as part of the model, not as an invisible preprocessing detail.

15.3 Ewald idea

Ewald summation splits the slowly convergent Coulomb sum into a short-range real-space term and a smooth reciprocal-space term, plus self and boundary corrections. The splitting parameter trades work between the two representations without changing the converged result. Practical codes choose tolerances and reciprocal grids automatically or through user settings.

15.4 Neutrality and framework chemistry

A periodic charged cell without a physically justified neutralisation strategy is problematic. Check oxidation states, counterions, missing linkers, protonation, and CIF occupancies before assigning charges. A mathematically valid Ewald sum cannot rescue a chemically invalid structure.

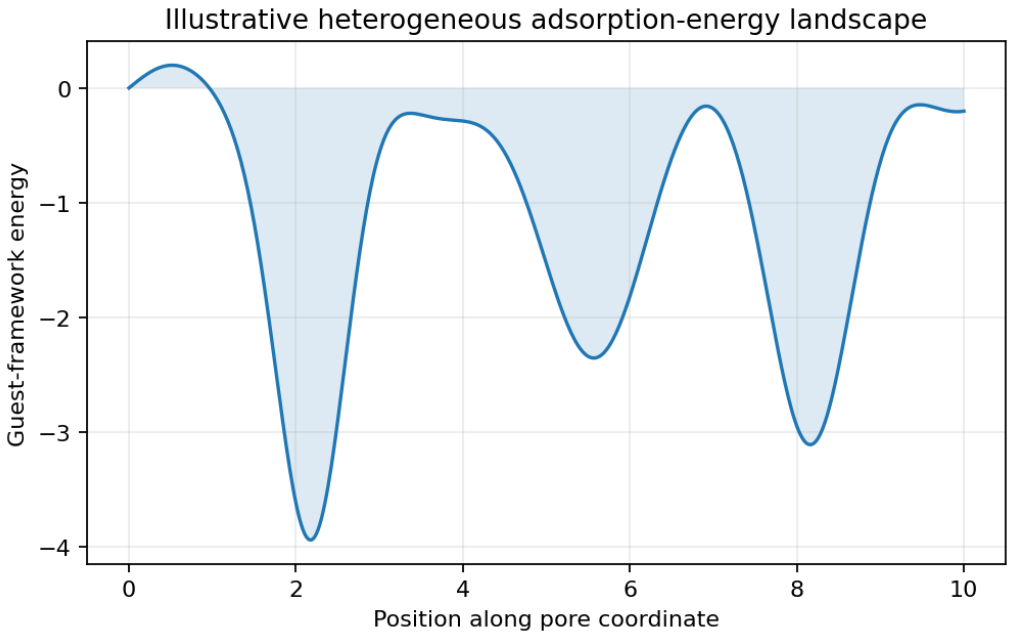


Figure 15.1: Electrostatics and dispersion combine to create a heterogeneous adsorption-energy landscape.

CO2 warning

Because CO2 has a strong quadrupole, predictions can be highly sensitive to framework partial charges and the CO2 model. Validate low-pressure behaviour before trusting high-throughput rankings.

16 Guest models: CO2, CH4, N2, H2, and water

Adsorbate models determine molecular geometry, Lennard–Jones sites, charges, and sometimes flexibility. A named gas is not enough information for reproducibility.

16.1 CO2

CO2 is commonly represented as a rigid linear three-site molecule with Lennard–Jones sites and partial charges chosen to reproduce bulk or phase-equilibrium properties. TraPPE-like and EPM2-like parameterisations are widely used. They are similar in spirit but not identical. Switching models can change both pure-component uptake and mixture selectivity.

16.2 Methane and nitrogen

Methane is often represented as a single united-atom Lennard–Jones site in adsorption screening, while nitrogen usually requires an electrostatic representation of its quadrupole. For CO2/CH4 and CO2/N2 separations, guest model consistency is important because selectivity is a ratio of competing adsorption free energies.

16.3 Hydrogen

Hydrogen has weak dispersion and significant quantum effects at low temperature. Classical models can be adequate in some ambient-temperature screening contexts, but cryogenic adsorption may require quantum corrections or specialised potentials.

16.4 Water

Water adsorption is especially challenging because hydrogen bonding, framework flexibility, defects, protonation, and cooperative clustering can dominate. A water GCMC result in a perfect hydrophobic crystal is not automatically a prediction of humidity stability. Many practical water simulations require advanced insertion methods because dense hydrogen-bonded clusters make conventional insertion inefficient.

Minimum reporting standard

State the exact guest model, rigid/flexible treatment, intramolecular geometry, charge model, and source. “CO2 was simulated is not reproducible.

17 Framework preparation: CIFs, cleaning, atom typing, and supercells

Most failed adsorption simulations begin with structure preparation rather than Monte Carlo mathematics. A systematic preprocessing workflow prevents silent errors.

17.1 Inspect the CIF

Check that the structure is connected, chemically sensible, and free of unresolved disorder. Identify solvent molecules, counterions, partial occupancies, duplicated atoms, missing hydrogens, and alternate conformations. Decide explicitly whether solvent should be removed. Experimental structures may represent solvated states that are not equivalent to activated adsorbents.

17.2 Activation model

Removing guest solvent from coordinates does not necessarily reproduce the activated experimental structure. Some frameworks contract, rotate linkers, lose coordination, or transform during activation. Where experimental activated structures exist, they are preferable. Otherwise document that activation is approximated by guest removal.

17.3 Atom typing and charges

Every framework atom must map to a nonbonded atom type with defined parameters, and electrostatic calculations require charges. Automated typing is convenient but should be checked manually for unusual metals or functional groups. Unknown atom types should cause a hard failure, not a silent fallback.

17.4 Unit-cell replication

Replicate the crystallographic cell so the simulation box is compatible with the nonbonded cutoff and periodic boundaries. A small primitive cell can be perfectly valid crystallographically yet unsuitable numerically for the chosen cutoff. Record replication factors in the output metadata.

Preflight checklist

1. Chemical formula plausible.
2. No unintended solvent or duplicate atoms.
3. Hydrogens/protonation appropriate.
4. Framework neutral or physically compensated.

5. Every atom typed.
6. Charges present if electrostatics used.
7. Cell replicated for the cutoff.
8. Accessible pores checked for obviously blocked regions.

18 Periodic boundary conditions and finite-size effects

A simulated MOF is usually an infinite periodic crystal represented by a finite repeating box. Understanding periodicity is essential for correct geometry and energy evaluation.

18.1 Periodic images

When a molecule leaves one side of the simulation box, its periodic image enters from the opposite side. This eliminates external surfaces and approximates bulk crystal behaviour. The framework is therefore not a finite nanoparticle unless a special nonperiodic model is constructed.

18.2 Minimum image and cutoff geometry

For simple short-range interactions, the nearest periodic image should be unambiguous within the cutoff. If the box is too small, a guest may interact incorrectly with multiple images of the same atom. Replication is cheap for a rigid framework because coordinates repeat and framework–guest grids can often accelerate evaluation.

18.3 Finite-size effects beyond cutoffs

Even when the cutoff criterion is satisfied, long-wavelength framework fluctuations, cooperative transitions, clustering, and mixture demixing can depend on box size. Compare at least two supercell sizes when studying collective phenomena or when a reviewer could reasonably question finite-size convergence.

Simple test

Double one box dimension and repeat a representative state point. If uptake or energy changes outside statistical uncertainty, the original cell was not demonstrably converged.

19 Rigid versus flexible frameworks

Rigid-host GCMC dominates high-throughput screening because it is efficient and often useful. But some MOFs cannot be understood without framework motion.

19.1 Why rigidity is attractive

Freezing the host removes expensive force evaluations and avoids the difficult task of obtaining a bonded force field for metal–ligand networks. For many robust frameworks away from structural transitions, rigid-host predictions can capture the dominant adsorption thermodynamics.

19.2 When rigidity fails

Breathing, gate opening, linker rotation that changes aperture size, adsorption-induced contraction, soft paddlewheel motion, and defect reorganisation can couple strongly to uptake. In these cases a rigid structure may show an apparently sharp pore exclusion where the real material opens, or may predict accessible volume that collapses experimentally.

19.3 Flexible simulation options

Flexible-host studies may use NPT Monte Carlo, hybrid MC/MD, molecular dynamics, osmotic ensembles, or enhanced-sampling methods. They require bonded interactions, stable integration/sampling, and a reference structure that the force field preserves. The uncertainty usually shifts from "is the adsorption force field correct?" to "is the entire host–guest Hamiltonian correct?".

Screening strategy

Use rigid GCMC as a baseline, then escalate only the materials or conditions for which flexibility is physically plausible and decision-relevant.

20 Pure-component adsorption isotherms: a complete workflow

This chapter turns the preceding theory into a robust end-to-end workflow for a single gas in a single MOF.

20.1 Define the scientific state space

Choose temperature, pressure/fugacity range, uptake basis, and whether the target is absolute or excess adsorption. Use logarithmic pressure spacing to resolve Henry and knee regions, supplemented by linear spacing near process operating points.

20.2 Preflight the structure and model

Validate the CIF, atom types, charges, guest model, mixing rules, cutoff, and supercell. Run an energy sanity check by placing one molecule in the pore and ensuring energies are finite and physically plausible. A simulation that starts with astronomical overlaps is a setup failure, not an equilibration problem.

20.3 Equilibrate each pressure

Start from zero loading for low pressure. At neighbouring pressures, a previous configuration can accelerate equilibration, but using a ladder also risks propagating a bad state. Compare ascending and descending pressure paths for difficult systems. Monitor loading and energy traces.

20.4 Production and uncertainty

Average only after equilibration. Report mean uptake with a block-averaged uncertainty or an equivalent correlated-sample estimator. Check that longer runs do not materially shift the mean. Save move acceptance, energy components, and restart files.

20.5 Post-process consistently

Convert molecules per cell to mol/kg or mmol/g using the correct framework mass. For excess adsorption, use the agreed accessible/pore volume and bulk density convention. Archive raw output so later unit conversions do not require rerunning the simulation.

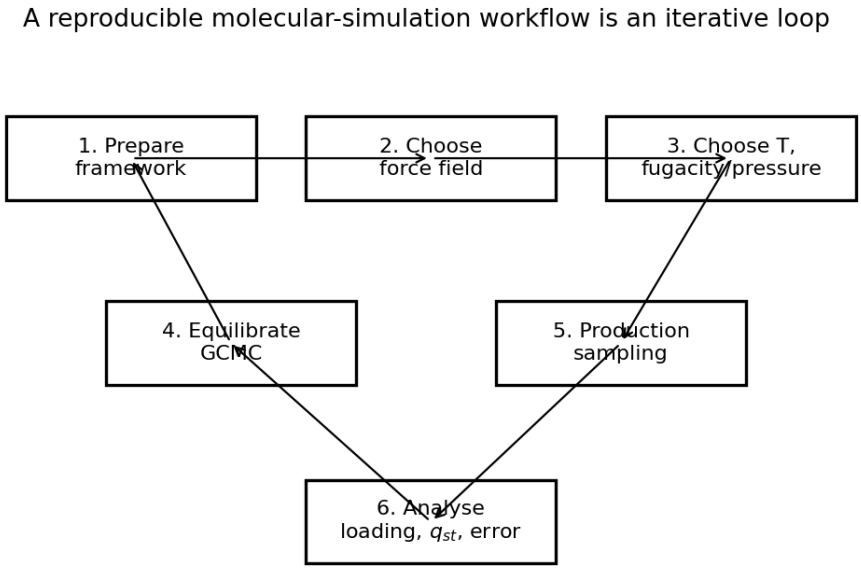


Figure 20.1: A reproducible molecular-simulation workflow is iterative: preparation, model selection, state specification, equilibration, production, and analysis feed back into each other.

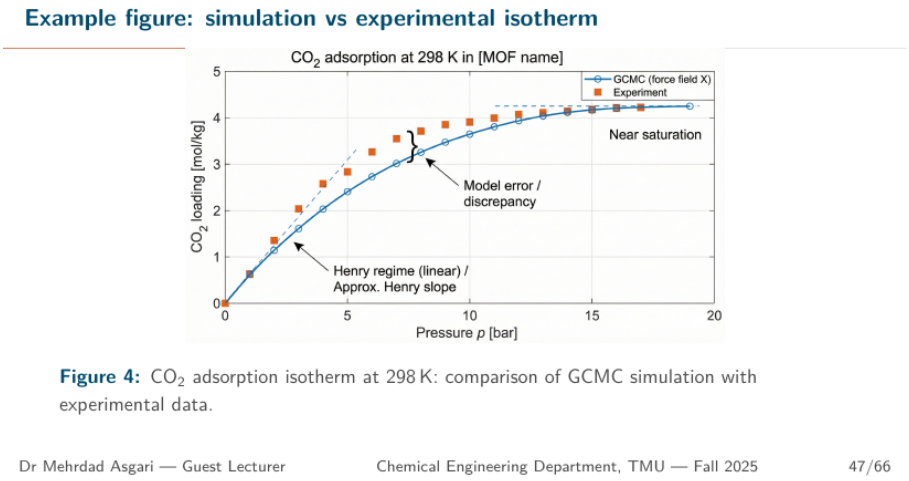


Figure 20.2: Original teaching slide: simulation–experiment comparison as a validation step.

21 Mixture adsorption and competitive GCMC

Real separations involve mixtures. Competitive adsorption can qualitatively reorder materials relative to pure-component screening.

21.1 Multicomponent grand canonical ensemble

Each species has its own chemical potential μ_i or fugacity f_i. GCMC performs insertion/deletion/translation/rotation moves for each component. The sampled composition inside the pore can differ dramatically from the gas-phase composition.

21.2 Adsorption selectivity

For a binary mixture, an equilibrium selectivity is often defined as

$$S_{1/2} = \frac{x_1/x_2}{y_1/y_2},$$

where x_i is adsorbed-phase mole fraction and y_i gas-phase mole fraction. Report the feed composition and pressure because selectivity is generally state dependent.

21.3 Why pure-gas ratios can fail

The ratio of Henry constants is valid only at infinite dilution. Ratios of pure-component loadings at finite pressure ignore competition for pore volume and strong sites. Mixture GCMC captures this competition explicitly, subject to the force field.

21.4 Sampling challenges

A strongly adsorbed species may be easy to insert but difficult to delete, while a weak component may have few successful insertions. Identity-change moves and configurational-bias methods can improve sampling for certain systems. Always inspect component-wise acceptance statistics.

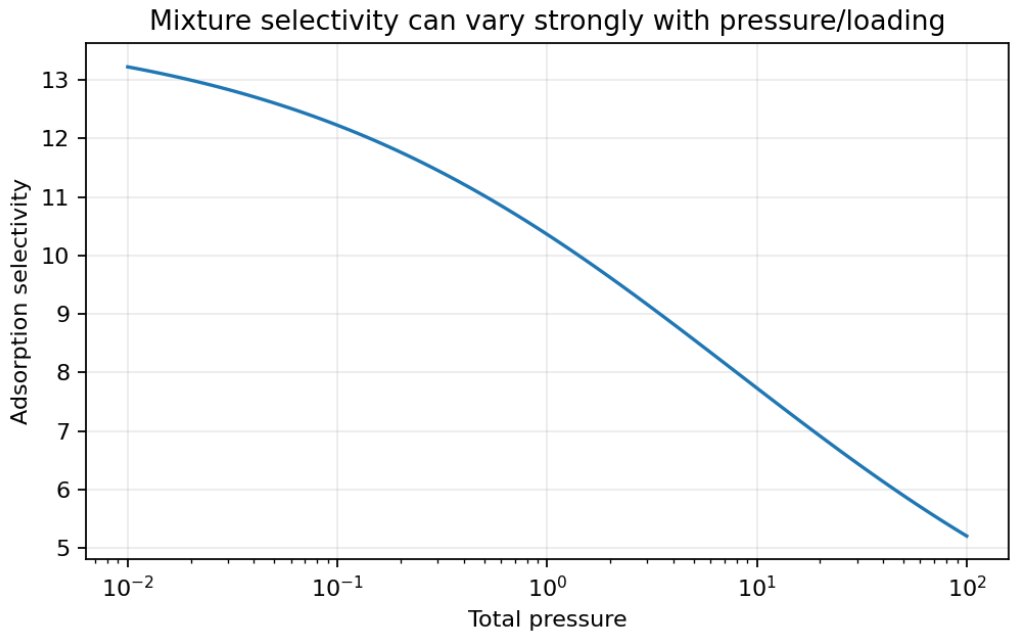


Figure 21.1: Illustrative pressure dependence of adsorption selectivity.

Mixture validation

If only pure-component experiments are available, mixture predictions carry additional uncertainty. Label them as predictions and, where possible, compare to breakthrough or mixture-isotherm data.

22 IAST and RAST: mixture prediction from pure-component isotherms

Ideal Adsorbed Solution Theory is a valuable bridge between pure-component isotherms and mixture adsorption. Real Adsorbed Solution Theory extends the framework when the adsorbed phase is non-ideal.

22.1 Spreading pressure

IAST treats the adsorbed phase analogously to an ideal solution at equal spreading pressure. For each pure component, the reduced spreading pressure is obtained from an integral of the pure isotherm,

$$\pi_i^*(p_i^*) \propto \int_0^{p_i^*} \frac{n_i^0(p)}{p}\, dp.$$

At mixture equilibrium the fictitious pure-component pressures p_i^* are chosen so that spreading pressures match and gas/adsorbed compositions satisfy the IAST relations.

22.2 What IAST needs

IAST needs smooth, physically reasonable pure-component isotherm representations over the required pressure range. Poor low-pressure fits contaminate dilute-mixture predictions; poor high-pressure extrapolation contaminates concentrated conditions. Dual-site Langmuir is popular because its spreading-pressure integral is analytic, but numerical integration of other smooth fits is also possible.

22.3 When IAST works well

IAST often performs well when components experience similar pore regions and adsorbed-phase non-ideality is modest. It is computationally much cheaper than mixture GCMC, making it useful for process optimisation and screening.

22.4 Why RAST may be needed

RAST introduces adsorbed-phase activity coefficients to represent non-ideal mixing. It can be useful when unlike interactions, size exclusion, site segregation, or strong heterogeneity cause IAST failure. The price is additional parameters and a need for mixture data or a physically motivated non-ideality model.

22.5 A sensible hierarchy

Use pure-component GCMC to obtain validated isotherms; use IAST as a cheap first mixture estimate; compare selected points to explicit mixture GCMC; invoke RAST only when the evidence shows systematic non-ideality that IAST cannot represent. This hierarchy keeps complexity proportional to evidence.

Do not confuse levels

IAST/RAST are thermodynamic mixture models built on pure-component isotherms. They are not substitutes for force-field validation, and they do not model transport or cyclic process dynamics.

23 Absolute, excess, and accessible-volume conventions

High-pressure adsorption can appear inconsistent across simulation and experiment simply because different definitions are being compared.

23.1 Absolute loading

Absolute adsorption counts the number of molecules assigned to the pore/adsorbed region by the simulation definition. GCMC naturally provides an average number of molecules in the simulation cell.

23.2 Excess loading

Excess adsorption subtracts the amount of bulk gas that would occupy a chosen pore volume at the same bulk density,

$$n_{\mathrm{ex}} = n_{\mathrm{abs}} - \rho_{\mathrm{bulk}} V_{\mathrm{pore}}.$$

The result depends on how V_{pore} is defined. Different helium-void or geometric definitions can shift the excess curve.

23.3 Why excess can decrease

As pressure rises, absolute loading approaches a finite pore-filling limit, while the bulk-gas reference term continues to increase. Excess adsorption can therefore pass through a maximum and decrease even though the actual number of molecules inside the pore continues to rise.

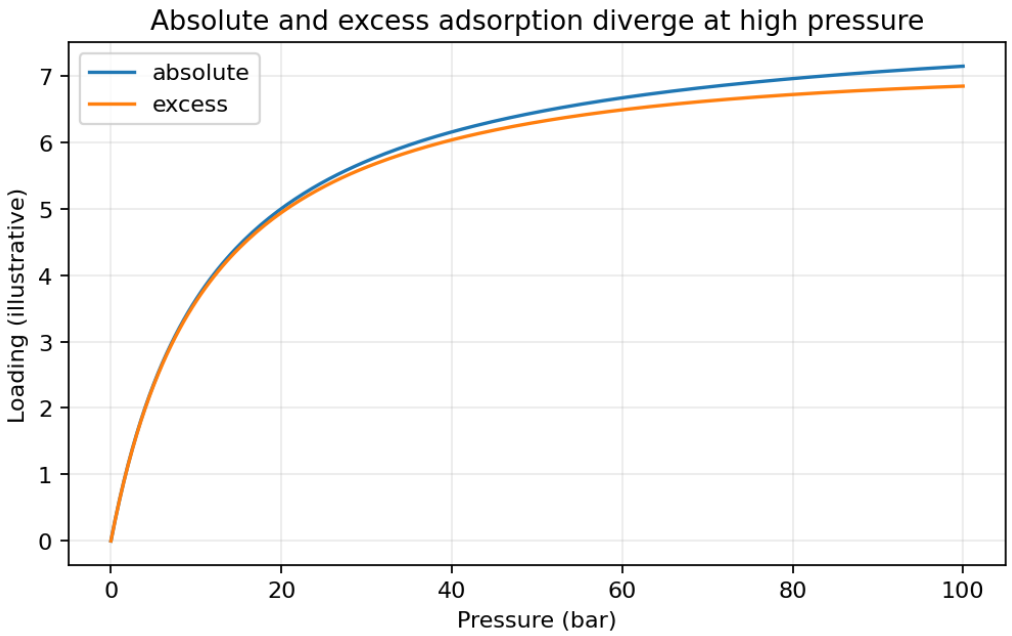


Figure 23.1: Illustrative relationship between absolute and excess loading at high pressure.

Comparison rule

Never compare simulated absolute uptake with experimental excess uptake without a documented conversion.

24 Density maps, adsorption sites, and microscopic interpretation

An isotherm tells how much adsorbs. Spatial analysis tells where and why. This is often the most chemically interesting part of a molecular-simulation study.

24.1 Three-dimensional density

During production, guest positions can be accumulated on a spatial grid. High-density regions reveal preferred adsorption sites. At low loading these often correspond to open metal sites, polar functional groups, narrow pockets, or regions where dispersion fields from opposing walls overlap.

24.2 Site occupancy

A density maximum is not automatically a discrete binding site. To define site occupancy, specify a spatial region or clustering rule and test sensitivity. Site populations can then be followed with pressure to show sequential filling.

24.3 Radial distribution functions

RDFs measure how likely a guest atom is to occur at distance r from a framework atom or another guest, relative to an uncorrelated reference. Peaks identify preferred coordination distances. Integrating a peak can give a coordination number when the reference density and geometry are handled correctly.

24.4 Energy decomposition

Compare density features with framework–guest electrostatic and dispersion contributions. A strong site may be dominated by electrostatics, dispersion, or both. This mechanistic attribution is useful when designing linker functionalisation or interpreting descriptor importance in machine-learning models.

25 Statistical convergence and uncertainty estimation

A reported loading without an uncertainty estimate is incomplete. Monte Carlo produces correlated stochastic data, so naive standard deviations can be misleading.

25.1 Stationarity first

Before estimating uncertainty, establish that the production trace is stationary. A drifting mean indicates insufficient equilibration or slow transitions between metastable states. Split the trajectory into early and late blocks and compare means.

25.2 Autocorrelation

Successive MC samples are correlated. If the integrated autocorrelation time is τ, M raw samples correspond to far fewer than M independent observations. The effective sample size is approximately $M_{\mathrm{eff}} \sim M/(2\tau)$ for common conventions.

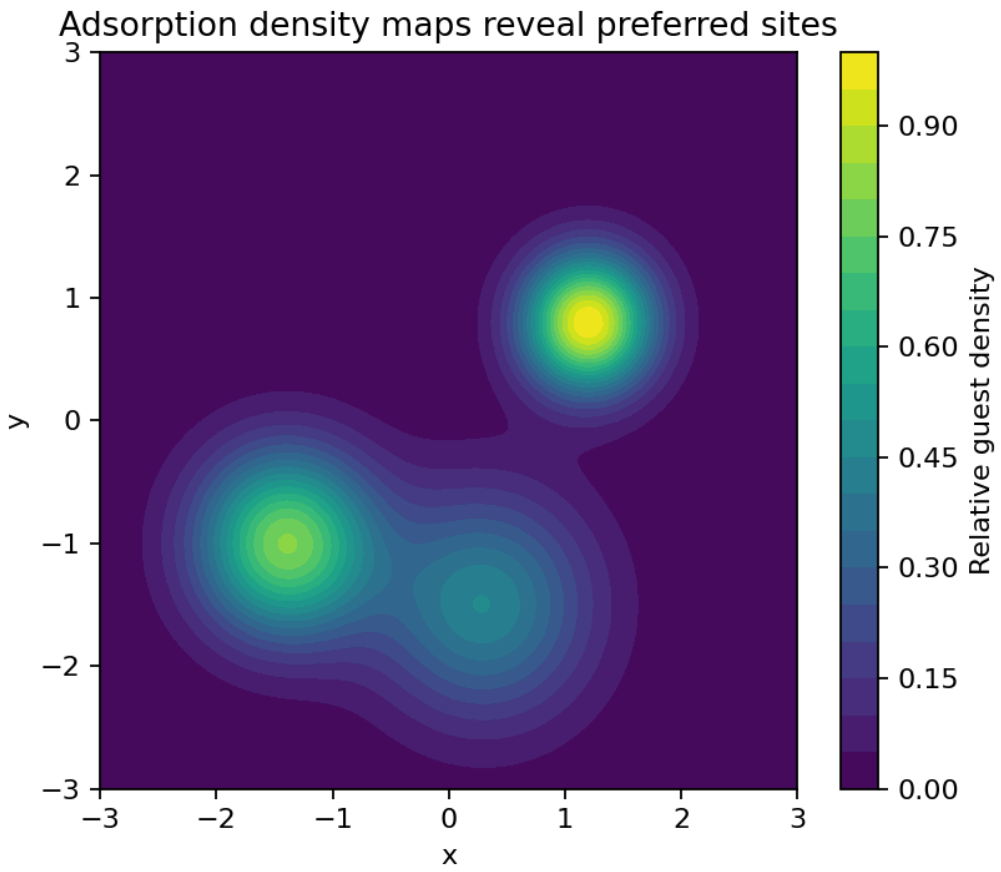


Figure 24.1: Synthetic density map illustrating how preferred adsorption regions appear as high-probability pockets.

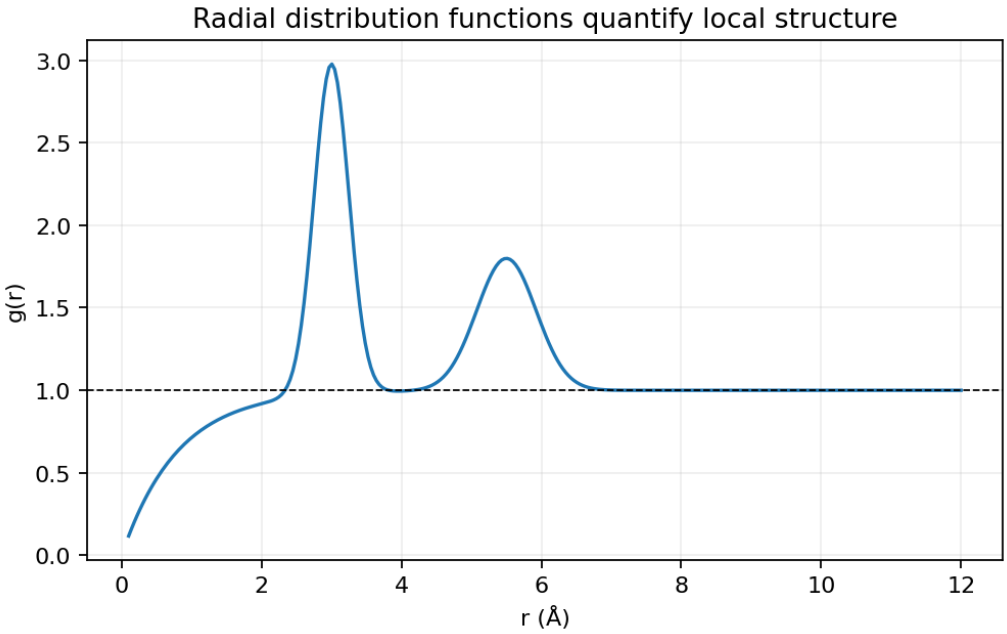


Figure 24.2: Illustrative radial distribution function for identifying local coordination shells.

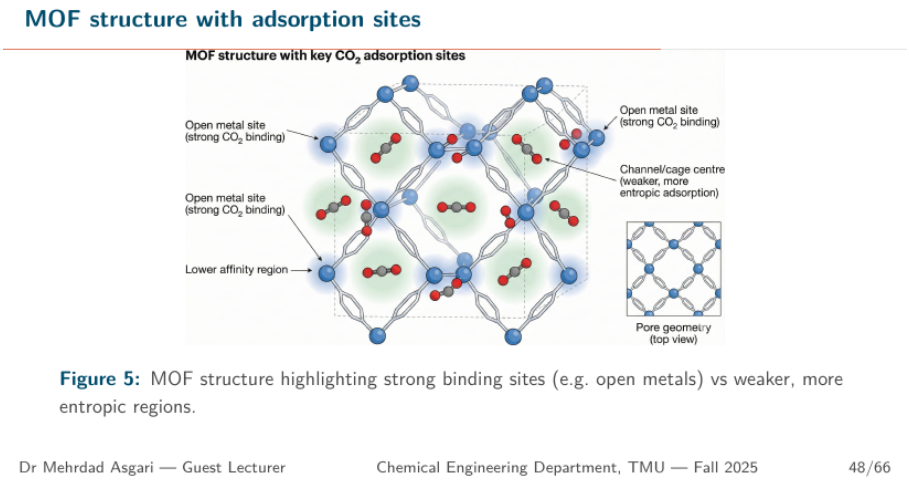


Figure 5: MOF structure highlighting strong binding sites (e.g. open metals) vs weaker, more entropic regions.

Figure 24.3: Original teaching slide: strong and weak adsorption environments in a MOF.

25.3 Block averaging

Divide the stationary trajectory into blocks longer than the correlation time, compute the mean of each block, and estimate the standard error from variation among block means. Increase block length until the estimated uncertainty stabilises.

25.4 Replicate runs

Independent seeds provide a useful diagnostic for difficult systems. Agreement within estimated uncertainty increases confidence that the chain is sampling the same stationary distribution. Disagreement suggests insufficient sampling, metastability, or a setup inconsistency.

25.5 Convergence criteria

Do not use one universal number of cycles for every material and pressure. Low-pressure Widom calculations, high-loading GCMC, mixture adsorption, and water clustering have very different correlation times. Define stopping criteria based on uncertainty of the target observable.

Research reporting

Report initialization cycles, production cycles, sampling frequency, uncertainty method, and at least one convergence check.

26 Validation against experiment

Validation is not a ceremonial final plot. It is the main way to learn which modelling assumptions are adequate for the intended chemical domain.

26.1 Like-for-like comparison

Match temperature, gas purity/composition, pressure definition, uptake basis, activation state, and crystal form. If the experiment is an excess isotherm, convert simulation consistently. If the experimental sample contains binder or nonporous mass, know whether uptake was normalised to total sample or active adsorbent.

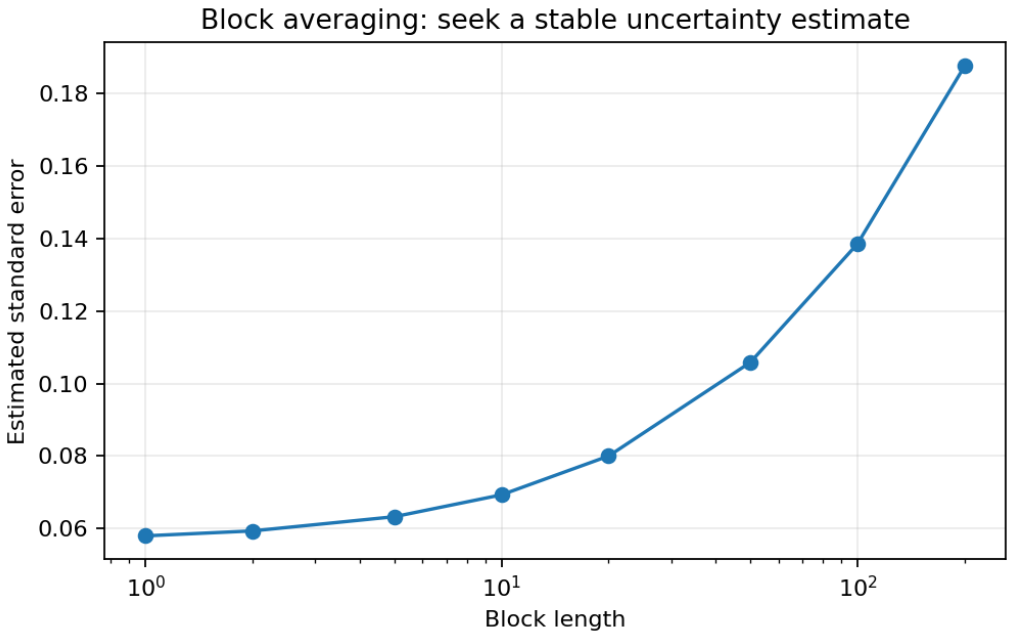


Figure 25.1: Block averaging is used to obtain an uncertainty estimate that accounts for correlation.

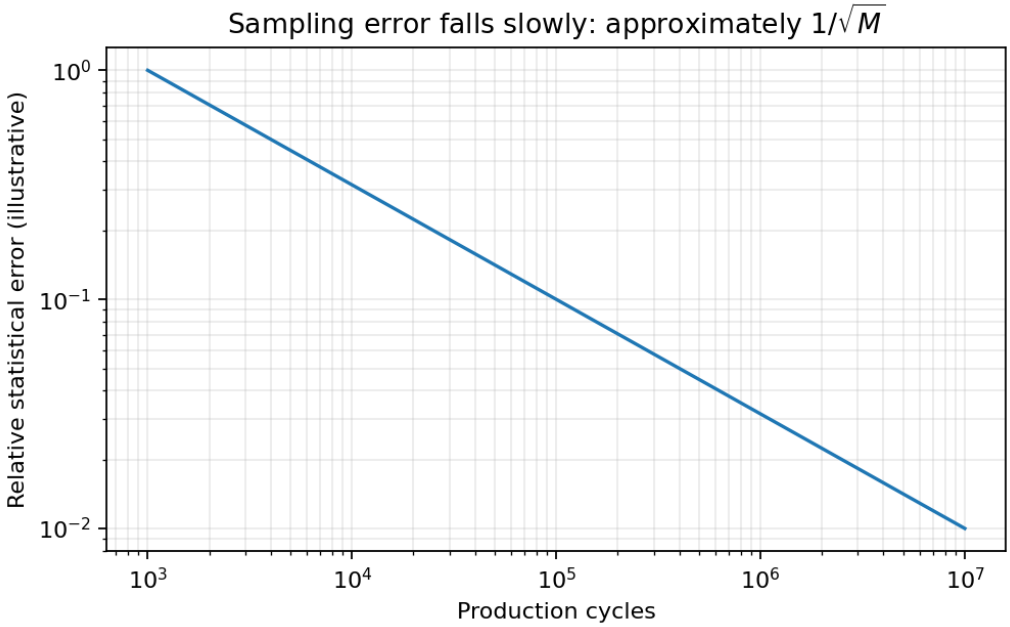


Figure 25.2: Statistical error generally falls only as the inverse square root of effective sample count.

26.2 Shape before score

Inspect where disagreement occurs. Low-pressure error points to affinity, charges, strong sites, or missing heterogeneity. High-pressure capacity error points to accessible pore volume, framework density, or absolute/excess conversion. A shifted knee can indicate the wrong energetic scale. A single RMSE hides this diagnostic structure.

26.3 Avoid retuning on the test set

If force-field parameters are fitted to one experimental isotherm, that material/condition is calibration evidence, not independent validation. Validate on other temperatures, gases, or frameworks whenever possible.

26.4 What good agreement means

Agreement does not prove the force field is universally correct. Compensating errors are possible. Use orthogonal evidence such as heats of adsorption, site locations, mixture data, or different temperatures to test mechanism.

Example figure: simulation vs experimental isotherm

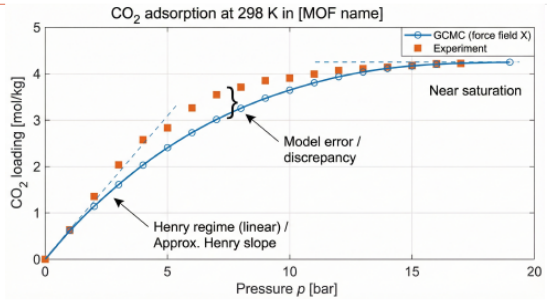


Figure 4: CO_2 adsorption isotherm at 298 K: comparison of GCMC simulation with experimental data.

Dr Mehrdad Asgari — Guest Lecturer Chemical Engineering Department, TMU — Fall 2025 47/66

Figure 26.1: The supplied seminar emphasises simulation–experiment comparison as part of the standard workflow.

Validation ladder

Structure sanity $ightarrow$ one-molecule energies $ightarrow$ Henry coefficient $ightarrow$ pure isotherm $ightarrow$ heat/site data $ightarrow$ mixture adsorption $ightarrow$ process-relevant validation.

27 Working capacity, selectivity, and process-relevant molecular descriptors

Molecular simulation becomes most useful for engineering when microscopic observables are translated into state-to-state quantities relevant to a separation process.

27.1 Working capacity

For a pressure swing,

$$\Delta q = q(P_{\mathrm{ads}}, T) - q(P_{\mathrm{des}}, T).$$

For a temperature swing the two state points differ in temperature as well as pressure. High uptake alone does not imply high working capacity if the material retains the gas too strongly at regeneration conditions.

27.2 Selectivity versus recovery

Equilibrium selectivity describes partitioning at equilibrium. Process recovery depends on cycle sequence, co-adsorption, gas holdup, mass transfer, and purge/recycle decisions. Treat selectivity as a descriptor rather than a complete process objective.

27.3 Volumetric metrics

Gravimetric capacity should be paired with crystal or packed density when vessel size matters. A low-density MOF may have impressive mmol/g uptake but mediocre mol/L capacity. Pellet voidage and binder further modify the process-scale volumetric capacity.

27.4 Heat and regenerability

Strong adsorption can be beneficial at dilute feed but may increase thermal or vacuum regeneration duty. A balanced screening set often includes low-pressure affinity, working capacity, selectivity, heat of adsorption, and density rather than ranking on one quantity.

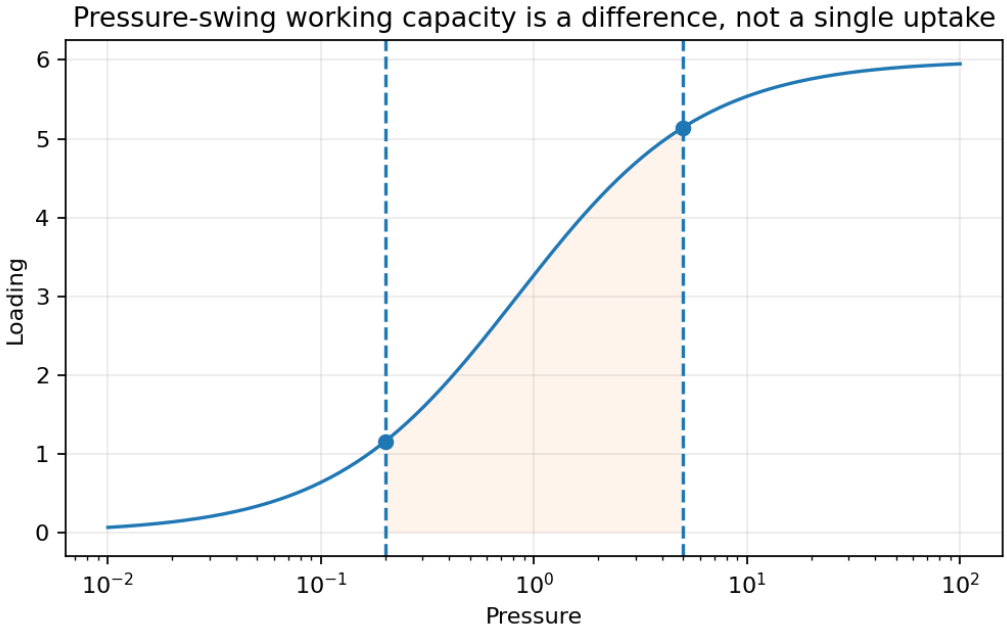


Figure 27.1: Working capacity is the difference between adsorption and regeneration state points.

28 Molecular dynamics and diffusion: what GCMC does not provide

GCMC samples equilibrium configurations without representing physical time. Transport requires a dynamical method or another kinetic model.

28.1 MC time is not physical time

Monte Carlo moves are artificial configuration updates chosen for sampling efficiency. One MC cycle is not a second, and an insertion move is not a physical diffusion event. Therefore ordinary GCMC cannot directly produce self-diffusion coefficients or transport rates.

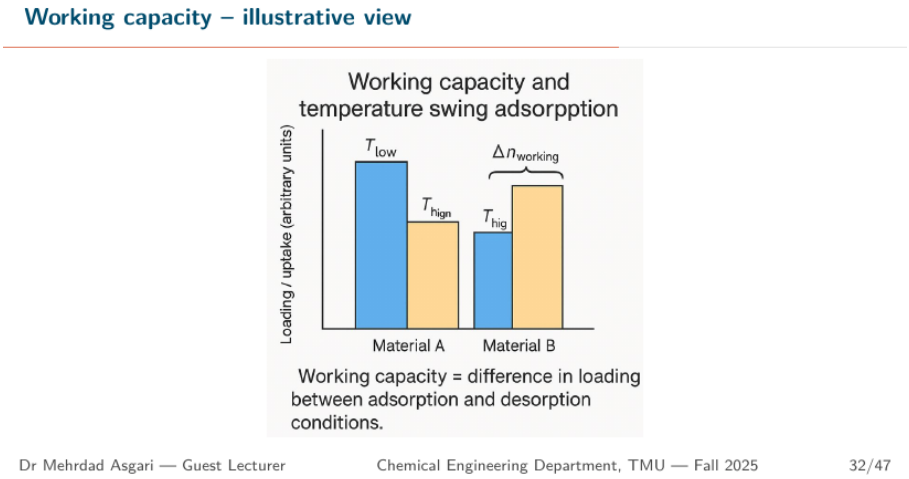


Figure 27.2: Original teaching slide: the motivation for state-to-state working capacity.

28.2 Molecular dynamics

MD integrates Newton's equations of motion using a force field. From trajectories one can calculate mean-squared displacement, velocity autocorrelation, and self-diffusion. In confined pores, long simulations may be needed because diffusion is slow and anisotropic.

28.3 Transport versus equilibrium

A material can have excellent equilibrium selectivity but poor diffusion. Conversely, kinetic sieving can create separation even when equilibrium selectivity is modest. Process screening should therefore ask whether the intended technology is equilibrium-controlled, kinetic, or mixed.

28.4 Coupled workflows

A practical multiscale workflow may use GCMC for equilibrium loading and thermodynamic factors, MD for diffusivity, and a continuum or cyclic model for process dynamics. Keep the statistical meaning of each level separate.

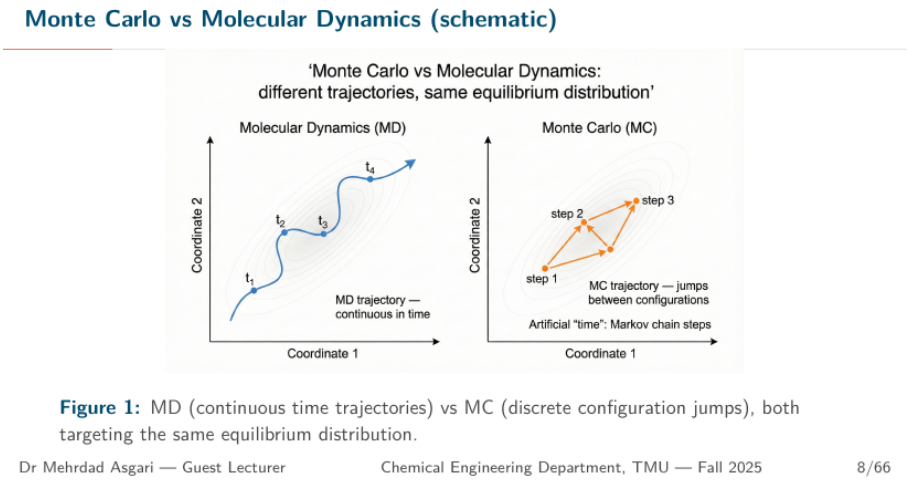


Figure 1: MD (continuous time trajectories) vs MC (discrete configuration jumps), both targeting the same equilibrium distribution.

Figure 28.1: Original seminar comparison of MD trajectories and MC configuration jumps.

29 Advanced Monte Carlo moves and difficult adsorption problems

Conventional insertion/deletion is simple but can become inefficient for dense pores, chain molecules, or strongly associating fluids.

29.1 Reinsertion and rotation

Reinsertion moves remove a molecule and place it elsewhere, improving spatial mixing without changing particle number. Rotation moves are essential for anisotropic molecules such as CO2 or N2 because orientation relative to pore electric fields matters.

29.2 Configurational-bias Monte Carlo

CBMC builds a molecule segment by segment using biased trial positions and corrects the bias through Rosenbluth weights. It is particularly useful for chain molecules, where random insertion of the entire molecule would almost always overlap.

29.3 Continuous fractional component methods

CFCMC introduces a fractional molecule whose coupling to the system is gradually changed. This can improve particle exchange in dense phases or tight pores. The method is more advanced but important for understanding modern RASPA capabilities.

29.4 Identity-change moves

For mixtures, changing the identity of a molecule can accelerate composition sampling when direct insertion of one component is difficult. The acceptance rule must include the correct chemical-potential and proposal factors.

Escalation rule

Use the simplest move set that demonstrably converges. Add advanced moves because sampling diagnostics demand them, not because they sound sophisticated.

30 Open metal sites, polarization, and other hard chemistry

MOFs with coordinatively unsaturated metals are scientifically important and computationally demanding because simple generic potentials may miss specific interactions.

30.1 Why open metal sites are hard

A guest such as CO2 can interact strongly with a localized metal centre through electrostatics, induction, quadrupole interactions, and specific coordination. Pairwise Lennard–Jones plus fixed point charges may not reproduce the correct orientation or binding energy.

30.2 Possible strategies

Options include force fields fitted to quantum-chemical interaction energies, off-centre charges or dummy sites, polarization models, machine-learned potentials, or direct electronic-structure calculations for representative adsorption configurations. Each strategy increases cost and decreases transferability unless carefully designed.

30.3 Validation targets

For strong sites, validate not only uptake but also adsorption enthalpy, preferred geometry, spectroscopic signatures, or crystallographic guest positions where available. A force field that matches the isotherm for the wrong microscopic reason may fail under mixture or temperature changes.

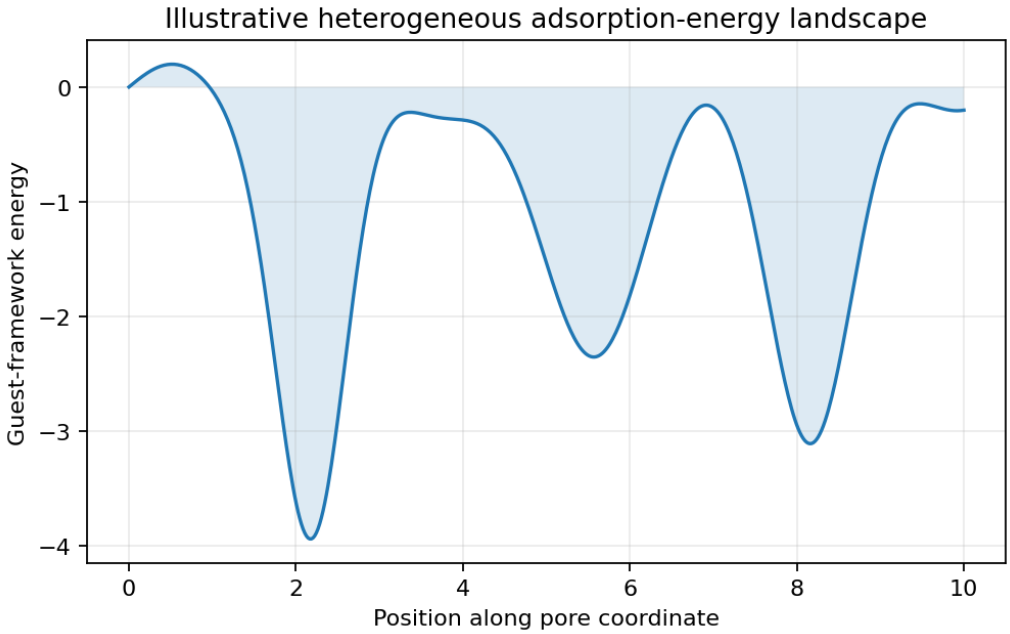


Figure 30.1: Heterogeneous energy landscapes are especially important when a small number of strong sites dominate low-pressure adsorption.

Screening implication

If an entire database is screened with a generic force field, treat open-metal-site frameworks as a potential high-uncertainty subgroup and validate separately.

31 High-throughput molecular simulation and data management

Scaling from one MOF to thousands changes the main problem from running a simulation to managing assumptions, provenance, failures, and reproducibility.

31.1 Standardise inputs

Use a consistent structure-cleaning pipeline, atom-typing policy, charge source, guest models, cutoffs, and state grid. Record a manifest for every calculation containing structure identifier, input hash, software version, force field, random seed, and run status.

31.2 Fail loudly

Do not silently skip unknown atom types, missing charges, impossible cells, or nonconverged simulations. Assign explicit failure codes. A machine-learning model trained on silently filtered successes can inherit severe selection bias.

31.3 Checkpoint and resume

Long screening campaigns should write intermediate results atomically and resume from completed state points. Store raw outputs for a small audit subset and compact parsed summaries for the full dataset.

31.4 Quality-control metrics

Automated checks can include finite energies, reasonable accessible volume, insertion/deletion acceptance, stationarity, relative uncertainty, monotonicity flags, mass/unit conversion checks, and replicate agreement for sampled cases.

31.5 From simulation to ML

Molecular-simulation outputs can be labels for machine learning, but model error in the simulation becomes label error. Report the simulation fidelity explicitly and use uncertainty or domain flags so a downstream ML model does not present force-field assumptions as experimental truth.

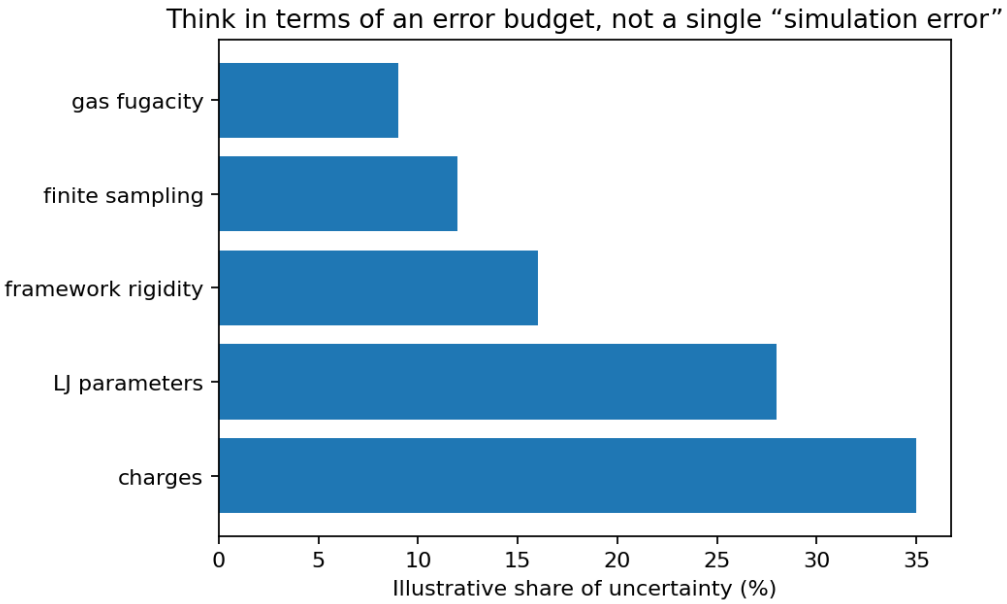


Figure 31.1: A useful high-throughput mindset is to maintain an explicit error budget rather than treating simulation as exact labels.

Data principle

The expensive part of a large simulation campaign is often not CPU time but the cost of losing provenance.

32 RASPA2: the classic workflow and input anatomy

Many established adsorption workflows and published examples use RASPA2-style text input. The supplied seminar slides use this syntax because it exposes the mapping between thermodynamic ideas and simulation keywords.

32.1 Core files

A traditional RASPA2 calculation typically uses a `simulation.input` file, a framework CIF, molecule definitions, pseudo-atom definitions, and force-field parameter files. The simulation input declares the ensemble, cycles, framework, unit-cell replication, temperature, pressure, and move probabilities.

32.2 Conceptual input

The following intentionally compact example mirrors the teaching slide. Exact names depend on the installed force-field library and RASPA version.

```
SimulationType MonteCarlo
NumberOfCycles 50000
NumberOfInitializationCycles 10000
PrintEvery 1000
Forcefield GenericMOFs
Framework 0
FrameworkName IRMOF-1
UnitCells 2 2 2
ExternalTemperature 298.0
ExternalPressure 100000.0
UseChargesFromCIFFile yes
Component 0 MoleculeName CO2
MoleculeDefinition TraPPE
TranslationProbability 0.3
RotationProbability 0.3
SwapProbability 0.4
CreateNumberOfMolecules 0
```

32.3 Keyword mapping

`ExternalTemperature` fixes T. `ExternalPressure` specifies the reservoir state that is converted to chemical potential/fugacity according to the code settings. `SwapProbability` enables particle exchange. `UnitCells` controls periodic replication. The force field and charge settings define the Hamiltonian being sampled.

32.4 Outputs to inspect

Do not extract only the final loading. Inspect initialization versus production, average loading, energy components, move acceptance, fugacity coefficient if reported, density, heat estimates, warnings, and timing. Preserve the complete output for validation cases.

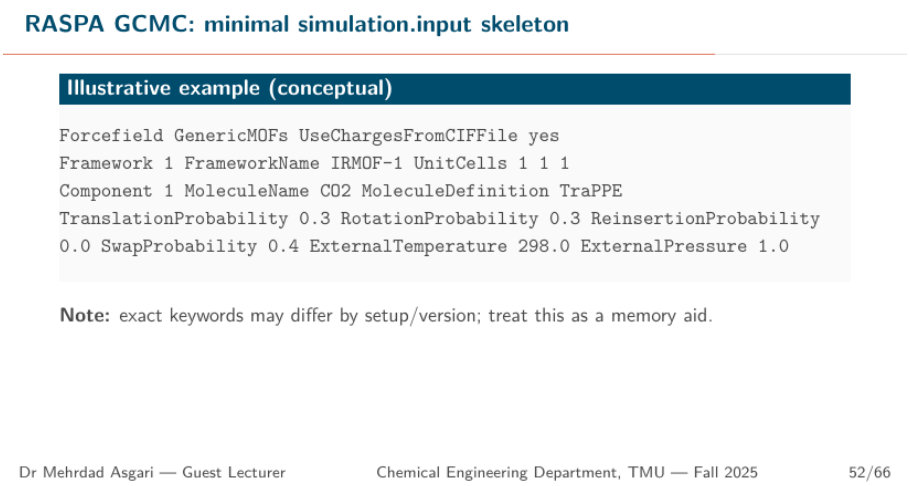


Figure 32.1: Original seminar slide: minimal RASPA-style GCMC input used as a memory aid.

Version caution

RASPA2 is mature and widely used, but its repository is archived. New students should understand its syntax because much literature uses it, while also learning RASPA3.

33 RASPA3: modern JSON inputs, outputs, and Python integration

RASPA3 is a substantial redesign. It uses JSON input/output and exposes the simulation engine to Python, making it attractive for reproducible and programmatic workflows.

33.1 What changed conceptually

The underlying statistical mechanics remains the same. The major practical changes are input/output structure and software architecture. According to the official RASPA3 documentation, input and output are JSON-based, fugacity coefficients and excess adsorption are handled automatically in supported workflows, and the C++ engine is exposed through a Python interface.

33.2 A minimal JSON-style pattern

A current RASPA3 example has the structure below. Treat it as a pattern rather than a universal template; always start from the example shipped with your installed release.

```
{
  "SimulationType": "MonteCarlo",
  "NumberOfProductionCycles": 500000,
  "NumberOfInitializationCycles": 10000,
  "Systems": [{
    "Type": "Framework",
    "Name": "MFI_SI",
    "NumberOfUnitCells": [2,2,2],
    "ExternalTemperature": 300.0,
    "ExternalPressure": 100000.0,
    "ChargeMethod": "None"
  }],
```

```
  "Components": [{
    "Name": "methane",
    "FugacityCoefficient": 1.0,
    "TranslationProbability": 0.5,
    "ReinsertionProbability": 0.5,
    "SwapProbability": 1.0,
    "WidomProbability": 1.0,
    "CreateNumberOfMolecules": 0
  }]
}
```

33.3 Why JSON helps

JSON makes nested systems/components explicit and is easy to generate and validate from Python. It also reduces fragile line-oriented parsing. A high-throughput workflow can create a validated JSON object, write it to disk, run RASPA3, and parse structured JSON output without regular-expression scraping.

33.4 Python integration

The Python interface enables notebooks, parameter sweeps, unit tests, and coupling to data-science workflows. For teaching, it allows students to construct a system, run a small simulation, and immediately plot results. For production screening, keep a separation between the orchestration layer and immutable calculation inputs so every run remains reproducible.

Current software note

RASPA3 is actively developed. Official releases and documentation should be treated as the source of truth for syntax. The conceptual equations in this handbook are stable even when keywords evolve.

34 Installing and testing RASPA on a workstation or HPC system

Installation is successful only when a known example reproduces expected behaviour. Students should learn a minimal validation sequence before launching research calculations.

34.1 Choose a version deliberately

If joining an existing group project, use the same version as the validated workflow unless there is a clear reason to migrate. RASPA2 and RASPA3 inputs are not interchangeable. For new training, RASPA3 is a sensible target while retaining RASPA2 literacy for legacy literature and datasets.

34.2 Environment isolation

Use a dedicated conda environment or module stack. Record the package version, compiler/build if relevant, and operating system. On Windows, a native RASPA3 package or a Linux environment such as WSL may be used depending on the group's validated setup. On clusters, use environment modules or containers where permitted.

34.3 Smoke test

Run an official basic example unchanged. Confirm that it completes, creates output, and produces plausible averages. Then duplicate the example into a scratch directory and change only one parameter, such as temperature or pressure. If the response is qualitatively sensible, the execution path is probably correct.

34.4 HPC considerations

Monte Carlo calculations are often embarrassingly parallel across materials, pressures, and random seeds. Rather than forcing one simulation to use many cores, submit arrays of independent state points unless the code/version has validated internal parallelism for the target method. This is usually more efficient and easier to resume.

Rule for beginners

Never debug chemistry and installation at the same time. First reproduce a shipped example, then change one thing at a time.

35 A first hands-on CO2 GCMC project

The fastest route to competence is a small project with a clearly defined deliverable. This chapter adapts the mini-project logic of the supplied seminar into a reproducible exercise.

35.1 Goal

Predict a short CO2 isotherm at 298 K in one well-behaved rigid MOF using a validated teaching force field. Use pressures 0.01, 0.1, 1, 5, and 10 bar. The exact framework is less important than choosing one supported by the installed force-field library.

35.2 Task A: preflight

Visualise the CIF. Record formula, cell lengths, framework mass, atom types, charge source, and supercell replication. Run a pore/accessibility check if available. Create a one-page data sheet before simulation.

35.3 Task B: low-cost affinity

Run Widom insertion or a low-pressure GCMC calculation. Record Henry coefficient or the approximate low-pressure slope and the adsorption energy distribution. Interpret the strongest site.

35.4 Task C: isotherm

Run the five pressure points with a common initialization/production policy. Save mean loading, uncertainty, energy, and acceptance ratios. Plot loading versus pressure on both linear and log-pressure axes.

35.5 Task D: interpretation

Explain the Henry region, knee, and approach to saturation. Identify which structural feature controls low-pressure affinity and which feature controls high-pressure capacity. State at least three modelling assumptions that could change the result.

Deliverables

1. CIF and input files.
2. Exact software/version.
3. Table of state points and uncertainties.
4. Isotherm figure.
5. One adsorption-site figure.
6. One-page interpretation.
7. README with a one-command rerun procedure.

36 A CO2/CH4 mixture project and comparison with IAST

This project introduces explicit mixture GCMC and teaches students to distinguish a cheap thermodynamic approximation from a molecular mixture calculation.

36.1 Pure-component stage

Generate validated pure CO2 and CH4 isotherms at the target temperature. Fit each with a smooth model such as single- or dual-site Langmuir, using only as many parameters as the data support.

36.2 IAST stage

Use the fitted pure isotherms to predict an equimolar or application-relevant mixture across pressure. Plot component loadings and selectivity. Check numerical stability at dilute composition and low pressure.

36.3 Mixture-GCMC stage

Run explicit binary GCMC at a small set of state points. Use reservoir fugacities consistent with the chosen gas equation of state. Compare GCMC selectivity and loadings to IAST.

36.4 Interpret discrepancies

If IAST and mixture GCMC disagree, examine site competition, component size, pressure-dependent selectivity, and guest–guest interactions. Do not declare RAST necessary until you have ruled out poor pure-isotherm fits and simulation convergence.

Research extension

Repeat across a chemically diverse mini-set of MOFs and ask: which structural descriptors predict where IAST fails? This can become a data-driven project rather than a single demonstration.

37 Linking molecular simulation to process simulation

Molecular simulation becomes especially valuable when used to provide physically grounded inputs to adsorption-cycle or membrane/process models.

37.1 Isotherm models for process simulators

Dynamic PSA/VSA/TSA models need fast repeated evaluation of multicomponent equilibrium. Raw GCMC points are therefore usually fitted to analytic models, splines, IAST/RAST representations, or surrogate models. The fit must be smooth over the entire process pressure/composition/temperature domain.

37.2 Thermal effects

Energy balances require heats of adsorption. A constant heat is simple but may be inaccurate for heterogeneous sites. If $q_{st}(q)$ varies strongly with loading, a loading-dependent representation can materially change temperature fronts and cycle performance.

37.3 Mass transfer

GCMC supplies equilibrium, not kinetic mass-transfer coefficients. Those may come from MD, experiments, or fitted LDF parameters. Do not infer kinetic constants from equilibrium acceptance probabilities.

37.4 Uncertainty propagation

A process optimum can be sensitive to molecular-level inputs. Instead of passing one fitted isotherm as exact, propagate uncertainty in force field, charge model, fit parameters, or heats through selected process cases. This identifies which molecular calculations deserve higher fidelity.

Multiscale principle

The best molecular descriptor is not necessarily the best process material. Process constraints can reverse material rankings.

38 Common failure modes and a troubleshooting logic

Troubleshooting becomes efficient when symptoms are mapped to classes of causes rather than handled by trial and error.

38.1 Zero uptake everywhere

Possible causes include pressure/fugacity units, a blocked pore, inaccessible framework, missing swap moves, enormous repulsive interactions, wrong unit-cell replication, or a guest model not found. First test one Widom insertion and inspect insertion energies.

38.2 Astronomical energies or immediate crashes

Look for overlapping atoms, duplicated CIF sites, wrong length units, missing atom types, incorrect charges, or a molecule definition with incompatible geometry. Visualise the initial cell.

38.3 Unphysical high uptake

Check framework mass conversion, duplicated periodic atoms, charge magnitudes, excessive attraction, pressure units, and whether reported numbers are molecules per cell rather than mmol/g.

38.4 Poor insertion acceptance at high loading

This may be physical. Increase production, use reinsertion or biasing methods, or start from a neighbouring pressure. If the result is still noisy, quantify uncertainty rather than hiding it.

38.5 Nonmonotonic pure isotherm

Small stochastic fluctuations are possible; large drops usually indicate insufficient convergence, phase-like metastability, unit conversion errors, or inconsistent initial states. Repeat with longer runs and independent seeds.

38.6 Simulation–experiment mismatch

Classify the mismatch by pressure region. Then test uptake convention, activation state, force field, charges, framework rigidity, defects, accessible pore volume, and experimental conditions in that order.

Debugging rule

Change one assumption at a time and preserve the baseline. Otherwise you cannot learn which change fixed the problem.

39 Reproducibility, provenance, and research-quality reporting

A molecular simulation should be rerunnable by another researcher from the archived inputs, not reconstructed from a paragraph months later.

39.1 Minimum metadata

Archive framework identifier and CIF hash, software and version, full input, force-field source, charge source, guest model, unit cells, cutoff/electrostatics settings, temperature, pressure/fugacity, cycle counts, random seed, and post-processing script.

39.2 Immutable raw outputs

Keep raw outputs separate from analysis products. Parse them into a tidy table but do not overwrite them. A future code update may change a parser or reveal a need for an observable you did not initially extract.

39.3 Version control

Track scripts, templates, and analysis in Git. Large raw trajectories may live outside Git with checksums and a manifest. Tag the exact commit used for a paper.

39.4 Reproducible figures

Every figure should have machine-readable source data or be reproducible from archived outputs by a script. This is especially important for high-throughput screening where manual plotting can silently mix conditions.

39.5 Negative and failed runs

Preserve failure logs and unsupported cases. Removing them from the dataset without record can bias conclusions about chemical-domain coverage.

Publication checklist

Can a reader determine exactly what was simulated? Can they reproduce one representative state point? Can they distinguish raw simulation output from fitted or derived quantities? Can they see which cases failed?

40 Designing a research project around molecular simulation

The final skill is not running GCMC but asking a question that benefits from it. This chapter offers project archetypes suitable for students with different backgrounds.

40.1 Project type 1: descriptor physics

Use a curated set of MOFs and calculate Henry coefficients, heats, capacities, and site-density features. Ask which geometric and chemical descriptors control each regime. A descriptor ladder can move from geometry-only to chemistry-aware to energy-based descriptors and test where each level adds predictive value.

40.2 Project type 2: force-field fidelity map

Select families with and without open metal sites, polar functional groups, or flexibility. Compare generic and specialised force fields against experimental isotherms/enthalpies. Map where a low-cost model is trustworthy and where it should abstain or escalate.

40.3 Project type 3: mixture-model validity

Compare IAST to explicit mixture GCMC across chemical families, pressures, and compositions. Learn a domain-of-validity model that predicts when explicit mixture simulation is worth the cost.

40.4 Project type 4: molecular-to-process ranking

Compute equilibrium data for a manageable material set, fit consistent process-ready isotherms, and propagate them through a simplified PSA/VSA model. Quantify rank reversals between molecular metrics and process performance.

40.5 Project type 5: active simulation

Instead of simulating every material at every state point, use uncertainty-aware active learning to select the next calculation that most reduces ranking uncertainty or process decision regret. This combines molecular simulation and data-driven techniques in a natural way.

A good student project

Has one falsifiable scientific question, one defined material domain, a validation plan, a bounded compute budget, and a clear statement of what would count as failure.

A1 Mathematical primer for molecular simulation

A1.1 Exponentials and Boltzmann factors

The exponential function appears because statistical weights multiply when independent energy contributions add. If two states differ by energy ΔU, the ratio of Boltzmann weights is $\exp(-\beta\Delta U)$. At room temperature, $RT \approx 2.48\,\mathrm{kJ\,mol^{-1}}$, so a seemingly modest energy difference of $5\,\mathrm{kJ\,mol^{-1}}$ changes a Boltzmann factor by roughly an order of magnitude. This is why small force-field errors can produce large low-pressure adsorption errors.

A1.2 Logarithms and free energies

Free energies convert products of statistical weights into sums through logarithms. If Z is a partition function, $F = -k_{\mathrm{B}}T \ln Z$. Chemical potentials are free-energy derivatives and therefore contain logarithms of activity/fugacity. Whenever a pressure axis spans many orders of magnitude, a logarithmic plot is usually more informative.

A1.3 Probability, averages, variance, covariance

For samples A_k, the arithmetic mean is $\bar{A} = M^{-1}\sum_k A_k$. The variance is the mean squared deviation from the mean. Covariance measures whether two quantities fluctuate together,

$$\mathrm{cov}(A,B) = \langle AB\rangle - \langle A\rangle\langle B\rangle.$$

Grand-canonical fluctuation formulae often contain $\mathrm{cov}(U,N)$ and $\mathrm{var}(N)$.

A1.4 Derivatives and response functions

Adsorption slopes, heat capacities, compressibilities, and other response functions are derivatives of state variables or free energies. Numerically, derivatives amplify noise. Prefer analytic derivatives of fitted physically sensible models or use uncertainty-aware finite differences.

A1.5 Dimensional analysis

Before trusting a formula, check units. βU must be dimensionless. A Henry coefficient must carry whatever units convert pressure/fugacity into the chosen loading basis. A simulation code may use Kelvin as an energy unit through U/k_{B}; another may report kJ/mol. Unit conversion is a frequent source of hundred- or thousand-fold errors.

A2 Derivation sketch: GCMC insertion and deletion

Consider a grand-canonical configurational probability proportional to

$$\pi_N(\mathbf{r}^N) \propto \frac{1}{N!\Lambda^{3N}} \exp[\beta\mu N - \beta U(\mathbf{r}^N)].$$

For an insertion proposal that chooses a uniform position in volume V, the proposal-density ratio between deletion and insertion contributes $V/(N+1)$. Combining this with the target-probability ratio yields the familiar acceptance factor

$$\frac{V}{(N+1)\Lambda^3}\exp(\beta\mu)\exp(-\beta\Delta U).$$

The deletion rule is its reciprocal in the reverse direction. For molecules, rotational and internal partition functions as well as biased construction weights modify the algebra. Software implementations may absorb constants into activity or fugacity definitions. The pedagogical lesson is the same: acceptance contains a reservoir term, a combinatorial/volume term, and an energetic term.

A3 Units and conversion tables

Quantity	Common units	Notes
Pressure	Pa, kPa, bar, atm	$1\,\mathrm{bar} = 10^5\,\mathrm{Pa}$
Temperature	K	Never use Celsius in thermodynamic equations without conversion
Energy	K, kJ/mol, kcal/mol	"Kelvin energy" usually means U/k_B
Loading	molecule/cell, mol/kg, mmol/g	Convert using framework formula mass and replication
Density	$\mathrm{kg/m^3}$, $\mathrm{g/cm^3}$	Distinguish crystal and packed density
Henry coefficient	many conventions	Always state loading and pressure/fugacity basis
Diffusion	$\mathrm{m^2/s}$	Requires physical dynamics, not ordinary MC

A4 RASPA2 quick-reference checklist

Before the run

Framework CIF visually inspected; atom labels match force-field typing; charges present if used; molecule definition selected; cutoff compatible with supercell; temperature and pressure units confirmed; initialization and production cycles declared; insertion/deletion moves enabled.

After the run

No fatal warnings; loadings finite; energy finite; production trace stationary; insertion/deletion statistics plausible; uncertainty estimated; unit conversion checked; raw output archived; software version recorded.

A5 RASPA3 quick-reference checklist

Start from the closest official example distributed with the exact installed release. Keep JSON under version control and validate it with a parser before submission. Prefer explicit local paths or a documented data directory for force fields/molecule definitions. Parse structured JSON output rather than scraping terminal text. Record the release version. RASPA3 documentation describes JSON input/output and a Python interface; these features make automated validation straightforward but do not remove the need to understand the statistical mechanics.

A6 Glossary

Term	Meaning
Adsorption isotherm	Equilibrium loading as a function of gas pressure or fugacity at fixed temperature.
Chemical potential	Free-energy cost of adding particles; equal across phases at diffusive equilibrium.
Fugacity	Effective pressure entering the chemical potential of a real gas.
Henry coefficient	Infinite-dilution slope relating loading to pressure/fugacity.
Langmuir model	Saturating single-site adsorption model with capacity and affinity parameters.
Dual-site Langmuir	Sum of two Langmuir populations, often used for heterogeneous adsorption.
GCMC	Grand Canonical Monte Carlo; fixed μ, V, T with fluctuating particle number.
Metropolis rule	Acceptance rule that samples a target Boltzmann-type distribution.
Detailed balance	Pairwise balance condition sufficient for the desired stationary distribution.
Widom insertion	Ghost-particle method for excess chemical potential and Henry-regime properties.
Isosteric heat	Heat/enthalpy quantity inferred at constant adsorbed loading.
Force field	Mathematical potential-energy model and associated parameters.
Mixing rule	Rule generating unlike-pair nonbonded parameters from pure atom-type parameters.
Ewald summation	Periodic long-range electrostatic summation split into real and reciprocal space.
IAST	Ideal Adsorbed Solution Theory for mixture adsorption from pure-component isotherms.
RAST	Real Adsorbed Solution Theory with adsorbed-phase non-ideality.
Absolute adsorption	Number of molecules assigned to the pore/adsorbed region.
Excess adsorption	Absolute adsorption minus a bulk-fluid reference amount in the pore volume.
Working capacity	Difference in loading between adsorption and regeneration state points.
Block averaging	Statistical method for uncertainty estimation in correlated trajectories.
Autocorrelation time	Characteristic correlation scale determining effective sample size.
Framework rigidity	Approximation in which host coordinates are held fixed during simulation.
Open metal site	Coordinatively unsaturated metal centre that may bind guests strongly.
Configurational-bias MC	Biased molecule construction method with exact statistical correction.
CFCMC	Continuous Fractional Component Monte Carlo, a staged particle-coupling method.

A7 Twenty-five practice worksheets

The following worksheets are designed for group training. Each is intentionally one page so students can complete it during or after a meeting.

Worksheet 1: Isotherm reading

Task

Annotate an isotherm with Henry region, knee, and saturation. Explain what physical information each region contains.

Notes / answer:

Worksheet 2: Unit conversion

Task

Convert molecules per unit cell to mmol/g using a framework formula mass and replication factor. Write every step.

Notes / answer:

Worksheet 3: Chemical potential

Task

Explain why pressure is not the natural internal variable of a rigid pore and why chemical potential is.

Notes / answer:

Worksheet 4: Langmuir fitting

Task

Given synthetic data, decide whether a single-site Langmuir fit is identifiable. List diagnostics.

Notes / answer:

Worksheet 5: Dual-site caution

Task

Construct two different DSL parameter sets that could give similar curves over a narrow pressure range. Explain non-identifiability.

Notes / answer:

Worksheet 6: Temperature

Task

Sketch how an exothermic adsorption isotherm changes from 273 to 323 K and explain why.

Notes / answer:

Worksheet 7: Heat definitions

Task

Write a paragraph distinguishing calorimetric differential heat and isosteric heat.

Notes / answer:

Worksheet 8: Ensembles

Task

Match NVE, NVT, NPT, and muVT to four physical systems and justify the choices.

Notes / answer:

Worksheet 9: Metropolis

Task

For three trial energy changes, compute qualitative acceptance at high and low temperature.

Notes / answer:

Worksheet 10: GCMC insertion

Task

Break the insertion acceptance rule into entropic, reservoir, and energetic contributions.

Notes / answer:

Worksheet 11: Widom

Task

Describe why a single rare favourable insertion can dominate an exponential average.

Notes / answer:

Worksheet 12: Force field

Task

List every assumption needed to compute CO2-framework energy in a charged MOF.

Notes / answer:

Worksheet 13: Cutoff

Task

Given a 12.8 A cutoff and cell lengths, choose replication factors and justify them.

Notes / answer:

Worksheet 14: Charges

Task

Compare two plausible charge-assignment methods and list what should be validated experimentally.

Notes / answer:

Worksheet 15: Rigid host

Task

Choose a MOF where rigidity is probably safe and one where it may fail; justify using structural behaviour.

Notes / answer:

Worksheet 16: Pure isotherm QC

Task

Design five automated checks for a high-throughput pure-component GCMC campaign.

Notes / answer:

Worksheet 17: Mixture selectivity

Task

Explain why pure-component uptake ratios are not mixture selectivities at finite loading.

Notes / answer:

Worksheet 18: IAST

Task

Draw an IAST workflow from pure-component GCMC to mixture prediction.

Notes / answer:

Worksheet 19: Absolute vs excess

Task

Explain how excess uptake can decrease while absolute uptake still rises.

Notes / answer:

Worksheet 20: Density map

Task

Interpret a hypothetical density map and propose which framework atoms define the strongest site.

Notes / answer:

Worksheet 21: Uncertainty

Task

Given a correlated loading trace, design an equilibration and block-averaging procedure.

Notes / answer:

Worksheet 22: Validation

Task

Create a like-for-like checklist for comparing a simulated CO2 isotherm to experiment.

Notes / answer:

Worksheet 23: Working capacity

Task

Rank three hypothetical materials using working capacity rather than adsorption-pressure uptake.

Notes / answer:

Worksheet 24: RASPA input

Task

Take a minimal RASPA input and map every keyword to a physical assumption.

Notes / answer:

Worksheet 25: Research proposal

Task

Write a one-page molecular-simulation project with hypothesis, material domain, validation, compute budget, and stop criteria.

Notes / answer:

A8 Suggested answers and discussion points for worksheets

1. Henry region reflects infinite-dilution affinity; the knee reflects filling of strong sites and onset of crowding; saturation reflects finite pore/site capacity.
2. Convert the average molecule count to moles using Avogadro's number and divide by framework mass represented in the simulated supercell. The replication cancels only if both numerator and denominator use the same supercell.
3. A rigid pore has no freely adjustable bulk volume; particle exchange is governed by equality of chemical potentials.
4. Check pressure coverage, residuals, parameter covariance, physical saturation capacity, and predictive stability under leave-out tests.
5. Four-parameter fits are often underdetermined on a narrow pressure range; parameters may exchange roles while producing nearly identical total uptake.
6. Exothermic adsorption generally weakens as temperature increases, shifting the isotherm to higher pressure and lowering uptake at fixed pressure.
7. Boundary conditions and thermodynamic paths differ; state the definition and sign convention.
8. NVE: isolated; NVT: closed fluid in a thermal bath; NPT: closed fluid under pressure bath; muVT: open adsorbing pore connected to a gas reservoir.
9. Downhill moves accept; uphill acceptance is $e^{-\beta\Delta U}$ and is larger at higher temperature.
10. The $V/(N+1)$ factor represents configurational/combinatorial availability, $e^{\beta\mu}$ the reservoir drive, and $e^{-\beta\Delta U}$ the energetic penalty/reward.
11. Exponential averaging gives very large weights to favourable energies; rare-event convergence must be checked.
12. Framework coordinates, atom types, LJ parameters, mixing rules, charges, Coulomb treatment, cutoff, and guest model are minimum requirements.
13. Replicate until box dimensions are compatible with the cutoff/minimum-image treatment; verify with the exact software convention.
14. Compare predicted low-pressure uptake, heat, site geometry, and transferability across materials rather than choosing by convenience.
15. Rigid host is safer for robust frameworks far from transitions and risky for breathing/gate-opening materials.
16. Examples: finite energies, stationary loading, uncertainty threshold, sensible acceptance, no missing atom types, reproducible replicate seed.
17. Mixture adsorption involves competition for sites and pore volume; pure-component curves do not encode that competition.
18. Fit validated pure isotherms, integrate spreading pressures, solve equal-spreading-pressure relations, recover mixture composition and loading, then validate selected points with mixture GCMC.
19. Excess subtracts a growing bulk-fluid reference amount from absolute pore occupancy.
20. Locate density maxima, inspect nearest framework atoms, decompose interaction energy, and verify occupancy versus pressure.
21. Remove equilibration, estimate autocorrelation/block size, compute block means, report standard error, and repeat with longer production.
22. Match temperature, gas, pressure, uptake convention, activation state, sample basis, and experimental uncertainty before scoring error.
23. Use $q(P_{ads}, T) - q(P_{des}, T)$, not the highest single uptake.

24. Map temperature, pressure/fugacity, framework, cell replication, force field, charges, and move probabilities to the thermodynamic/model assumptions.
25. A good proposal specifies one question, bounded domain, independent validation, compute plan, and explicit evidence that would falsify the hypothesis.

A9 Twenty-five worked training examples

These examples are intended for guided discussion after the conceptual chapters. Each example is placed on a separate page so it can be used as a handout.

Worked example 1: Langmuir loading calculation

Problem and discussion

A material follows $q = q_s bp/(1 + bp)$ with $q_s = 5$ mmol g^{-1} and $b = 0.8$ bar^{-1}. Calculate q at 0.1, 1, and 10 bar. Then identify which point is closest to the Henry regime.

Discussion. At 0.1 bar, $bp = 0.08$ and the linear approximation is good; at 10 bar the material is near saturation. The calculation illustrates that "high affinity" changes the pressure at which nonlinearity appears.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 2: Henry coefficient from Langmuir parameters

Problem and discussion

For the same Langmuir model, show that $K_H = q_s b$ as $p \to 0$. Explain why fitting q_s and b separately from only low-pressure data is difficult.

Discussion. Only the product $q_s b$ is strongly constrained by the initial slope; independent capacity information requires data extending into curvature/saturation.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 3: Fugacity correction

Problem and discussion

At a state point with $p = 50$ bar and fugacity coefficient $\phi = 0.85$, compute f. Explain the consequence of using p instead of f in a GCMC activity.

Discussion. $f = 42.5$ bar. The chemical-potential error is $RT \ln(p/f) = RT \ln(1/0.85)$, which can shift loading substantially in a steep part of the isotherm.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 4: Metropolis acceptance

Problem and discussion

At 300 K, compare the qualitative acceptance of trial moves with $\Delta U = -5$, $+2$, and $+10$ kJ mol^{-1}.

Discussion. Negative ΔU accepts automatically. Positive changes accept with $e^{-\Delta U/RT}$, so modest energetic penalties can already reduce acceptance strongly.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 5: GCMC insertion anatomy

Problem and discussion

Take the insertion rule and label each factor as reservoir, configurational/combinatorial, or energetic. Describe what happens qualitatively when pressure rises, the pore becomes crowded, or temperature increases.

Discussion. Higher activity favours insertion; crowding makes ΔU more positive and insertion less likely; temperature weakens the discrimination of a fixed energy difference.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 6: Two-region Widom toy model

Problem and discussion

Imagine 5% of the accessible volume has insertion energy -10 kJ mol^{-1} and 95% has -2 kJ mol^{-1} at 298 K. Write the weighted Boltzmann average and decide which region dominates K_H.

Discussion. The exponential weighting makes the small strong region disproportionately important; this is the energetic–entropic competition behind Henry affinity.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 7: Isosteric-heat covariance

Problem and discussion

Explain why a covariance between U and N appears in a grand-canonical heat estimator. What would it mean if high-N states systematically have much lower energy?

Discussion. Particle-number fluctuations probe the energetic cost of adding particles. Strong negative correlation signals favourable adsorption energetics associated with increasing loading.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 8: Molecules per cell to mmol per gram

Problem and discussion

Write the general conversion from $\langle N \rangle$ molecules in a supercell to mmol g^{-1} using Avogadro's number and the molar mass of framework contained in that supercell.

Discussion. Keep numerator and framework mass on the same replicated-cell basis. This simple accounting check prevents many apparent "giant uptake" errors.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 9: Absolute to excess adsorption

Problem and discussion

Given $n_{abs} = 8$ mmol g^{-1}, bulk-fluid density expressed on a compatible molar basis, and a pore volume per gram, compute $n_{ex} = n_{abs} - \rho_{bulk} V_{pore}$. List the assumptions hidden in V_{pore}.

Discussion. The pore-volume convention is part of the reported observable, not a cosmetic post-processing choice.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 10: Mixture selectivity

Problem and discussion

For $y_{CO2} = 0.15$ and an adsorbed phase with $x_{CO2} = 0.60$, calculate $S_{CO2/CH4} = (x_{CO2}/x_{CH4})/(y_{CO2}/y_{CH4})$. Explain why the same material can have a different selectivity at another pressure.

Discussion. Site saturation and competitive occupancy change x_i with pressure even at fixed gas composition.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 11: IAST data-quality check

Problem and discussion

You have excellent high-pressure pure isotherms but only one very noisy point below 0.1 bar. Would you trust an IAST prediction for a dilute CO2 mixture? Why?

Discussion. No. The spreading-pressure integral and dilute selectivity are sensitive to the low-pressure representation; the missing Henry region should be repaired first.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 12: Cutoff and supercell

Problem and discussion

A framework has cell lengths 9, 14, and 18 Å. Your validated short-range cutoff is 12 Å. Propose replication factors that make each box length comfortably exceed twice the cutoff.

Discussion. A simple starting choice is $3\times2\times2$, giving 27, 28, and 36 Å; confirm against the exact software minimum-image convention.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 13: Charge-model sensitivity

Problem and discussion

Two charge schemes give similar high-pressure uptake but very different low-pressure uptake. What does this pattern suggest?

Discussion. High-pressure filling is increasingly capacity/packing dominated, while the Henry region is highly sensitive to local electrostatic affinity. Low-pressure experimental validation is therefore decisive.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 14: Block averaging design

Problem and discussion

A trace contains 200,000 production samples with visible correlations over about 100 samples. Propose a block length and number of blocks for a first uncertainty estimate.

Discussion. A block length comfortably larger than the correlation scale, e.g. 500–1000 samples, gives hundreds of block means; then verify that the standard error plateaus as the block size increases.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 15: Ascending versus descending pressure

Problem and discussion

Why might a difficult flexible or strongly clustering system give different apparent isotherms when pressure is scanned upward versus downward? What should you do?

Discussion. Metastability or slow structural/cluster transitions can create history dependence. Use independent starts, longer sampling, enhanced methods, and do not call a hysteresis loop an equilibrium result without evidence.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 16: Force-field error versus sampling error

Problem and discussion

A 10× longer run reduces the standard error from 3% to 1%, but simulation remains 25% above experiment. What should you investigate next?

Discussion. Model discrepancy dominates: uptake convention, structure, charges, force field, rigidity, or experiment matching. More sampling will not remove a 25% systematic bias.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 17: Working-capacity ranking

Problem and discussion

Material A adsorbs 8 mmol g^{-1} at adsorption pressure and 6 at regeneration pressure; Material B adsorbs 6 and 1. Which has the larger working capacity? Why can uptake-only ranking mislead?

Discussion. A gives 2; B gives 5 mmol g^{-1}. Strong retention can make a high-capacity material difficult to regenerate.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 18: Heat versus regenerability

Problem and discussion

Two MOFs have the same working capacity, but one has much higher q_{st}. List two reasons the lower-heat material may be process-attractive and one reason the higher-heat material may still be preferred.

Discussion. Lower heat can reduce thermal duty and temperature swings; higher affinity may be crucial for dilute feeds or selectivity.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 19: High-throughput failure audit

Problem and discussion

Out of 10,000 structures, 700 fail charge assignment and 300 fail atom typing. Why is deleting those 1,000 rows dangerous?

Discussion. Failure may correlate with chemistry (metals, functional groups, complexity), creating selection bias. Preserve failure categories and describe the supported domain.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 20: Validation portfolio

Problem and discussion

You can afford experimental validation for only 12 materials. Design a selection that is more informative than simply choosing the top 12 predicted performers.

Discussion. Cover chemical families, predicted performance regimes, uncertainty/domain distance, strong-site and weak-site cases, and at least a few challenging or near-boundary examples.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 21: HPC array design

Problem and discussion

You need 200 MOFs at 20 pressures and two gases. Explain why 8,000 independent state-point jobs may be easier to scale than one monolithic parallel job.

Discussion. State points are naturally embarrassingly parallel, can be retried independently, checkpointed, load-balanced, and audited with explicit status files.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 22: RASPA version migration

Problem and discussion

A RASPA2 workflow is validated against published data, but a student wants to migrate to RASPA3. Design a safe migration test.

Discussion. Freeze the physical model, reproduce one low-, mid-, and high-pressure state plus Widom/energy checks, compare observables and conventions, then expand only after agreement.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 23: Density-map interpretation

Problem and discussion

A density map shows two pockets at low pressure that merge into a broad pore-filling region at high pressure. What physical story would you test?

Discussion. Strong-site occupation precedes secondary-site and pore-volume filling. Quantify site occupancies and correlate with loading-dependent heat.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 24: When to escalate to mixture GCMC

Problem and discussion

IAST and experiment agree at low pressure but diverge at high pressure. What additional calculations would be most informative?

Discussion. Explicit mixture GCMC at the divergence state points, sensitivity to pure-isotherm fits, and analysis of site competition/guest–guest interactions.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

Worked example 25: A one-page reproducibility record

Problem and discussion

Draft the metadata header that should accompany every simulation campaign: structure source/hash, software/version, force field, charges, guest model, state grid, cutoffs/electrostatics, cycles, seeds, parser version, and validation status.

Discussion. Treat this as a machine-readable manifest as well as a human README.

Extension question: What modelling assumption in this example would you test first in a real MOF study, and why?

PART I

Expanded Explanations and Guided Examples

A9.1 Expanded tutorial 1: Why simulation?

Core idea

Molecular simulation connects an atomistic representation of a porous crystal to equilibrium adsorption observables. The important scientific step is not running a code but defining which microscopic assumptions are sufficient for the question being asked.

A9.1.1 Start from the physical question

A reliable way to learn why simulation? is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.1.2 Mathematical reading

The compact relation to keep in mind is

$$\text{structure} + \text{force field} + \text{ensemble} + \text{sampling} \rightarrow \langle A \rangle .$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.1.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.1.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.1.5 Worked example

Problem

Two MOFs have equal pore volume but different low-pressure CO2 uptake. Explain why geometry alone cannot determine the Henry coefficient.

Solution. Because Henry behaviour depends on the Boltzmann-weighted energy landscape. Polar groups, open metal sites, or confinement can change guest-framework energies without changing total pore volume.

A9.1.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.1.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.1.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of why simulation? in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.2 Expanded tutorial 2: Experimental isotherms

Core idea

An adsorption isotherm is an equilibrium relation between a reservoir state and a reported adsorbed amount. Absolute, excess, gravimetric, volumetric, and STP-volume bases are different conventions and must not be mixed silently.

A9.2.1 Start from the physical question

A reliable way to learn experimental isotherms is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.2.2 Mathematical reading

The compact relation to keep in mind is

$$N_{\mathrm{ex}} = N_{\mathrm{abs}} - \rho_{\mathrm{bulk}} V_{\mathrm{acc}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.2.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.2.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.2.5 Worked example

Problem

A simulation gives 6.0 mol/kg absolute uptake and a bulk-gas reference of 0.8 mol/kg in the accessible pore volume. Find excess uptake.

Solution. The excess uptake is 5.2 mol/kg.

A9.2.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.2.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.2.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of experimental isotherms in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.3 Expanded tutorial 3: Chemical potential

Core idea

Chemical potential is the natural variable for particle exchange. For a rigid pore connected to a reservoir, adsorption equilibrium is expressed by equality of temperature and chemical potential rather than by assigning a scalar mechanical pressure inside the solid.

A9.3.1 Start from the physical question

A reliable way to learn chemical potential is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.3.2 Mathematical reading

The compact relation to keep in mind is

$$\mu(T, f) = \mu^{\circ}(T) + RT \ln(f/f^{\circ}).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.3.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.3.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.3.5 Worked example

Problem

At 298 K, what is the ideal-gas chemical-potential increase from 0.1 to 10 bar?

Solution. It is RT ln(100), about 11.4 kJ/mol.

A9.3.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.3.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.3.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of chemical potential in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.4 Expanded tutorial 4: Henry regime

Core idea

At infinite dilution, adsorbate-adsorbate interactions vanish and uptake becomes linear in reservoir activity. The Henry coefficient is therefore an affinity descriptor, not a saturation-capacity descriptor.

A9.4.1 Start from the physical question

A reliable way to learn henry regime is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.4.2 Mathematical reading

The compact relation to keep in mind is

$$q \simeq K_H f.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.4.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.4.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.4.5 Worked example

Problem

If q=0.04 mmol/g at 0.02 bar and the point is in the Henry regime, estimate K_H.

Solution. K_H is about 2 mmol g^{-1} bar^{-1}.

A9.4.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.4.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.4.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of henry regime in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.5 Expanded tutorial 5: Langmuir model

Core idea

The Langmuir equation is a compact saturation model based on a finite population of equivalent independent sites. Its value is interpretability; its danger is overinterpreting fitted parameters when the pressure range does not identify them.

A9.5.1 Start from the physical question

A reliable way to learn langmuir model is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.5.2 Mathematical reading

The compact relation to keep in mind is

$$q = q_s \frac{bp}{1 + bp}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.5.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.5.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.5.5 Worked example

Problem

For q_s=5 mmol/g and b=0.5 bar^{-1}, calculate q at 2 bar.

Solution. bp=1, so q=q_s/2=2.5 mmol/g.

A9.5.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.5.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.5.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of langmuir model in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.6 Expanded tutorial 6: Dual-site Langmuir

Core idea

A dual-site Langmuir model represents two site populations with different capacities and affinities. Four parameters can fit many shapes, so identifiability and parameter correlation must be checked.

A9.6.1 Start from the physical question

A reliable way to learn dual-site langmuir is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.6.2 Mathematical reading

The compact relation to keep in mind is

$$q = \sum_{j=1}^{2} q_{s,j} \frac{b_j p}{1 + b_j p}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.6.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.6.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.6.5 Worked example

Problem

If q_s1=1, b1=8, q_s2=4, b2=0.25, find the low-pressure slope.

Solution. K_H=q_s1 b1+q_s2 b2=8+1=9 in consistent units.

A9.6.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.6.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.6.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of dual-site langmuir in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.7 Expanded tutorial 7: Isotherm fitting

Core idea

Fitting should preserve the experimental error structure and the physical limits of the model. Residual plots, uncertainty, parameter correlation, and prediction over the process-relevant pressure range matter more than a near-unity R-squared.

A9.7.1 Start from the physical question

A reliable way to learn isotherm fitting is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.7.2 Mathematical reading

The compact relation to keep in mind is

$$\hat{\theta} = \arg\min_{\theta} \sum_{i} w_i [q_i - q(p_i; \theta)]^2.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.7.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity

boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.7.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.7.5 Worked example

Problem

Why can linearising the Langmuir equation give a different fit from nonlinear regression?

Solution. The transformation changes the error model and weights different pressure regions unequally.

A9.7.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.7.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.7.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of isotherm fitting in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.8 Expanded tutorial 8: Temperature dependence

Core idea

Adsorption is usually exothermic, so increasing temperature reduces equilibrium uptake at fixed fugacity. Multi-temperature data probe the balance of energetic stabilisation and entropy.

A9.8.1 Start from the physical question

A reliable way to learn temperature dependence is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.8.2 Mathematical reading

The compact relation to keep in mind is

$$\left(\frac{\partial \ln p}{\partial (1/T)}\right)_n = -q_{st}/R.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.8.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.8.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.8.5 Worked example

Problem

At fixed loading the pressure doubles when temperature is raised. What does that qualitatively imply?

Solution. The adsorbed state is stabilised enthalpically; higher temperature requires higher pressure to maintain the same loading.

A9.8.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.8.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.8.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of temperature dependence in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.9 Expanded tutorial 9: Heat of adsorption

Core idea

Differential, isosteric, calorimetric, and simulation fluctuation heats are related but depend on boundary conditions and sign conventions. Always state whether a positive number denotes exothermic heat release.

A9.9.1 Start from the physical question

A reliable way to learn heat of adsorption is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.9.2 Mathematical reading

The compact relation to keep in mind is

$$q_{st} \sim -\left(\frac{\partial U}{\partial N}\right)_{T,V} + \text{reference terms.}$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.9.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity

boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.9.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.9.5 Worked example

Problem

Why can q_st decrease with loading?

Solution. Strong sites fill first; later molecules occupy weaker regions, lowering the incremental binding enthalpy.

A9.9.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.9.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.9.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of heat of adsorption in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.10 Expanded tutorial 10: Microstates and probabilities

Core idea

Statistical mechanics treats a macroscopic state as an ensemble of microscopic configurations. Molecular simulation estimates observables by sampling configurations with the correct probability weight.

A9.10.1 Start from the physical question

A reliable way to learn microstates and probabilities is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.10.2 Mathematical reading

The compact relation to keep in mind is

$$\langle A \rangle = \frac{\sum_\alpha A_\alpha e^{-\beta E_\alpha}}{\sum_\alpha e^{-\beta E_\alpha}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.10.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.10.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.10.5 Worked example

Problem

Two states differ by 2 kBT. What is the probability ratio high/low?

Solution. exp(-2), about 0.135.

A9.10.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.10.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.10.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of microstates and probabilities in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.11 Expanded tutorial 11: Ensembles

Core idea

An ensemble is a statement about constraints and exchange with the surroundings. NVT fixes particle number; grand canonical muVT fixes chemical potential instead and allows particle number to fluctuate.

A9.11.1 Start from the physical question

A reliable way to learn ensembles is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.11.2 Mathematical reading

The compact relation to keep in mind is

$$\pi(N, \mathbf{r}^N) \propto \frac{e^{-\beta(U-\mu N)}}{N!\Lambda^{3N}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.11.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.11.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.11.5 Worked example

Problem

Which ensemble is natural for a rigid MOF in equilibrium with a gas reservoir?

Solution. The grand-canonical muVT ensemble.

A9.11.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.11.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.11.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of ensembles in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.12 Expanded tutorial 12: Metropolis MC

Core idea

Metropolis Monte Carlo is an importance-sampling method. Trial moves need not resemble real dynamics; they must instead produce the desired equilibrium distribution through a correct proposal and acceptance rule.

A9.12.1 Start from the physical question

A reliable way to learn metropolis mc is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.12.2 Mathematical reading

The compact relation to keep in mind is

$$A = \min[1, e^{-\beta \Delta U}].$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.12.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.12.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.12.5 Worked example

Problem

At 300 K, estimate acceptance of a +2.5 kJ/mol symmetric trial move.

Solution. RT is about 2.49 kJ/mol, so acceptance is exp(-1.00), about 0.37.

A9.12.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.12.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.12.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of metropolis mc in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.13 Expanded tutorial 13: Detailed balance

Core idea

Detailed balance pairs forward and reverse transitions so that equilibrium probability flow is balanced. It is a standard sufficient condition for the target distribution to be stationary.

A9.13.1 Start from the physical question

A reliable way to learn detailed balance is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.13.2 Mathematical reading

The compact relation to keep in mind is

$$\pi(o)P(o \to n) = \pi(n)P(n \to o).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.13.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.13.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.13.5 Worked example

Problem

Why must insertion and deletion be treated as reverse moves?

Solution. Their proposal and acceptance factors must balance the grand-canonical probability ratio; otherwise the sampled N distribution is biased.

A9.13.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.13.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.13.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of detailed balance in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.14 Expanded tutorial 14: GCMC insertion/deletion

Core idea

Grand-canonical Monte Carlo changes particle number through insertion and deletion, while translations, rotations, and reinsertion explore coordinates at fixed N. The average loading emerges from number fluctuations.

A9.14.1 Start from the physical question

A reliable way to learn gcmc insertion/deletion is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.14.2 Mathematical reading

The compact relation to keep in mind is

$$A_{\mathrm{ins}} \propto \frac{V}{N+1} e^{\beta\mu} e^{-\beta\Delta U}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.14.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.14.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.14.5 Worked example

Problem

Why can insertion acceptance become extremely small at high loading?

Solution. Crowding produces large positive overlap energies, so the Boltzmann factor suppresses insertion.

A9.14.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.14.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.14.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of gcmc insertion/deletion in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.15 Expanded tutorial 15: Fugacity

Core idea

Pressure is a convenient experimental variable, but chemical potential is linked to fugacity. At low density f≈p; at elevated pressure the fugacity coefficient can materially alter the reservoir activity.

A9.15.1 Start from the physical question

A reliable way to learn fugacity is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.15.2 Mathematical reading

The compact relation to keep in mind is

$$f = \phi p.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.15.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or

scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.15.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.15.5 Worked example

Problem

At 50 bar with phi=0.86, calculate fugacity.

Solution. 43 bar.

A9.15.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.15.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to

estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.15.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of fugacity in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.16 Expanded tutorial 16: Widom insertion

Core idea

Widom test-particle insertion evaluates an excess chemical-potential average at infinite dilution. A small set of favourable insertions can dominate because of the exponential Boltzmann weight.

A9.16.1 Start from the physical question

A reliable way to learn widom insertion is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.16.2 Mathematical reading

The compact relation to keep in mind is

$$\mu^{ex} = -k_B T \ln\langle e^{-\beta \Delta U} \rangle .$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.16.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.16.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.16.5 Worked example

Problem

Why should you inspect the distribution of insertion weights, not only the mean?

Solution. A mean can be dominated by rare favourable events; insufficient rare-event sampling can create severe bias.

A9.16.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.16.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.16.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of widom insertion in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.17 Expanded tutorial 17: Isosteric heat in GCMC

Core idea

Energy-number covariance provides a grand-canonical route to differential adsorption enthalpy. Covariances converge more slowly than means and require careful reference-state treatment.

A9.17.1 Start from the physical question

A reliable way to learn isosteric heat in gcmc is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.17.2 Mathematical reading

The compact relation to keep in mind is

$$\frac{\partial U}{\partial \langle N \rangle} \sim \frac{\langle UN \rangle - \langle U \rangle \langle N \rangle}{\langle N^2 \rangle - \langle N \rangle^2}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.17.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.17.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.17.5 Worked example

Problem

Why can loading be converged while q_st is still noisy?

Solution. q_st uses second-order fluctuation statistics, which generally require more sampling.

A9.17.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.17.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.17.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of isosteric heat in gcmc in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.18 Expanded tutorial 18: Force fields

Core idea

The force field is the physical hypothesis that assigns energy to each configuration. A perfectly sampled but inappropriate force field produces a precise answer to the wrong model.

A9.18.1 Start from the physical question

A reliable way to learn force fields is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.18.2 Mathematical reading

The compact relation to keep in mind is

$$U = U_{\mathrm{vdW}} + U_{\mathrm{elec}} + U_{\mathrm{bonded}} + \cdots .$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.18.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.18.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.18.5 Worked example

Problem

Low-pressure CO2 is overpredicted but saturation is reasonable. Which model ingredients deserve early scrutiny?

Solution. Framework charges/electrostatics and local guest-framework nonbonded parameters.

A9.18.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.18.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.18.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of force fields in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.19 Expanded tutorial 19: Lennard-Jones

Core idea

The Lennard-Jones 12-6 potential combines steep repulsion with attractive dispersion. Pore confinement makes the sum of many framework-atom contributions important.

A9.19.1 Start from the physical question

A reliable way to learn lennard-jones is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.19.2 Mathematical reading

The compact relation to keep in mind is

$$U_{LJ} = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^{6}].$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.19.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.19.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.19.5 Worked example

Problem

At what separation is the LJ minimum?

Solution. $r = 2^{1/6}\sigma$.

A9.19.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.19.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.19.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of lennard-jones in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.20 Expanded tutorial 20: Cutoffs and supercells

Core idea

Finite cutoffs reduce cost but create geometric and energetic conventions. Simulation-cell dimensions should be compatible with the cutoff and with the correlation length of the system.

A9.20.1 Start from the physical question

A reliable way to learn cutoffs and supercells is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.20.2 Mathematical reading

The compact relation to keep in mind is

$$L_{\perp,\min} \gtrsim 2r_c.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.20.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.20.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.20.5 Worked example

Problem

With a 12 A cutoff, what minimum perpendicular box length is a common starting guideline?

Solution. Greater than roughly 24 A.

A9.20.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.20.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.20.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of cutoffs and supercells in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.21 Expanded tutorial 21: Electrostatics

Core idea

Coulomb interactions are long ranged. In periodic porous crystals, Ewald-style methods are commonly used so that the periodic lattice contribution is treated consistently.

A9.21.1 Start from the physical question

A reliable way to learn electrostatics is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.21.2 Mathematical reading

The compact relation to keep in mind is

$$U_{ij} = \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.21.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.21.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.21.5 Worked example

Problem

Why is simple short-range truncation of electrostatics risky?

Solution. The 1/r decay is slow, so neglected long-range contributions can remain large and distort energetics.

A9.21.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.21.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.21.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of electrostatics in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.22 Expanded tutorial 22: Partial charges

Core idea

Framework partial charges are model parameters and can dominate low-pressure adsorption of polar and quadrupolar gases. Charge neutrality and charge provenance should be audited.

A9.22.1 Start from the physical question

A reliable way to learn partial charges is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.22.2 Mathematical reading

The compact relation to keep in mind is

$$\sum_i q_i = Q_{\text{cell}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.22.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.22.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.22.5 Worked example

Problem

A neutral unit cell's charges sum to +0.2e. What should you do?

Solution. Investigate the source/rounding/typing and correct the charge model rather than ignoring the imbalance.

A9.22.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.22.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.22.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of partial charges in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.23 Expanded tutorial 23: Guest models

Core idea

Guest models define interaction sites, charges, geometry, and flexibility. Methane can often be represented simply; CO2 and N2 require orientation-dependent models; water and cryogenic hydrogen require additional care.

A9.23.1 Start from the physical question

A reliable way to learn guest models is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.23.2 Mathematical reading

The compact relation to keep in mind is

$$U_{AB} = \sum_{i \in A} \sum_{j \in B} [U_{LJ,ij} + U_{C,ij}].$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.23.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity

boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.23.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.23.5 Worked example

Problem

Why is a spherical one-site CO2 model usually inadequate in a polar MOF?

Solution. It cannot represent the quadrupolar orientation-dependent electrostatics.

A9.23.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.23.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.23.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of guest models in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.24 Expanded tutorial 24: CIF preparation

Core idea

A crystallographic CIF may contain disorder, solvent, partial occupancies, missing hydrogens, and labels that are not simulation-ready. Cleaning must be chemically informed and reversible.

A9.24.1 Start from the physical question

A reliable way to learn cif preparation is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.24.2 Mathematical reading

The compact relation to keep in mind is

$$\mathbf{H}_{\text{super}} = \mathbf{H}\,\mathrm{diag}(n_a, n_b, n_c).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.24.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.24.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.24.5 Worked example

Problem

Cell widths are 11, 14, 18 A with a 12 A cutoff. Give a simple replication.

Solution. 3x2x2 gives 33, 28, 36 A.

A9.24.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.24.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.24.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of cif preparation in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.25 Expanded tutorial 25: Periodic boundaries

Core idea

Periodic boundary conditions represent a bulk infinite crystal by tiling a finite cell. They remove external surfaces but can create self-image artefacts if the box is too small.

A9.25.1 Start from the physical question

A reliable way to learn periodic boundaries is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.25.2 Mathematical reading

The compact relation to keep in mind is

$$\mathbf{r} \equiv \mathbf{r} + \mathbf{H}\mathbf{n}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.25.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or

scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.25.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.25.5 Worked example

Problem

Why may a large flexible guest require a larger box even if 2rc is satisfied?

Solution. Its conformational or correlation length may extend beyond the simple cutoff criterion.

A9.25.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.25.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to

estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.25.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of periodic boundaries in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.26 Expanded tutorial 26: Rigid versus flexible MOFs

Core idea

Rigid-framework GCMC is efficient and often reliable for stiff materials, but cannot reproduce breathing, gate opening, or adsorption-induced phase changes.

A9.26.1 Start from the physical question

A reliable way to learn rigid versus flexible mofs is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.26.2 Mathematical reading

The compact relation to keep in mind is

$$\Xi \propto \sum_N \int d\mathbf{R}\, d\mathbf{r}^N e^{-\beta[U(\mathbf{R},\mathbf{r}^N)-\mu N]}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.26.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.26.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.26.5 Worked example

Problem

An experiment shows a sharp step and large unit-cell expansion. What is missing from rigid GCMC?

Solution. The host structural transition and the associated framework free-energy coupling.

A9.26.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.26.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.26.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of rigid versus flexible mofs in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.27 Expanded tutorial 27: Pure isotherm workflow

Core idea

A production isotherm is a collection of validated state points. Use a pressure grid that resolves the Henry region, curvature, and saturation or process window, and attach convergence evidence to every point.

A9.27.1 Start from the physical question

A reliable way to learn pure isotherm workflow is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.27.2 Mathematical reading

The compact relation to keep in mind is

$$p_i = 10^{a+i(b-a)/(m-1)}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.27.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.27.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.27.5 Worked example

Problem

Why is log-spaced pressure usually better than linear spacing across six decades?

Solution. It allocates points to low-pressure affinity and broad dynamic range instead of clustering most points at high pressure.

A9.27.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.27.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.27.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of pure isotherm workflow in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.28 Expanded tutorial 28: Mixture GCMC

Core idea

Explicit mixture GCMC samples simultaneous competition among species. Adsorbed composition and selectivity depend on total pressure, feed composition, and cross interactions.

A9.28.1 Start from the physical question

A reliable way to learn mixture gcmc is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.28.2 Mathematical reading

The compact relation to keep in mind is

$$S_{A/B} = (x_A/x_B)/(y_A/y_B).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.28.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.28.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.28.5 Worked example

Problem

A 10/90 feed produces 70/30 adsorbed composition. Compute selectivity.

Solution. 21.

A9.28.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.28.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.28.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of mixture gcmc in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.29 Expanded tutorial 29: IAST

Core idea

IAST predicts mixture adsorption from pure isotherms through equal spreading pressure. It is fast and useful, but sensitive to fit quality and can fail when the adsorbed phase is strongly nonideal.

A9.29.1 Start from the physical question

A reliable way to learn iast is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.29.2 Mathematical reading

The compact relation to keep in mind is

$$\pi_i = RT \int_0^{p_i^0} n_i(p)\, dp/p.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.29.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.29.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.29.5 Worked example

Problem

Why can low-pressure fit error affect IAST at moderate pressure?

Solution. The spreading-pressure integral contains n/p and accumulates information from the low-pressure region.

A9.29.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.29.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.29.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of iast in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.30 Expanded tutorial 30: RAST

Core idea

RAST extends ideal adsorbed-solution ideas by introducing nonideality through activity coefficients or related corrections. The extra flexibility requires extra parameters or mixture information.

A9.30.1 Start from the physical question

A reliable way to learn rast is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.30.2 Mathematical reading

The compact relation to keep in mind is

$$\mu_i^{ads} = \mu_i^{*,ads} + RT \ln(x_i \gamma_i).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.30.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.30.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.30.5 Worked example

Problem

Why is RAST not automatically better than IAST?

Solution. Additional nonideality parameters can be poorly constrained; complexity only helps if supported by data.

A9.30.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.30.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.30.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of rast in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.31 Expanded tutorial 31: Absolute versus excess

Core idea

Absolute loading counts assigned pore molecules; excess loading subtracts a bulk-fluid reference occupying the accessible pore volume. The difference grows at high fluid density.

A9.31.1 Start from the physical question

A reliable way to learn absolute versus excess is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.31.2 Mathematical reading

The compact relation to keep in mind is

$$N_{ex} = N_{abs} - \rho_b V_{acc}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.31.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.31.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.31.5 Worked example

Problem

Absolute loading is 100 molecules/cell and the bulk reference is 18. Find excess.

Solution. 82 molecules/cell.

A9.31.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.31.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.31.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of absolute versus excess in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.32 Expanded tutorial 32: Density maps

Core idea

Spatial density maps show where adsorbates spend statistical time. They can reveal strong sites and filling sequences but must be based on ensemble averages, not single snapshots.

A9.32.1 Start from the physical question

A reliable way to learn density maps is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.32.2 Mathematical reading

The compact relation to keep in mind is

$$\rho_k \approx \frac{1}{M\Delta V}\sum_m n_k^{(m)}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.32.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.32.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.32.5 Worked example

Problem

What does emergence of a second density pocket at higher loading suggest?

Solution. Primary sites may be saturating and secondary sites are beginning to fill.

A9.32.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.32.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.32.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of density maps in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.33 Expanded tutorial 33: RDFs

Core idea

Radial distribution functions describe pair correlations around a reference species or site. Peaks suggest preferred separations but do not alone establish a chemical bond.

A9.33.1 Start from the physical question

A reliable way to learn rdfs is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.33.2 Mathematical reading

The compact relation to keep in mind is

$$g(r) = \rho(r)/\rho_{\mathrm{bulk}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.33.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.33.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.33.5 Worked example

Problem

Why should RDF interpretation be combined with geometry and energy analysis?

Solution. A peak indicates probability enhancement, not its physical cause.

A9.33.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.33.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.33.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of rdfs in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.34 Expanded tutorial 34: Convergence

Core idea

A long run is not necessarily a converged run. Stationarity, correlation time, block behaviour, and replicate agreement are more informative than cycle count alone.

A9.34.1 Start from the physical question

A reliable way to learn convergence is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.34.2 Mathematical reading

The compact relation to keep in mind is

$$N_{\mathrm{eff}} \approx N/(2\tau_{\mathrm{int}}).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.34.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.34.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.34.5 Worked example

Problem

For N=200000 and tau=50, estimate N_eff.

Solution. About 2000.

A9.34.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.34.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.34.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of convergence in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.35 Expanded tutorial 35: Block averaging

Core idea

Block averaging estimates uncertainty when samples are correlated. Block length should exceed the dominant correlation scale and the resulting standard error should stabilise with increasing block size.

A9.35.1 Start from the physical question

A reliable way to learn block averaging is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.35.2 Mathematical reading

The compact relation to keep in mind is

$$\bar{A}_b = \frac{1}{m} \sum_{i \in b} A_i.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.35.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.35.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.35.5 Worked example

Problem

If correlation extends about 100 samples, is a block length of 20 adequate?

Solution. No. The blocks are still strongly correlated; use larger blocks and check for an error plateau.

A9.35.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.35.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.35.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of block averaging in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.36 Expanded tutorial 36: Validation

Core idea

Simulation-experiment comparison should first align material state, temperature, pressure/fugacity, uptake basis, and activation. Error shape often provides more mechanism than one global score.

A9.36.1 Start from the physical question

A reliable way to learn validation is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.36.2 Mathematical reading

The compact relation to keep in mind is

$$\delta_i = (q_i^{sim} - q_i^{exp})/q_i^{exp}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.36.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.36.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.36.5 Worked example

Problem

Simulation is high only at low pressure. Name a likely model class to inspect.

Solution. Affinity terms such as charges, strong-site structure, or guest-framework parameters.

A9.36.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.36.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.36.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of validation in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.37 Expanded tutorial 37: Working capacity

Core idea

Process-relevant performance depends on differences between adsorption and regeneration states, not simply the maximum loading at one condition.

A9.37.1 Start from the physical question

A reliable way to learn working capacity is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.37.2 Mathematical reading

The compact relation to keep in mind is

$$\Delta q = q(P_{ads}, T_{ads}) - q(P_{reg}, T_{reg}).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.37.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.37.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.37.5 Worked example

Problem

Material A loads 8 to 6; B loads 6 to 1. Which has larger working capacity?

Solution. B: 5 versus 2 mmol/g.

A9.37.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.37.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.37.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of working capacity in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.38 Expanded tutorial 38: Volumetric metrics

Core idea

Gravimetric performance can be misleading for very low-density frameworks. Packed-bed or crystal density converts per-mass uptake into per-volume capacity relevant to equipment size.

A9.38.1 Start from the physical question

A reliable way to learn volumetric metrics is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.38.2 Mathematical reading

The compact relation to keep in mind is

$$q_V = q_m \rho_{\mathrm{pack}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.38.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.38.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.38.5 Worked example

Problem

6 mol/kg at 0.55 kg/L gives what volumetric loading?

Solution. 3.3 mol/L.

A9.38.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.38.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.38.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of volumetric metrics in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.39 Expanded tutorial 39: MD and diffusion

Core idea

Monte Carlo has no direct physical clock. Molecular dynamics integrates equations of motion and can provide transport information when the model and trajectory length are adequate.

A9.39.1 Start from the physical question

A reliable way to learn md and diffusion is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.39.2 Mathematical reading

The compact relation to keep in mind is

$$D = \lim_{t\to\infty} \langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle / (6t).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.39.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.39.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.39.5 Worked example

Problem

Why cannot GCMC swap acceptance be converted into a diffusivity?

Solution. Swap moves are artificial equilibrium-sampling proposals, not physical molecular trajectories.

A9.39.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.39.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.39.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of md and diffusion in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.40 Expanded tutorial 40: Configurational bias

Core idea

Configurational-bias Monte Carlo proposes flexible or crowded configurations intelligently and corrects the bias through Rosenbluth weights, improving efficiency without changing the target ensemble.

A9.40.1 Start from the physical question

A reliable way to learn configurational bias is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.40.2 Mathematical reading

The compact relation to keep in mind is

$$W = \sum_j e^{-\beta U_j}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.40.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.40.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.40.5 Worked example

Problem

Does CBMC change the physical force field?

Solution. No. It changes the proposal/sampling strategy, not the potential energy model.

A9.40.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.40.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.40.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of configurational bias in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.41 Expanded tutorial 41: Open metal sites

Core idea

Open metal sites often require more chemical specificity than generic fixed-charge Lennard-Jones force fields can provide. Polarization, coordination, or quantum-derived potentials may be needed.

A9.41.1 Start from the physical question

A reliable way to learn open metal sites is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.41.2 Mathematical reading

The compact relation to keep in mind is

$$U_{\mathrm{model}} = U_{\mathrm{generic}} + \Delta U_{\mathrm{site}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.41.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.41.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.41.5 Worked example

Problem

What is a useful first validation target for an open-metal-site model?

Solution. Low-loading affinity or heat and adsorption-site geometry against suitable experiment or higher-level calculation.

A9.41.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.41.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.41.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of open metal sites in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.42 Expanded tutorial 42: High-throughput GCMC

Core idea

At scale, provenance and failure handling become scientific necessities. A pipeline should record hashes, versions, state points, run status, convergence, and parser provenance.

A9.42.1 Start from the physical question

A reliable way to learn high-throughput gcmc is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.42.2 Mathematical reading

The compact relation to keep in mind is

$$N_{\text{jobs}} = MPTC.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.42.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.42.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.42.5 Worked example

Problem

500 materials x 16 pressures x 2 temperatures x 3 compositions gives how many jobs?

Solution. 48,000.

A9.42.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.42.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.42.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of high-throughput gcmc in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.43 Expanded tutorial 43: RASPA2 files

Core idea

RASPA2 separates simulation control, framework structure, pseudo-atoms, force fields, and molecule definitions. Understanding which file controls which part of the model is essential for reproducibility.

A9.43.1 Start from the physical question

A reliable way to learn raspa2 files is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.43.2 Mathematical reading

The compact relation to keep in mind is

$$\text{input} + \text{CIF} + \text{FF} + \text{molecule} \rightarrow \text{run}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.43.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.43.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.43.5 Worked example

Problem

Which file normally contains the framework coordinates?

Solution. The CIF identified by FrameworkName.

A9.43.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.43.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.43.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of raspa2 files in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.44 Expanded tutorial 44: RASPA2 units

Core idea

The supplied manual uses angstrom, kelvin, atomic mass, picosecond, and atomic charge as base units internally and specifies external pressure input in pascal.

A9.44.1 Start from the physical question

A reliable way to learn raspa2 units is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.44.2 Mathematical reading

The compact relation to keep in mind is

$$1\,\mathrm{bar} = 10^5\,\mathrm{Pa}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.44.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.44.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.44.5 Worked example

Problem

What does ExternalPressure 1.0 request in RASPA2?

Solution. 1 Pa, not 1 bar.

A9.44.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.44.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.44.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of raspa2 units in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.45 Expanded tutorial 45: RASPA2 move keywords

Core idea

Translation and rotation explore existing molecules; reinsertion changes position or conformation more globally; swap changes particle number; Widom measures insertion statistics without permanently adding the probe.

A9.45.1 Start from the physical question

A reliable way to learn raspa2 move keywords is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.45.2 Mathematical reading

The compact relation to keep in mind is

$$P_{\text{moves}} \propto \{P_t, P_r, P_{reins}, P_{swap}, P_W\}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.45.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.45.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.45.5 Worked example

Problem

Which move is essential for ordinary GCMC adsorption?

Solution. Swap, because insertion/deletion exchange is required.

A9.45.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.45.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.45.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of raspa2 move keywords in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.46 Expanded tutorial 46: RASPA2 fugacity

Core idea

When no explicit component fugacity coefficient is supplied, the manual states that RASPA can compute it with Peng-Robinson using molecular critical data. Supplying FugacityCoefficient 1.0 forces f=p.

A9.46.1 Start from the physical question

A reliable way to learn raspa2 fugacity is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.46.2 Mathematical reading

The compact relation to keep in mind is

$$f_i = \phi_i y_i P.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.46.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.46.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.46.5 Worked example

Problem

Why is setting phi=1 a modelling choice rather than a numerical convenience?

Solution. It assumes ideal-gas-like fugacity mapping and may be inaccurate at high pressure.

A9.46.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.46.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.46.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of raspa2 fugacity in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.47 Expanded tutorial 47: RASPA2 excess adsorption

Core idea

HeliumVoidFraction is used by RASPA to define accessible pore volume for excess adsorption reporting. It should be calculated consistently for the actual framework model.

A9.47.1 Start from the physical question

A reliable way to learn raspa2 excess adsorption is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.47.2 Mathematical reading

The compact relation to keep in mind is

$$N_{ex} = N_{abs} - \rho_b V_{\mathrm{He}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.47.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.47.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.47.5 Worked example

Problem

Why must helium void fraction be recomputed after changing framework geometry?

Solution. The accessible volume changes with geometry.

A9.47.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.47.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.47.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of raspa2 excess adsorption in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.48 Expanded tutorial 48: RASPA2 restarts

Core idea

The manual distinguishes normal restart behaviour and binary crash recovery. Long HPC calculations should test restart logic before production.

A9.48.1 Start from the physical question

A reliable way to learn raspa2 restarts is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.48.2 Mathematical reading

The compact relation to keep in mind is

$$\text{state}_{t+\Delta t} \leftarrow \text{restart}(\text{state}_t).$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.48.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.48.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.48.5 Worked example

Problem

Why is restart capability not a substitute for replicate runs?

Solution. A restart continues one stochastic history; it does not provide independent sampling evidence.

A9.48.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.48.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.48.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of raspa2 restarts in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

A9.49 Expanded tutorial 49: HPC arrays

Core idea

Isotherm state points are naturally embarrassingly parallel. One state point per directory simplifies retries, provenance, and resource allocation.

A9.49.1 Start from the physical question

A reliable way to learn hpc arrays is to begin with the physical question rather than the software keyword. Ask what is controlled externally, what can fluctuate, and what observable is being estimated. This prevents a common failure mode in molecular simulation: obtaining a numerically precise result whose physical meaning is unclear. In MOF adsorption the reservoir state, framework model, guest model, and sampling ensemble should all be stated before discussing numerical settings.

The second question is scale. Some quantities describe infinite dilution, others describe finite loading, and others describe a process difference between two state points. A number can be correct in one regime and misleading in another. For example, a Henry coefficient is a low-loading affinity measure; it is not a saturation capacity. Likewise, a high-pressure uptake is not automatically a useful working capacity. Always attach the regime and state point to the interpretation.

A third question is whether the framework model represents the phenomenon being discussed. A rigid perfect crystal is often the best first model because it is reproducible, efficient, and chemically interpretable. It is not a universal representation of defects, water, finite surfaces, pellet porosity, or adsorption-induced flexibility. A strong student should be able to say both why the simple model is useful and where it may fail.

A9.49.2 Mathematical reading

The compact relation to keep in mind is

$$N_{\mathrm{array}} = N_{\mathrm{statepoints}}.$$

Do not memorise this expression in isolation. Identify the units of every term, then test a limiting case. Ask what happens when pressure approaches zero, temperature rises, the interaction energy vanishes, or the pore becomes crowded. Limiting-case reasoning is one of the fastest ways to detect a wrong sign, a mistaken pressure unit, an implausible fit, or an inconsistent standard state.

A second mathematical habit is to distinguish a microscopic instantaneous quantity from an ensemble average. Simulation outputs such as loading, energy, density, or heat are averages over a sampled distribution. A single configuration may be chemically suggestive, but it is not the thermodynamic result. When fluctuations enter the formula, as in heat or response functions, convergence requirements become more demanding than for a simple mean.

A third habit is dimensional analysis. Before substituting numbers, write the expected units of the result. Henry coefficients, adsorption capacities, chemical potentials, force-field energies, diffusion coefficients, and pressure all use different conventions across papers and software. Unit checking should be automatic, not an afterthought.

A9.49.3 Assumptions and model hierarchy

The standard introductory adsorption model is intentionally simplified: a periodic crystal, often rigid; pairwise force-field interactions; a classical guest model; and a defined thermodynamic reservoir. These assumptions are powerful because they make GCMC efficient and interpretable. They are also a validity boundary. Defects, moisture, inaccessible crystals, framework transformations, reactive adsorption, external surfaces, and pellet-scale effects are not automatically represented.

A good hierarchy is therefore: begin with the simplest model capable of answering the question; verify sampling; test sensitivity to the most important model assumption; and escalate only when the decision or scientific interpretation depends on that assumption. This hierarchy saves computational resources and makes disagreements easier to diagnose.

When a result is later used in a process model, add another validity layer. An equilibrium isotherm does not by itself provide mass-transfer coefficients, heat-transfer parameters, pressure drop, valve scheduling, or cyclic steady state. Molecular simulation is one part of a multiscale chain, not the entire process model.

A9.49.4 Diagnostic thinking

When a result looks wrong, do not immediately run longer. Check, in order: pressure and concentration units; structure and atom count; charge neutrality and atom typing; force-field provenance; supercell/cutoff compatibility; ensemble and move set; equilibration; and finally production length. Large errors often originate before sampling begins. More Monte Carlo cycles cannot repair a 1 Pa versus 1 bar error or an incorrect CIF.

A useful publication practice is to separate *sampling uncertainty* from *model uncertainty*. Sampling uncertainty can be reduced by more effective samples. Model uncertainty requires alternative structures, force fields, charge schemes, flexibility treatments, equations of state, or comparison with experiment. This distinction should appear explicitly in student reports and research papers.

Another useful practice is to keep a signed discrepancy record. If simulation is consistently high at low pressure but accurate at high pressure, that pattern suggests a different failure mode from an error that grows only near saturation. The direction and pressure dependence of error can be mechanistically informative.

A9.49.5 Worked example

Problem

Why is one giant monolithic job often less robust than many independent state-point jobs?

Solution. Independent jobs can be retried, load-balanced, and audited separately.

A9.49.6 How this connects to RASPA2 practice

The supplied RASPA 2.0.45 manual should be treated as the syntax reference for the examples in this book. The conceptual mapping is stable even when syntax changes: the framework file defines the host geometry, force-field and pseudo-atom files define the energy model, molecule files define the guest, reservoir keywords define the thermodynamic state, and Monte Carlo move probabilities define how the desired ensemble is explored. Students should learn to explain this mapping before copying an input deck.

The manual also makes an important reproducibility point indirectly through its file precedence rules: a file placed in the current working directory can override an installed definition. This is convenient for force-field development but dangerous if the local override is not archived. Every production run should therefore save or hash all effective model files, not only `simulation.input`.

When pressure is involved, remember that the supplied RASPA2 manual specifies `ExternalPressure` in pascal. A human-facing workflow should therefore store pressure both in convenient scientific units such as bar and in the exact pascal value written to the input. The output should be checked for the resulting pressure, fugacity coefficient, and fugacity.

A9.49.7 Connection to the Frenkel–Smit treatment

The standard Frenkel–Smit perspective emphasises that Monte Carlo is an algorithm for sampling a known statistical distribution, not a black-box molecular movie. The same viewpoint is adopted here. Whenever

a trial move, bias, histogram, or estimator is introduced, ask which ensemble average it is designed to estimate and whether the move leaves the target distribution invariant. That conceptual discipline is more transferable than memorising any one program's keyword list.

A second transferable principle is to distinguish the equilibrium distribution from sampling efficiency. Two correct algorithms can target the same distribution but differ dramatically in how quickly they explore it. Translation, rotation, reinsertion, configurational bias, and advanced grand-canonical moves should therefore be judged by both correctness and efficiency.

A9.49.8 Exercises without solutions

Questions for independent work

1. Restate the core idea of hpc arrays in one sentence without equations.
2. Identify the most dangerous unit or convention error for this topic.
3. Propose one low-cost numerical test and one experimental or higher-fidelity validation test.
4. Explain what would change if the framework were flexible instead of rigid.
5. Write a one-paragraph methods statement that would let another researcher reproduce this part of a study.
6. Give one example in which the same numerical result could support two different physical interpretations.

PART II

Visual Reasoning Atlas

A10 Reading adsorption physics from pictures

The figures in this chapter are deliberately simple. Their purpose is to train the eye to connect a plot or schematic to a physical statement, a simulation control, and a diagnostic question. A good molecular simulation researcher should be able to look at an isotherm, acceptance curve, insertion-energy distribution, or convergence trace and explain what microscopic mechanism could produce it.

A10.1 Chemical potential is the bridge between pressure and GCMC

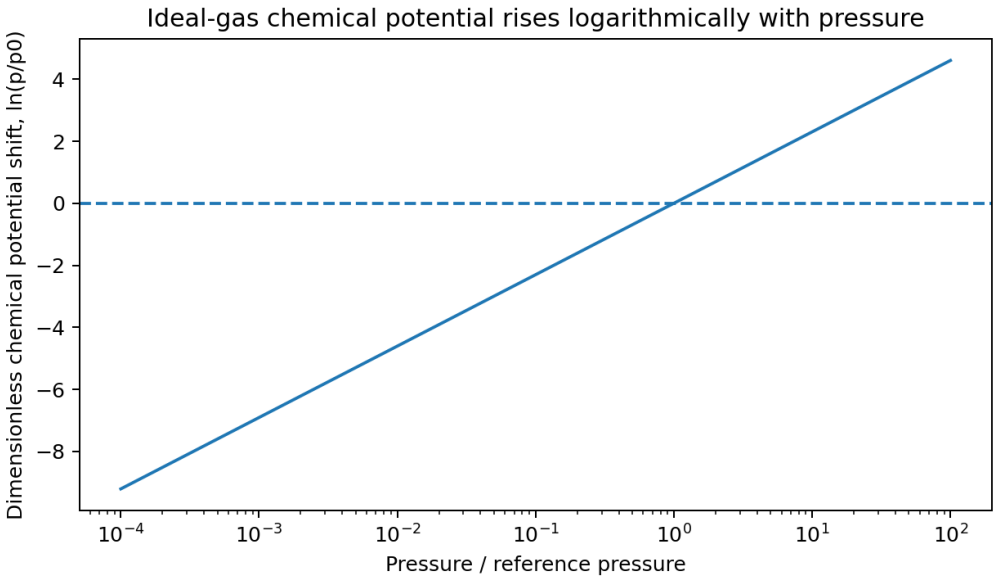


Figure A10.1: Schematic relation between reservoir pressure and chemical potential at fixed temperature. For an ideal gas, increasing pressure raises the chemical potential logarithmically. Real-gas work replaces pressure with fugacity.

At fixed temperature, the gas reservoir does not tell the pore how many molecules it must contain. It fixes the chemical potential. The pore responds by fluctuating among different particle numbers. This is the reason the grand-canonical ensemble is a natural model of adsorption in a rigid porous crystal. A pressure value therefore enters a GCMC calculation only after it has been translated into a chemical potential or fugacity consistent with the chosen gas model.

A useful limiting check is the low-pressure limit. When the gas is sufficiently dilute, fugacity approaches pressure and the reservoir becomes nearly ideal. At higher pressure, treating p and f as identical may introduce a systematic thermodynamic error even if the Monte Carlo sampling itself is perfect.

Worked interpretation

Suppose two calculations use the same framework, force field and temperature, but one uses $f = p$ while the other uses a real-gas fugacity coefficient $\phi < 1$. At the same reported pressure, the second calculation supplies the pore with the lower chemical potential because $f = \phi p$. A difference in loading can therefore arise before any change in the microscopic force field.

A10.2 Metropolis acceptance is a probability, not an optimizer

The Metropolis rule does not search for the single lowest-energy configuration. Instead, it constructs a Markov chain whose long-time distribution is the required thermodynamic distribution. This distinction is

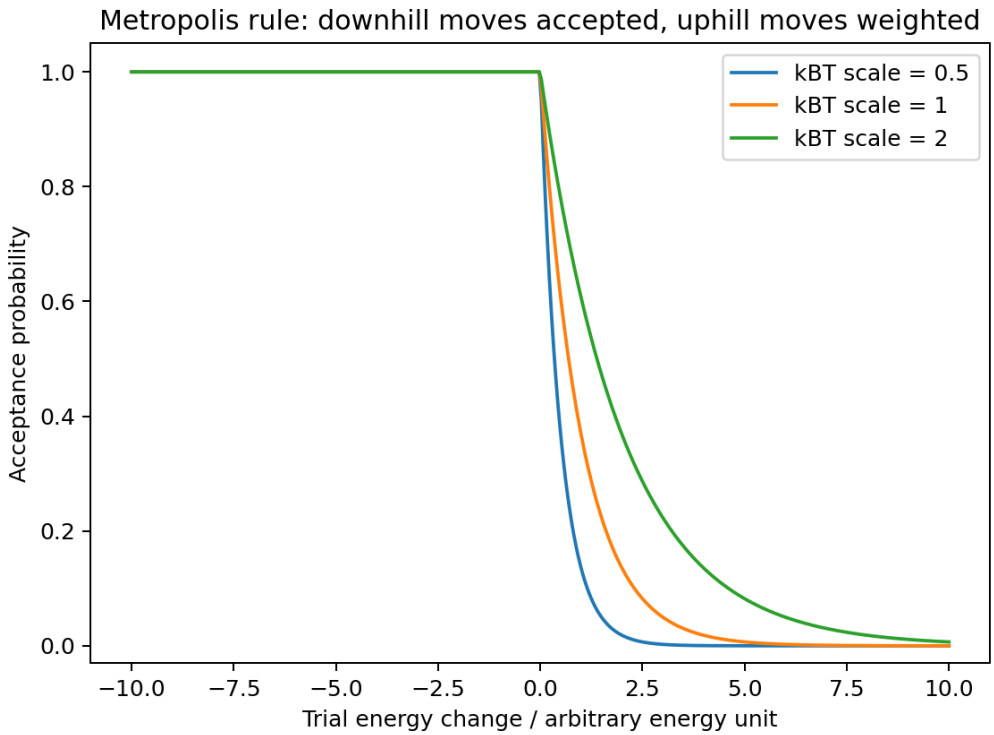


Figure A10.2: Schematic Metropolis acceptance probability as a function of an energy increase. Energy-lowering moves are accepted, while energy-increasing moves remain possible with Boltzmann probability.

fundamental. At finite temperature, the system must sample higher-energy configurations because their configurational multiplicity contributes entropy.

For a symmetric proposal in the canonical ensemble, a move with $\Delta U \leq 0$ is accepted, whereas a move with $\Delta U > 0$ is accepted with probability $\exp(-\beta \Delta U)$. In GCMC, insertion and deletion additionally carry particle-number, volume, thermal-wavelength and chemical-potential factors.

A10.3 Why Widom insertion is dominated by rare favourable events

A naive average insertion energy is not a free-energy estimator. Widom insertion uses an exponential average. This means that a relatively small number of favourable insertions can dominate the result. In a spacious pore at low loading this can be efficient. In a dense phase or strongly excluded region, most random insertions overlap and contribute negligible weight, making the estimator noisy.

This picture is a reminder to examine convergence of the weighted observable, not merely the number of trial insertions. Independent seeds and block statistics are especially valuable when the weight distribution is broad.

A10.4 Move acceptance changes with loading

The appropriate Monte Carlo move mixture is state dependent. At low loading, random insertion into a large pore may be easy. Near saturation, the probability of finding a low-overlap location falls sharply. A low insertion acceptance rate is therefore not automatically a software failure; it can be a physical consequence of crowding. The diagnostic question is whether the Markov chain still explores the relevant particle-number and configurational space efficiently.

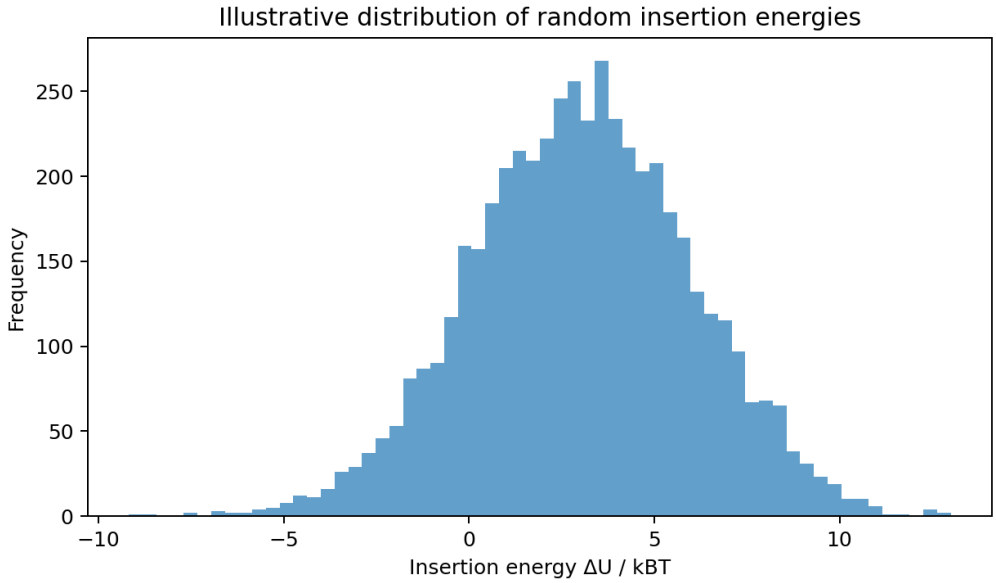


Figure A10.3: Illustrative insertion-energy distribution. Widom averages weight each insertion by $\exp(-\beta\Delta U)$, so the favourable tail can contribute disproportionately.

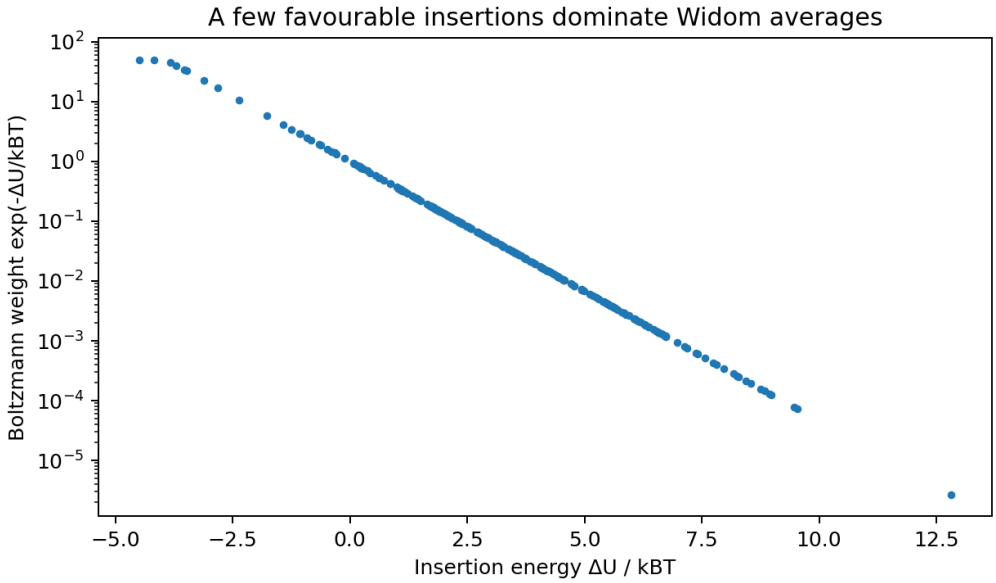


Figure A10.4: Illustrative cumulative contribution to a Widom average. A minority of high-weight trials may account for most of the estimator.

A10.5 Pressure versus fugacity

When a code reports both pressure and fugacity, record both. The pressure is the experimental control variable most readers recognize; the fugacity is the thermodynamic quantity that belongs in the chemical-potential relation. For mixtures, component fugacities are the reservoir quantities controlling insertion/deletion equilibrium.

A10.6 Periodic boundaries, cutoffs and supercell size

A periodic framework is intended to mimic an infinite crystal. However, the finite simulation cell and interaction cutoff can generate self-interaction artefacts if chosen carelessly. A practical geometric check is to

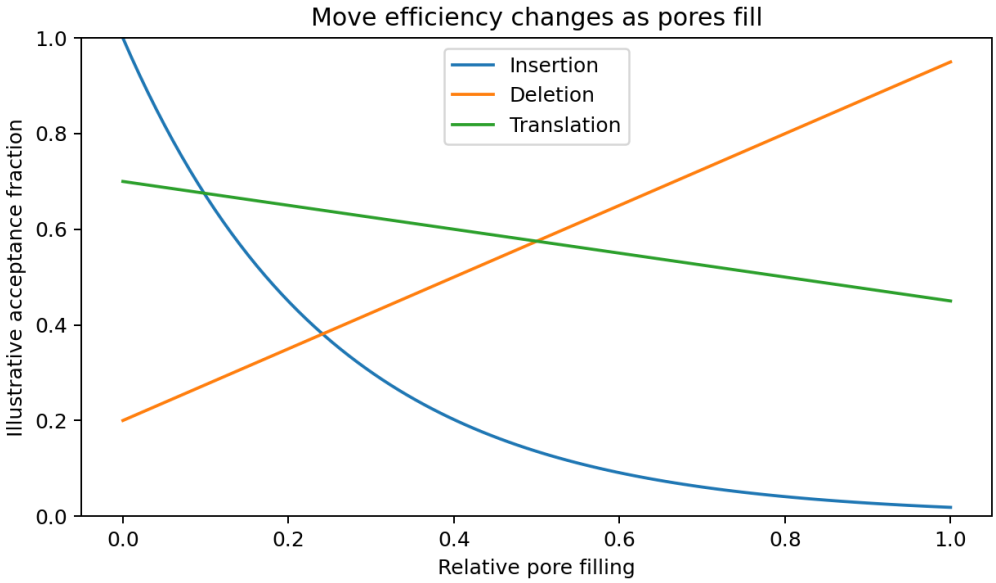


Figure A10.5: Qualitative evolution of move acceptance with pore loading. Insertion can become difficult as free volume disappears, while translation and deletion may retain different efficiencies.

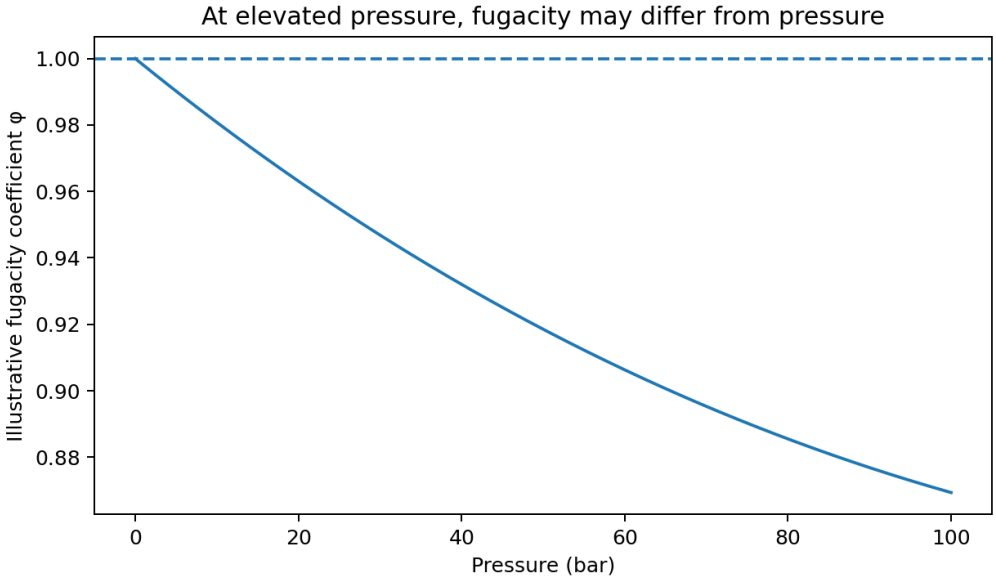


Figure A10.6: Illustrative departure of fugacity from pressure as non-ideality increases. The direction and size of the deviation depend on the gas, temperature and equation of state.

ensure that the periodic box dimensions are comfortably compatible with the cutoff and the minimum-image treatment used by the code. The supplied RASPA2 tutorial explicitly illustrates supercell selection in relation to a 12 Å van der Waals cutoff.

A10.7 Langmuir regimes are physically different regimes

At low pressure, a Langmuir isotherm is linear and its slope combines capacity and affinity. In the intermediate region, curvature contains information about filling. At high pressure, the model tends toward its saturation capacity. Fitting only one regime and extrapolating to another is therefore dangerous. Parameter identifiability must be judged against the pressure range represented by the data.

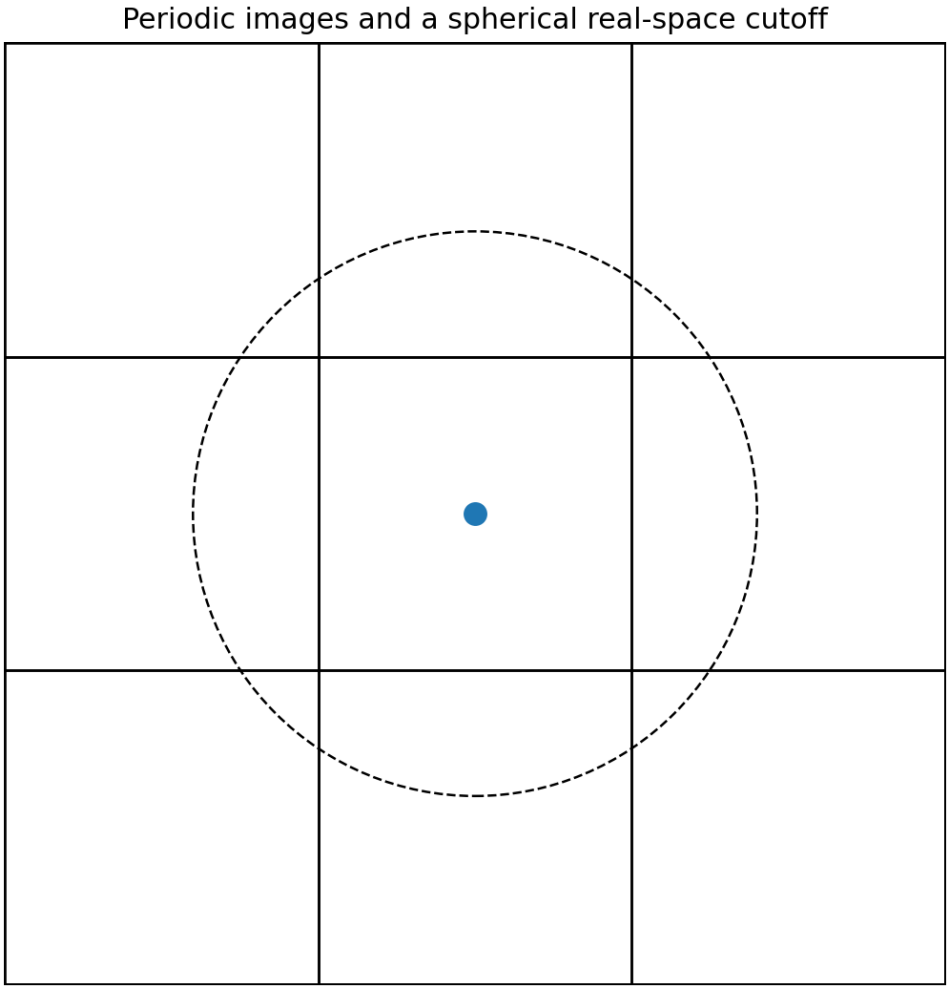


Figure A10.7: Schematic periodic system showing why the simulation cell must be chosen consistently with the non-bonded cutoff.

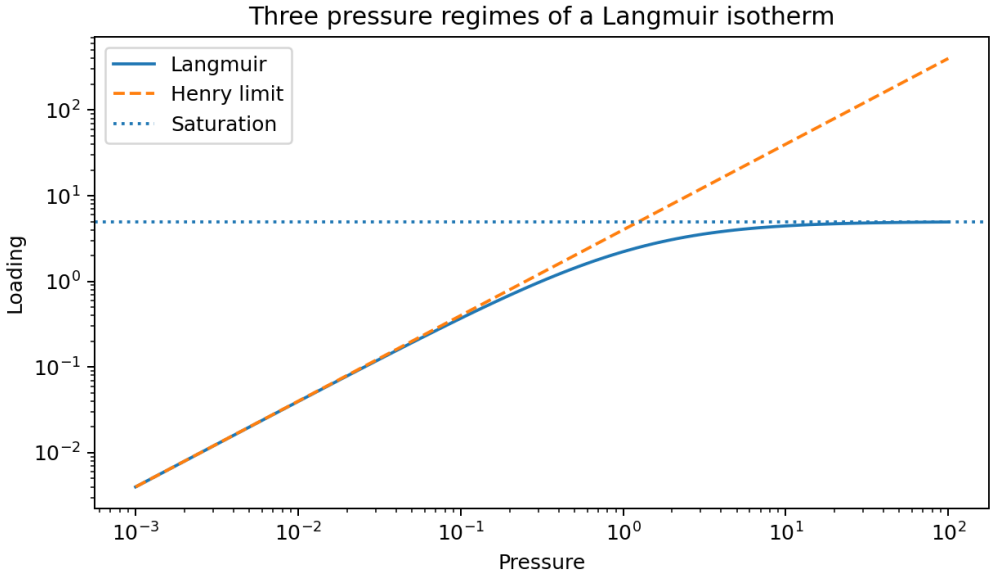


Figure A10.8: Low-pressure Henry region, curved intermediate region and saturation region of a Langmuir-type isotherm.

A10.8 Mixture selectivity is state dependent

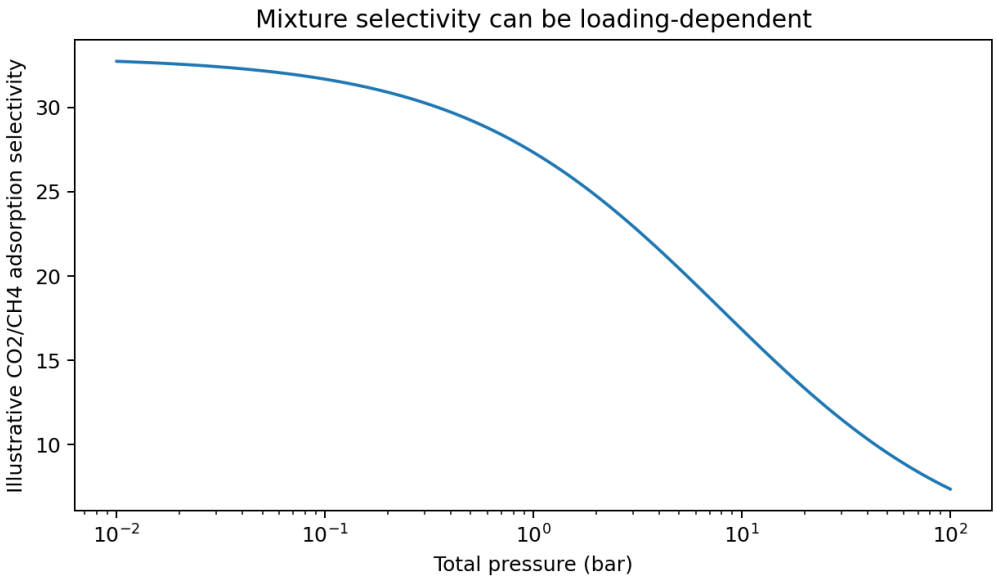


Figure A10.9: Illustrative pressure dependence of mixture selectivity. Competition and site saturation can make selectivity vary strongly with loading.

A single selectivity value does not describe an entire mixture isotherm. At infinite dilution, selectivity may largely reflect relative Henry coefficients. At finite loading, competition for strong sites, saturation and guest–guest interactions can change the ratio. For process work, report the state point together with the selectivity definition.

A10.9 Use IAST as a model, direct GCMC as a test

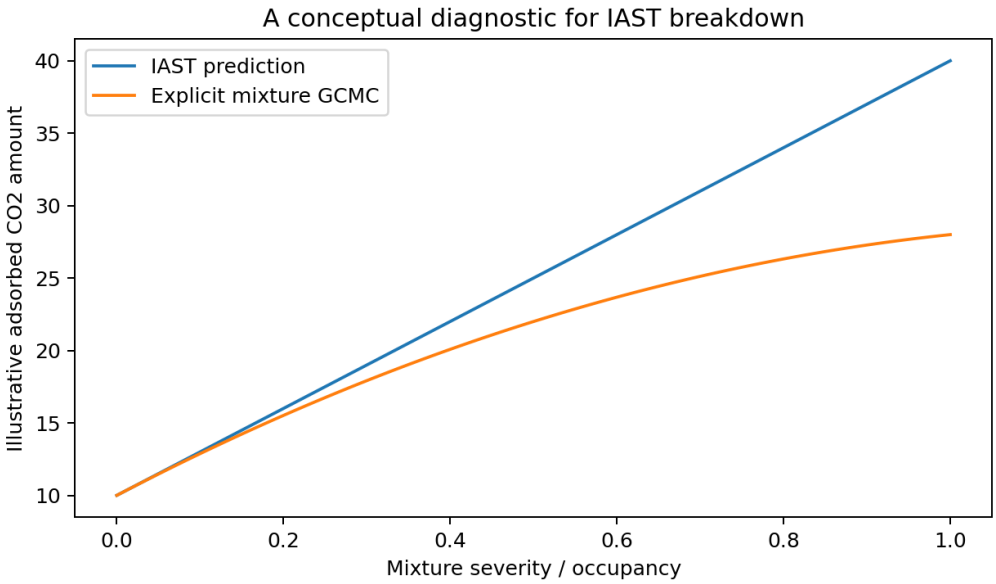


Figure A10.10: Illustrative comparison between an IAST prediction based on fitted pure-component isotherms and direct mixture GCMC.

IAST is attractive because it converts pure-component isotherms into mixture predictions cheaply. Direct mixture GCMC samples competition explicitly under the force field. Agreement increases confidence that

the ideal-adsorbed-solution approximation is adequate in that region. Disagreement is scientifically useful: it identifies where model form, pure-component fits, heterogeneity or non-ideal adsorbed-phase behaviour deserves attention.

A10.10 Convergence means reproducibility across time and seeds

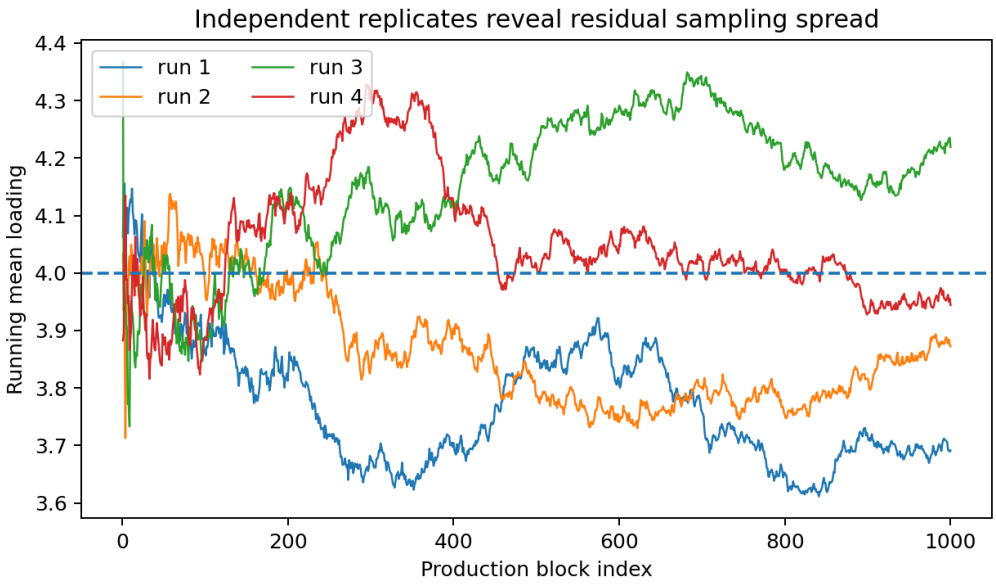


Figure A10.11: Illustrative replicate trajectories converging to a common loading distribution. Independent seeds help distinguish slow equilibration from ordinary statistical scatter.

A long single trajectory is not the only evidence of convergence. Compare early and late blocks, inspect running means, examine particle-number fluctuations, and repeat selected state points with independent random seeds. If independent runs settle into different plateaus, additional cycles may be needed or the state space may contain barriers that require better moves.

A10.11 Heat of adsorption can reveal site filling

A decreasing heat with loading is often consistent with energetic heterogeneity: the most favourable sites dominate at low loading and are progressively saturated. A nearly constant heat can be consistent with a more uniform energetic landscape, although compensating effects are possible. Always distinguish the sign convention for adsorption enthalpy from the positive magnitude often reported as a heat of adsorption.

A10.12 A RASPA run is a chain of physical decisions

Software should be the last step in a chain of decisions, not the first. The structure must be chemically interpretable; atom types and charges must match the force field; the supercell must be suitable for the cutoff; the guest model must be documented; the pressure/fugacity convention must be clear; and the simulation must demonstrate equilibration and adequate sampling. Only then does a reported loading become evidence rather than merely output.

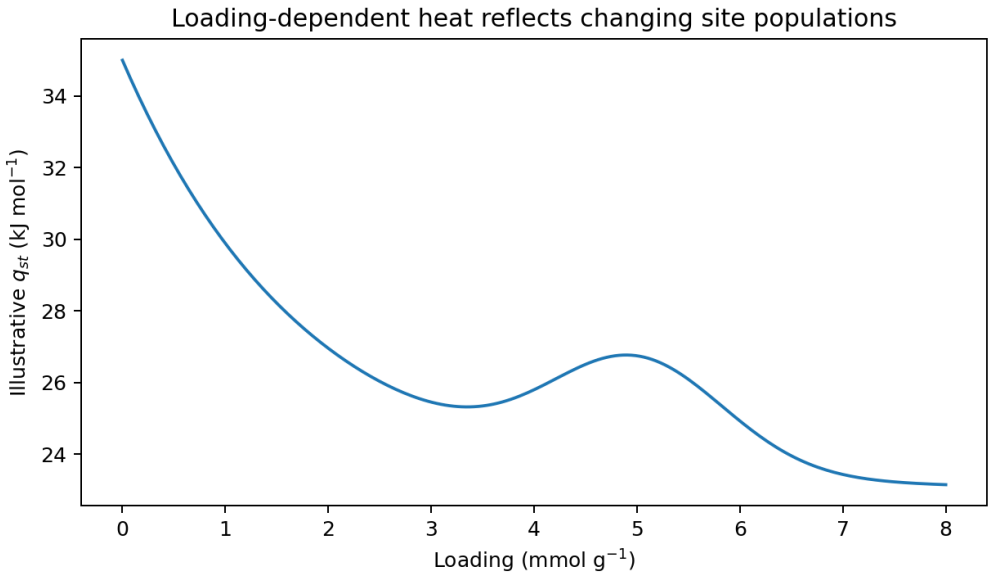


Figure A10.12: Illustrative isosteric-heat trend when strong sites fill before weaker regions of the pore.

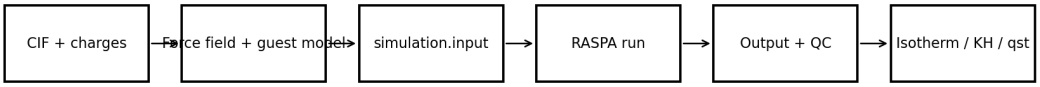


Figure A10.13: Practical RASPA2 workflow: validate structure and force field, declare thermodynamic state and moves, equilibrate, sample production statistics, then audit output and provenance.

PART III

Statistical-Mechanics Lessons for Reliable Monte Carlo

A11 Practical lessons adapted from the standard molecular-simulation framework

This chapter paraphrases a small set of methodological ideas emphasized in standard molecular-simulation texts, especially Frenkel and Smit, and translates them directly into MOF adsorption practice. It intentionally stays within the scope of equilibrium Monte Carlo, GCMC, Widom insertion, configurational bias, periodic systems and statistical error analysis.

A11.1 The ensemble is part of the model

Choosing an ensemble is not a software preference. It specifies which extensive quantities are allowed to exchange with idealized reservoirs and therefore which fluctuations are physically represented. In adsorption, a rigid pore in contact with a gas reservoir naturally exchanges particles and energy while keeping its volume fixed. This motivates the grand-canonical description. If framework breathing or a variable cell is central to the science, a rigid μVT model no longer represents the complete thermodynamic problem.

Research habit

Before running a job, write one sentence stating the ensemble and why it matches the physical experiment. If that sentence is difficult to write, the simulation design is not yet clear.

A11.2 Sampling a distribution is different from minimizing energy

Monte Carlo is designed to reproduce probability distributions. Low-energy configurations matter strongly, but entropy gives statistical weight to the number of configurations available. A minimization answers a zero-temperature structural question; an equilibrium Monte Carlo simulation estimates finite-temperature averages. Confusing these two goals leads to incorrect interpretations of adsorption sites and occupancies.

A11.3 Detailed balance is a route to the desired stationary distribution

A Markov chain is useful only if its stationary distribution is the target ensemble and if the relevant state space is accessible. Detailed balance provides a convenient sufficient condition for constructing acceptance rules. For particle insertion and deletion, the forward and reverse proposals do not have identical combinatorial weights, which is why grand-canonical acceptance expressions contain factors involving V, N, chemical potential and thermal wavelength in addition to the Boltzmann energy factor.

A11.4 Importance sampling spends effort where probability is high

Uniform enumeration of molecular configurations is impossible for realistic systems. Metropolis sampling concentrates visits according to the target distribution. Configurational-bias Monte Carlo goes further for flexible or difficult molecules by generating trial conformations and selecting among them with Rosenbluth weights. The gain is efficiency, but the selection bias must be exactly accounted for in the acceptance probability.

A11.5 Correlation reduces the number of independent samples

Saving one million Monte Carlo steps does not imply one million independent measurements. Nearby states along a Markov chain are correlated. Statistical uncertainty should therefore be estimated with blocks or

another correlation-aware method. The effective sample size depends on the integrated autocorrelation time of the observable, which can differ greatly between loading, energy, and rare-event quantities.

Solved example: effective information content

A production trajectory contains 2.0×10^6 recorded MC measurements, but the loading has an integrated correlation time corresponding roughly to 200 recorded steps. A rough effective sample count is therefore on the order of 10^4, not 2.0×10^6. This explains why the standard error can remain visible despite a very large raw step count.

A11.6 Finite size and periodicity are modelling choices

Periodic boundary conditions replace external crystal surfaces by infinite repetition. This is appropriate for bulk-crystal adsorption but not for finite-particle surface chemistry. The cell must also be compatible with the non-bonded interaction treatment. A simulation can be internally converged yet systematically wrong if the periodic model does not represent the intended experiment.

A11.7 Widom insertion is a free-energy method

Widom insertion estimates an excess chemical potential from a Boltzmann-weighted insertion average. The exponential weighting is the essential idea. The method is especially natural at infinite dilution because the test particle does not perturb the sampled host configuration. It becomes inefficient when almost every insertion has severe overlap, motivating biased insertions or alternative free-energy methods in dense systems.

A11.8 Validation should separate sampling from model error

If independent simulations agree with each other but disagree with experiment, increasing the number of cycles may only make the wrong model more precise. Diagnose separately: (i) numerical/sampling convergence; (ii) structural representation; (iii) force-field and charge model; (iv) thermodynamic state and fugacity treatment; and (v) experimental comparability such as activation, defects, excess versus absolute uptake and sample density.

PART IV

RASPA2 Practical Workbook Matched to the Supplied Manual

A12 How to use the laboratories

These laboratories convert the supplied RASPA 2.0.45 manual into a guided student workflow. They use manual-matched RASPA2 concepts and syntax, but the exact structure, force-field, and molecule files must come from the student's validated installation.

A12.1 Five file roles

A routine rigid-MOF adsorption calculation uses `simulation.input`; a CIF matching `FrameworkName`; pseudo-atom definitions; force-field mixing and optional explicit pair definitions; and molecule-definition files. The manual states that working-directory definitions can take precedence over installed shared definitions, so local overrides must be archived.

A12.2 Pressure and fugacity

The supplied manual specifies `ExternalPressure` in pascal. One bar is therefore `100000.0`. If an explicit fugacity coefficient is not supplied, RASPA can compute one from Peng–Robinson when the molecule definition contains the required critical data. An explicit `FugacityCoefficient 1.0` is a physical modelling choice that imposes $f = p$.

A12.3 Initialization and production

The manual separates `NumberOfInitializationCycles` from `NumberOfCycles`. Do not copy tutorial cycle counts into research calculations without a convergence study.

A13 Laboratory 1: a fixed-N methane box

Learn the run structure before attempting adsorption.

A13.1 Input template

```
SimulationType MonteCarlo
NumberOfCycles 100000
NumberOfInitializationCycles 10000
PrintEvery 1000
Box 0
BoxLengths 30 30 30
ExternalTemperature 300.0
Component 0 MoleculeName methane
TranslationProbability 1.0
CreateNumberOfMolecules 100
```

A13.2 Physical interpretation

There is no swap move, so particle number remains fixed. The calculation is useful for learning RASPA output and translation move statistics, but it is not an adsorption isotherm.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A13.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A13.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A13.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A14 Laboratory 2: N2 GCMC in MIL-47

Reproduce the logic of the manual's MOF adsorption tutorial at one state point.

A14.1 Input template

```
SimulationType MonteCarlo
NumberOfCycles 10000
NumberOfInitializationCycles 10000
PrintEvery 100
RestartFile no
Forcefield ExampleMOFsForceField
UseChargesFromCIFFile yes
Framework 0
FrameworkName MIL-47
UnitCells 4 2 2
HeliumVoidFraction 0.61
ExternalTemperature 300.0
```

```
ExternalPressure 100000.0
Component 0 MoleculeName N2
MoleculeDefinition ExampleDefinitions
TranslationProbability 0.5
RotationProbability 0.5
ReinsertionProbability 0.5
SwapProbability 1.0
CreateNumberOfMolecules 0
```

A14.2 Physical interpretation

Swap attempts create particle exchange with the reservoir. At 300 K and 100000 Pa the requested pressure is 1 bar. The manual tutorial uses a 4 x 2 x 2 supercell and a helium void fraction, so check charge neutrality, box dimensions, absolute/excess loading, and swap statistics.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A14.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A14.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A14.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A15 Laboratory 3: scout the pressure range

Use short exploratory runs to find the Henry regime and onset of curvature.

A15.1 Input template

```
# Keep the validated MIL-47/N2 model from Laboratory 2
NumberOfCycles 3000
NumberOfInitializationCycles 2000
# run independent directories at:
# 100, 1000, 10000, 100000, 1000000 Pa
```

A15.2 Physical interpretation

These are reconnaissance calculations, not final isotherm points. Use them to design a log-spaced production grid with extra resolution where loading changes rapidly.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A15.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A15.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A15.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.

3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A16 Laboratory 4: Henry coefficient with Widom insertion

Estimate infinite-dilution affinity directly.

A16.1 Input template

```
SimulationType MonteCarlo
NumberOfCycles 50000
NumberOfInitializationCycles 5000
PrintEvery 1000
Forcefield ExampleMOFsForceField
UseChargesFromCIFFile yes
Framework 0
FrameworkName MIL-47
UnitCells 4 2 2
ExternalTemperature 300.0
Component 0 MoleculeName N2
MoleculeDefinition ExampleDefinitions
WidomProbability 1.0
CreateNumberOfMolecules 0
```

A16.2 Physical interpretation

The manual connects Widom/Rosenbluth insertion statistics to the Henry coefficient. Compare the result with a low-pressure GCMC slope, but only after verifying that the slope points are truly in the linear regime.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A16.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A16.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A16.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A17 Laboratory 5: infinite-dilution adsorption enthalpy

Use the same validated model to learn reference-state and sign conventions.

A17.1 Input template

```
# Same framework and guest model as Laboratory 4
SimulationType MonteCarlo
NumberOfCycles 50000
NumberOfInitializationCycles 5000
PrintPropertiesEvery 1000
Framework 0
FrameworkName MIL-47
UnitCells 4 2 2
ExternalTemperature 300.0
Component 0 MoleculeName N2
MoleculeDefinition ExampleDefinitions
WidomProbability 1.0
CreateNumberOfMolecules 0
```

A17.2 Physical interpretation

The manual tutorial relates the infinite-dilution enthalpy to host, guest, and host-plus-guest energy averages and notes the relation between adsorption enthalpy and positive heat-of-adsorption conventions. Read the output label carefully.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A17.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A17.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A17.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A18 Laboratory 6: methane in MFI

Study absolute/excess adsorption, fugacity handling, and histograms.

A18.1 Input template

```
SimulationType MonteCarlo
NumberOfCycles 25000
NumberOfInitializationCycles 2000
PrintEvery 1000
Forcefield ExampleZeolitesForceField
RemoveAtomNumberCodeFromLabel yes
Framework 0
FrameworkName MFI_SI
UnitCells 2 2 2
HeliumVoidFraction 0.29
ExternalTemperature 300.0
ExternalPressure 100000.0
```

```
ComputeNumberOfMoleculesHistogram yes
WriteNumberOfMoleculesHistogramEvery 5000
ComputeEnergyHistogram yes
WriteEnergyHistogramEvery 5000
Component 0 MoleculeName methane
MoleculeDefinition ExampleDefinitions
TranslationProbability 0.5
ReinsertionProbability 0.5
SwapProbability 1.0
CreateNumberOfMolecules 0
```

A18.2 Physical interpretation

This follows the manual's example logic. Inspect the reported fugacity coefficient, loading in several units, absolute/excess difference, histograms, and swap addition/deletion performance.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A18.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A18.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A18.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A19 Laboratory 7: CO2 in Cu-BTC with CIF charges

Audit charge neutrality and a deliberately specified fugacity coefficient.

A19.1 Input template

```
SimulationType MonteCarlo
NumberOfCycles 10000
NumberOfInitializationCycles 5000
PrintEvery 1000
RestartFile no
Forcefield ExampleMOFsForceField
UseChargesFromCIFFile yes
Framework 0
FrameworkName Cu-BTC
UnitCells 1 1 1
HeliumVoidFraction 0.29
ExternalTemperature 323.0
ExternalPressure 100000.0
Component 0 MoleculeName CO2
MoleculeDefinition ExampleDefinitions
FugacityCoefficient 1.0
TranslationProbability 0.5
RotationProbability 0.5
ReinsertionProbability 0.5
SwapProbability 1.0
CreateNumberOfMolecules 0
```

A19.2 Physical interpretation

Use this manual-matched pattern to confirm that the framework charge sum is zero and to understand that FugacityCoefficient 1.0 makes numerical fugacity equal numerical pressure.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A19.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A19.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A19.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A20 Laboratory 8: a binary CO2/CH4 mixture

Use documented component mole fractions to build an explicit competitive adsorption calculation.

A20.1 Input template

```
SimulationType MonteCarlo
NumberOfCycles 100000
NumberOfInitializationCycles 20000
PrintEvery 2000
Forcefield ExampleMOFsForceField
UseChargesFromCIFFile yes
Framework 0
FrameworkName YourMOF
UnitCells 2 2 2
HeliumVoidFraction 0.50
ExternalTemperature 298.0
ExternalPressure 1000000.0
Component 0 MoleculeName CO2
MoleculeDefinition ExampleDefinitions
MolFraction 0.10
TranslationProbability 0.5
RotationProbability 0.5
ReinsertionProbability 0.5
SwapProbability 1.0
CreateNumberOfMolecules 0
Component 1 MoleculeName methane
MoleculeDefinition ExampleDefinitions
MolFraction 0.90
TranslationProbability 0.5
ReinsertionProbability 0.5
SwapProbability 1.0
CreateNumberOfMolecules 0
```

A20.2 Physical interpretation

RASPA normalises the mole fractions. Inspect the printed component partial pressures/fugacities, adsorbed composition, selectivity, and component-specific swap efficiency. Compare with IAST from consistent pure-component isotherms.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A20.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A20.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A20.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A21 Laboratory 9: crash recovery

Test restart and binary crash recovery before a large HPC campaign.

A21.1 Input template

```
SimulationType MonteCarlo
```

```
NumberOfCycles 500000
NumberOfInitializationCycles 50000
PrintEvery 5000
ContinueAfterCrash yes
WriteBinaryRestartFileEvery 25000
# validated framework and component blocks follow
```

A21.2 Physical interpretation

The manual documents binary crash recovery. Test it deliberately on a short disposable run so that queue limits or node failures do not destroy long production calculations.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A21.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A21.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A21.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A22 Laboratory 10: a SLURM isotherm array

Run independent state points as auditable array jobs.

A22.1 Input template

```
#!/bin/bash
#SBATCH --job-name=raspa_iso
#SBATCH --array=0-7
#SBATCH --cpus-per-task=1
#SBATCH --time=08:00:00
#SBATCH --mem=2G
set -euo pipefail
export RASPA_DIR="$HOME/RASPA/simulations"
P=(100 1000 10000 100000 300000 1000000 3000000 10000000)
cd "P_${P[$SLURM_ARRAY_TASK_ID]}Pa"
"$RASPA_DIR/bin/simulate" > run.stdout 2> run.stderr
```

A22.2 Physical interpretation

Prepare each directory in advance. Do not let concurrent jobs edit a shared input file. Store pressure, input hash, code version, job ID, exit status, and convergence status in a manifest.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A22.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A22.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A22.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A23 Laboratory 11: output parsing and QC

Turn RASPA output into one machine-readable state-point row.

A23.1 Input template

```
# Parse:
# requested pressure and printed fugacity
# absolute and excess loading plus uncertainty
# enthalpy/energy if requested
# swap addition/deletion attempts and acceptance
# framework density and helium void fraction
# run completion marker
```

A23.2 Physical interpretation

A parser should fail loudly if a required field is absent. Unit-test it against frozen outputs. Do not silently substitute zero for missing loading or for a failed job.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A23.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.
- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A23.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A23.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

A24 Laboratory 12: complete student mini-project

Integrate Widom, pure GCMC, mixture GCMC, IAST, uncertainty, and interpretation.

A24.1 Input template

```
# A. validate CIF, atom types, charges, and force field
# B. Widom CO2 and CH4 at 298 K
# C. pure GCMC at 0.01, 0.1, 1, 5, 10 bar
# D. 10/90 CO2/CH4 mixture at 1 and 10 bar
# E. fit pure isotherms and run IAST
# F. repeat one state point with three independent runs
```

A24.2 Physical interpretation

The final report should state the scientific question, model provenance, convergence evidence, agreement or disagreement between IAST and mixture GCMC, and the dominant source of uncertainty.

Read the input as a physical declaration. Confirm the framework identity, supercell, charge source, force field, temperature, pressure in pascal, component model, and enabled moves before running. After the run, confirm that the output echoes the intended model and state point.

A24.3 Step-by-step checks

Before accepting a result

- Confirm requested pressure, printed pressure, fugacity coefficient, and fugacity.
- Check framework atom count, cell, density, and net charge where reported.
- Confirm all atom types and molecule definitions were found.
- Inspect move attempts and acceptance statistics.

- Check block values or independent runs for stationarity.
- Record absolute/excess basis and units.
- Archive the exact input, local overrides, full output, and a manifest row.

A24.4 Worked interpretation example

A common diagnostic

Suppose the job finishes but the loading is unexpectedly zero. Before extending the run, check pressure units, the guest component, swap probability, atom typing, pore accessibility, and whether the output indicates repeated insertion overlap or a missing force-field interaction. A longer run cannot fix an input that requests the wrong thermodynamic state.

A24.5 Independent extensions

Do these without looking for a solution

1. Change one thermodynamic variable only and predict the direction of the result before running.
2. Double production length and determine whether the uncertainty changes as expected.
3. Identify the single model assumption most likely to dominate systematic error.
4. Write a three-sentence interpretation suitable for a group meeting.
5. Record one failure mode and how you would detect it automatically in a high-throughput campaign.

PART V

Extended Solved Examples and Independent Exercises

A25 Sixty solved examples

The purpose of these examples is not to create a catalogue of arithmetic. Each solution is followed by a research interpretation: what assumption was made, which unit or convention can fail, and what simulation would be needed to turn the classroom result into a defensible research result.

A25.1 Solved example 1: Pressure units

Problem and solution

Problem. A user types ExternalPressure 5 intending 5 bar. What was actually requested?

Solution. 5 Pa, which is 100000 times smaller than 5 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.2 Solved example 2: Half saturation

Problem and solution

Problem. For Langmuir b=0.2 bar^{-1}, find half-saturation pressure.

Solution. p=1/b=5 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.3 Solved example 3: Henry slope

Problem and solution

Problem. q=0.03 mmol/g at 0.01 bar in Henry regime. Find K_H.

Solution. 3 mmol g^{-1} bar^{-1}.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.4 Solved example 4: Working capacity

Problem and solution

Problem. q_ads=7.2 and q_reg=1.6 mmol/g. Find delta q.

Solution. 5.6 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.5 Solved example 5: Volumetric loading

Problem and solution

Problem. 5 mol/kg at packed density 0.6 kg/L. Find mol/L.

Solution. 3.0 mol/L.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.6 Solved example 6: Selectivity

Problem and solution

Problem. Feed 10/90, adsorbed 80/20. Find selectivity.

Solution. 36.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.7 Solved example 7: Fugacity

Problem and solution

Problem. p=70 bar and phi=0.8. Find f.

Solution. 56 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.8 Solved example 8: Chemical potential

Problem and solution

Problem. Find ideal-gas Delta mu for pressure ratio 10 at 298 K.

Solution. RT ln 10 is approximately 5.71 kJ/mol.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a

second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.9 Solved example 9: Metropolis

Problem and solution

Problem. At 298 K, Delta U=+4 kJ/mol. Find symmetric acceptance.

Solution. exp(-4/RT) is about 0.20.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.10 Solved example 10: Autocorrelation

Problem and solution

Problem. N=400000, tau_int=100. Estimate N_eff.

Solution. about 2000.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.11 Solved example 11: Excess loading

Problem and solution

Problem. Absolute=9 mol/kg; bulk reference=1.5. Find excess.

Solution. 7.5 mol/kg.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.12 Solved example 12: Supercell

Problem and solution

Problem. Lengths 10,13,17 A, cutoff 12 A. Suggest replication.

Solution. 3x2x2 gives 30,26,34 A.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.13 Solved example 13: Charge replication

Problem and solution

Problem. Unit cell net charge +0.04e; replicate 3x2x2. Net charge?

Solution. +0.48e.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.14 Solved example 14: Job count

Problem and solution

Problem. 800 MOFs x 15 pressures x 2 gases. Jobs?

Solution. 24,000.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric.

If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.15 Solved example 15: Core hours

Problem and solution

Problem. 24,000 jobs x 20 min one core. Core-hours?

Solution. 8,000 core-hours.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.16 Solved example 16: Dual-site Henry

Problem and solution

Problem. q1=1,b1=5,q2=4,b2=0.25. K_H?

Solution. 6 in consistent units.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.17 Solved example 17: LJ minimum

Problem and solution

Problem. Where is the LJ 12-6 minimum?

Solution. $r = 2^{1/6}\sigma$.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.18 Solved example 18: Temperature trend

Problem and solution

Problem. At fixed p, uptake falls with increasing T. What sign of adsorption enthalpy is expected?

Solution. Exothermic adsorption, negative enthalpy change under the usual thermodynamic sign convention.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.19 Solved example 19: Widom ratio

Problem and solution

Problem. Two insertion averages are 0.01 and 0.1. Henry ratio?

Solution. 10.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.20 Solved example 20: Molecules per unit cell

Problem and solution

Problem. 80 molecules in a 2x2x2 supercell. Molecules per unit cell?

Solution. 10.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.21 Solved example 21: Selectivity variant 21

Problem and solution

Problem. Feed A fraction is 0.04; adsorbed A fraction is 0.20. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.22 Solved example 22: Pressure conversion variant 22

Problem and solution

Problem. Convert 12.5 bar to pascal.

Solution. 1250000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a

second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.23 Solved example 23: Effective-sample variant 23

Problem and solution

Problem. A trace has 600000 samples and tau_int=40. Estimate N_eff.

Solution. Approximately 7500.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.24 Solved example 24: Fugacity variant 24

Problem and solution

Problem. At 70 bar with phi=0.80, calculate fugacity.

Solution. f=phi p=56.0 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.25 Solved example 25: Working-capacity variant 25

Problem and solution

Problem. Adsorption loading is 12.0 mmol/g and regeneration loading is 3.0 mmol/g. Compute working capacity.

Solution. 9.0 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric.

If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.26 Solved example 26: Selectivity variant 26

Problem and solution

Problem. Feed A fraction is 0.09; adsorbed A fraction is 0.45. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.27 Solved example 27: Pressure conversion variant 27

Problem and solution

Problem. Convert 2.5 bar to pascal.

Solution. 250000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.28 Solved example 28: Effective-sample variant 28

Problem and solution

Problem. A trace has 200000 samples and tau_int=20. Estimate N_eff.

Solution. Approximately 5000.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.29 Solved example 29: Fugacity variant 29

Problem and solution

Problem. At 30 bar with phi=0.76, calculate fugacity.

Solution. f=phi p=22.8 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.30 Solved example 30: Working-capacity variant 30

Problem and solution

Problem. Adsorption loading is 8.0 mmol/g and regeneration loading is 2.0 mmol/g. Compute working capacity.

Solution. 6.0 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.31 Solved example 31: Selectivity variant 31

Problem and solution

Problem. Feed A fraction is 0.05; adsorbed A fraction is 0.25. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.32 Solved example 32: Pressure conversion variant 32

Problem and solution

Problem. Convert 15.0 bar to pascal.

Solution. 1500000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.33 Solved example 33: Effective-sample variant 33

Problem and solution

Problem. A trace has 700000 samples and tau_int=70. Estimate N_eff.

Solution. Approximately 5000.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric.

If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.34 Solved example 34: Fugacity variant 34

Problem and solution

Problem. At 80 bar with phi=0.86, calculate fugacity.

Solution. f=phi p=68.8 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.35 Solved example 35: Working-capacity variant 35

Problem and solution

Problem. Adsorption loading is 13.0 mmol/g and regeneration loading is 1.0 mmol/g. Compute working capacity.

Solution. 12.0 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.36 Solved example 36: Selectivity variant 36

Problem and solution

Problem. Feed A fraction is 0.01; adsorbed A fraction is 0.05. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.37 Solved example 37: Pressure conversion variant 37

Problem and solution

Problem. Convert 5.0 bar to pascal.

Solution. 500000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.38 Solved example 38: Effective-sample variant 38

Problem and solution

Problem. A trace has 300000 samples and tau_int=50. Estimate N_eff.

Solution. Approximately 3000.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a

second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.39 Solved example 39: Fugacity variant 39

Problem and solution

Problem. At 40 bar with phi=0.82, calculate fugacity.

Solution. f=phi p=32.8 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.40 Solved example 40: Working-capacity variant 40

Problem and solution

Problem. Adsorption loading is 9.0 mmol/g and regeneration loading is 3.5 mmol/g. Compute working capacity.

Solution. 5.5 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.41 Solved example 41: Selectivity variant 41

Problem and solution

Problem. Feed A fraction is 0.06; adsorbed A fraction is 0.30. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric.

If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.42 Solved example 42: Pressure conversion variant 42

Problem and solution

Problem. Convert 17.5 bar to pascal.

Solution. 1750000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.43 Solved example 43: Effective-sample variant 43

Problem and solution

Problem. A trace has 800000 samples and tau_int=30. Estimate N_eff.

Solution. Approximately 13333.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.44 Solved example 44: Fugacity variant 44

Problem and solution

Problem. At 90 bar with phi=0.78, calculate fugacity.

Solution. f=phi p=70.2 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.45 Solved example 45: Working-capacity variant 45

Problem and solution

Problem. Adsorption loading is 5.0 mmol/g and regeneration loading is 2.5 mmol/g. Compute working capacity.

Solution. 2.5 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.46 Solved example 46: Selectivity variant 46

Problem and solution

Problem. Feed A fraction is 0.02; adsorbed A fraction is 0.10. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.47 Solved example 47: Pressure conversion variant 47

Problem and solution

Problem. Convert 7.5 bar to pascal.

Solution. 750000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.48 Solved example 48: Effective-sample variant 48

Problem and solution

Problem. A trace has 400000 samples and tau_int=80. Estimate N_eff.

Solution. Approximately 2500.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.49 Solved example 49: Fugacity variant 49

Problem and solution

Problem. At 50 bar with phi=0.74, calculate fugacity.

Solution. f=phi p=37.0 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric.

If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.50 Solved example 50: Working-capacity variant 50

Problem and solution

Problem. Adsorption loading is 10.0 mmol/g and regeneration loading is 1.5 mmol/g. Compute working capacity.

Solution. 8.5 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.51 Solved example 51: Selectivity variant 51

Problem and solution

Problem. Feed A fraction is 0.07; adsorbed A fraction is 0.35. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.52 Solved example 52: Pressure conversion variant 52

Problem and solution

Problem. Convert 20.0 bar to pascal.

Solution. 2000000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.53 Solved example 53: Effective-sample variant 53

Problem and solution

Problem. A trace has 900000 samples and tau_int=60. Estimate N_eff.

Solution. Approximately 7500.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.54 Solved example 54: Fugacity variant 54

Problem and solution

Problem. At 10 bar with phi=0.84, calculate fugacity.

Solution. f=phi p=8.4 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a

second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.55 Solved example 55: Working-capacity variant 55

Problem and solution

Problem. Adsorption loading is 6.0 mmol/g and regeneration loading is 4.0 mmol/g. Compute working capacity.

Solution. 2.0 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.56 Solved example 56: Selectivity variant 56

Problem and solution

Problem. Feed A fraction is 0.03; adsorbed A fraction is 0.15. Compute A/B selectivity.

Solution. Use S=(x_A/x_B)/(y_A/y_B); keep the state point attached to the resulting number.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.57 Solved example 57: Pressure conversion variant 57

Problem and solution

Problem. Convert 10.0 bar to pascal.

Solution. 1000000 Pa.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.58 Solved example 58: Effective-sample variant 58

Problem and solution

Problem. A trace has 500000 samples and tau_int=40. Estimate N_eff.

Solution. Approximately 6250.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.59 Solved example 59: Fugacity variant 59

Problem and solution

Problem. At 60 bar with phi=0.80, calculate fugacity.

Solution. f=phi p=48.0 bar.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A25.60 Solved example 60: Working-capacity variant 60

Problem and solution

Problem. Adsorption loading is 11.0 mmol/g and regeneration loading is 3.0 mmol/g. Compute working capacity.

Solution. 8.0 mmol/g.

Interpretation. Before using this result in research, identify the uptake basis, thermodynamic state, force-field/model assumptions, and whether the quantity is an equilibrium descriptor or a process metric. If the result comes from RASPA, save the exact input and verify the printed pressure/fugacity, structure identity, and relevant move statistics.

Follow-up calculation. Repeat the calculation under one deliberately changed assumption and state whether the numerical conclusion should remain stable. Examples include changing temperature, using a second charge scheme, doubling the supercell, extending production, or comparing ideal pressure mapping with a real-gas fugacity correction.

A26 One hundred and twenty unsolved exercises

These problems intentionally omit solutions. They are suitable for weekly student assignments, group-meeting discussion, or oral examination. Students should state assumptions and units explicitly.

A26.1 Thermodynamics

1. Calculate an ideal-gas chemical-potential change for a specified pressure ratio. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Explain why chemical potential is preferable to an internal pore pressure for rigid adsorption. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Derive the low-pressure limit of a Langmuir model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Compare two isotherms with equal saturation capacity but different Henry coefficients. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Explain how temperature shifts an exothermic adsorption equilibrium. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Design a multi-temperature experiment for estimating q_st. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Calculate an ideal-gas chemical-potential change for a specified pressure ratio. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Explain why chemical potential is preferable to an internal pore pressure for rigid adsorption. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Derive the low-pressure limit of a Langmuir model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. Compare two isotherms with equal saturation capacity but different Henry coefficients. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Explain how temperature shifts an exothermic adsorption equilibrium. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Design a multi-temperature experiment for estimating q_st. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.2 Isotherm fitting

1. Fit a conceptual Langmuir curve and identify which pressure region constrains q_s. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Explain parameter correlation in a dual-site Langmuir fit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Compare nonlinear fitting with a linearised transform. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Design residual plots that would reveal systematic curvature. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Explain why a high R-squared does not prove a physically meaningful model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Propose an uncertainty analysis for extrapolated working capacity. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Fit a conceptual Langmuir curve and identify which pressure region constrains q_s. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Explain parameter correlation in a dual-site Langmuir fit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Compare nonlinear fitting with a linearised transform. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. Design residual plots that would reveal systematic curvature. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Explain why a high R-squared does not prove a physically meaningful model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Propose an uncertainty analysis for extrapolated working capacity. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.3 Monte Carlo

1. Derive a Metropolis ratio for a symmetric NVT translation. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Explain detailed balance using insertion and deletion as reverse moves. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Discuss ergodicity in a disconnected pore network. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Explain why Monte Carlo step count is not physical time. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Design a move-mix diagnostic at high loading. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Explain the purpose of configurational bias. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Derive a Metropolis ratio for a symmetric NVT translation. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Explain detailed balance using insertion and deletion as reverse moves. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Discuss ergodicity in a disconnected pore network. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

10. Explain why Monte Carlo step count is not physical time. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Design a move-mix diagnostic at high loading. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Explain the purpose of configurational bias. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.4 GCMC and Widom

1. Explain the origin of the volume and particle-number terms in an insertion acceptance factor. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Describe what happens to insertion acceptance as pores fill. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Compare low-pressure GCMC with Widom insertion. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Explain rare-event bias in a Widom average. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Design a Henry-coefficient convergence test. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Explain why deletion and insertion statistics should both be inspected. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Explain the origin of the volume and particle-number terms in an insertion acceptance factor. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Describe what happens to insertion acceptance as pores fill. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Compare low-pressure GCMC with Widom insertion. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. Explain rare-event bias in a Widom average. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Design a Henry-coefficient convergence test. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Explain why deletion and insertion statistics should both be inspected. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.5 Force fields

1. Separate guest-framework and guest-guest energy contributions. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Explain the roles of sigma and epsilon in a Lennard-Jones model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Design a charge-sensitivity test for CO2 adsorption. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Explain why open metal sites may need higher-fidelity models. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

5. Describe a cutoff-convergence test. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Explain why a force-field name alone is insufficient provenance. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Separate guest-framework and guest-guest energy contributions. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Explain the roles of sigma and epsilon in a Lennard-Jones model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Design a charge-sensitivity test for CO2 adsorption. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. Explain why open metal sites may need higher-fidelity models. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Describe a cutoff-convergence test. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Explain why a force-field name alone is insufficient provenance. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.6 Structures and PBC

1. Create a CIF preflight checklist. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Choose supercell replication for a given cutoff. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Explain periodic self-image artefacts. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Discuss when defects invalidate a perfect-crystal model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Explain why a flexible framework may require a larger cell. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Design an atom-typing coverage audit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Create a CIF preflight checklist. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Choose supercell replication for a given cutoff. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Explain periodic self-image artefacts. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. Discuss when defects invalidate a perfect-crystal model. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Explain why a flexible framework may require a larger cell. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Design an atom-typing coverage audit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.7 Mixtures and IAST

1. Compute selectivity from feed and adsorbed compositions. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Explain why pure-gas uptake ratios are not mixture selectivities. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Describe the spreading-pressure idea in IAST. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. List mechanisms of IAST failure. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Design an IAST versus mixture-GCMC validation set. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Explain what additional information RAST conceptually requires. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Compute selectivity from feed and adsorbed compositions. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Explain why pure-gas uptake ratios are not mixture selectivities. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Describe the spreading-pressure idea in IAST. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. List mechanisms of IAST failure. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Design an IAST versus mixture-GCMC validation set. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Explain what additional information RAST conceptually requires. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.8 Uncertainty and validation

1. Estimate effective sample size from an autocorrelation time. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Design a block-averaging workflow. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Explain why q_st may converge slower than loading. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Diagnose a low-pressure-only simulation-experiment mismatch. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Separate calibration from validation. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Design three independent validation cases. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Estimate effective sample size from an autocorrelation time. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Design a block-averaging workflow. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Explain why q_st may converge slower than loading. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

10. Diagnose a low-pressure-only simulation-experiment mismatch. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Separate calibration from validation. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Design three independent validation cases. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.9 RASPA2

1. Convert a human-facing pressure in bar to the RASPA2 input unit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Explain the roles of FrameworkName, UnitCells, and HeliumVoidFraction. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Explain the roles of TranslationProbability, RotationProbability, ReinsertionProbability, SwapProbability, and WidomProbability. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Describe how local force-field files can override installed definitions. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
5. Explain what FugacityCoefficient 1.0 means. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Design a parser for absolute and excess loading. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Convert a human-facing pressure in bar to the RASPA2 input unit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Explain the roles of FrameworkName, UnitCells, and HeliumVoidFraction. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Explain the roles of TranslationProbability, RotationProbability, ReinsertionProbability, SwapProbability, and WidomProbability. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. Describe how local force-field files can override installed definitions. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Explain what FugacityCoefficient 1.0 means. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Design a parser for absolute and excess loading. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A26.10 HPC and reproducibility

1. Calculate state-point count for a multidimensional screening grid. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
2. Design a SLURM array mapping. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
3. Create a job manifest schema. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
4. Explain why failed structures must remain in the audit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

5. Design restart and crash-recovery testing. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
6. Write a publication-quality provenance checklist. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
7. Calculate state-point count for a multidimensional screening grid. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
8. Design a SLURM array mapping. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
9. Create a job manifest schema. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
10. Explain why failed structures must remain in the audit. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
11. Design restart and crash-recovery testing. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.
12. Write a publication-quality provenance checklist. Include one quantitative check or equation where appropriate, one modelling assumption, and one possible failure mode.

A27 Hints for the independent exercises

> These are direction-of-travel hints rather than full solutions. Attempt the problem first.

- Write units before numbers and check a limiting case.
- For an isotherm, identify Henry, transition, and saturation regimes.
- For Monte Carlo, identify target distribution, proposal, reverse move, and acceptance probability.
- For RASPA2, remember that `ExternalPressure` is in pascal and inspect the echoed model in the output.
- For mixture adsorption, distinguish gas mole fraction, component fugacity, adsorbed mole fraction, and selectivity.
- For validation, align material state, temperature, pressure/fugacity, and uptake basis before computing an error.
- For uncertainty, separate statistical sampling error from model-form uncertainty.

A28 Selected further reading

- Frenkel and Smit, *Understanding Molecular Simulation*, 2nd ed. A foundational source for statistical mechanics and Monte Carlo algorithms.
- Dubbeldam, Calero, Ellis, and Snurr, "RASPA: molecular simulation software for adsorption and diffusion in flexible nanoporous materials," *Molecular Simulation* 42 (2016) 81–101.
- Ran et al., "RASPA3: A Monte Carlo code for computing adsorption and diffusion in nanoporous materials and thermodynamic properties of fluids," *Journal of Chemical Physics* 161 (2024) 114106.
- The current RASPA3 manual and examples in the official iRASPA/RASPA3 repository. Because the package is actively developed, these should be consulted for version-specific keywords.
- Original papers for the particular force fields, guest models, charge methods, and equations of state used in a research project.

References

Bibliography

[1] M. Asgari, *Adsorption Thermodynamics – Session 1: Equilibrium, Langmuir/Henry, Temperature Dependence, Heat of Adsorption*, TMU guest lecture slides, 2025.

[2] M. Asgari, *Adsorption Thermodynamics – Session 2: Statistical Thermodynamics and Grand Canonical Monte Carlo*, TMU guest lecture slides, 2025.

[3] D. Frenkel and B. Smit, *Understanding Molecular Simulation: From Algorithms to Applications*, 2nd ed., Academic Press, 2002.

[4] D. Dubbeldam, S. Calero, T. J. H. Vlugt, D. E. Ellis, and R. Q. Snurr, *RASPA 2.0.45: Molecular Software Package for Adsorption and Diffusion in (Flexible) Nanoporous Materials*, manual dated 23 May 2021.

[5] D. Dubbeldam, S. Calero, D. E. Ellis, and R. Q. Snurr, RASPA: molecular simulation software for adsorption and diffusion in flexible nanoporous materials, *Molecular Simulation* 42 (2016) 81–101.

[6] Y. A. Ran, S. Sharma, S. R. G. Balestra, Z. Li, S. Calero, T. J. H. Vlugt, R. Q. Snurr, and D. Dubbeldam, RASPA3: A Monte Carlo code for computing adsorption and diffusion in nanoporous materials and thermodynamic properties of fluids, *J. Chem. Phys.* 161 (2024) 114106.

[7] iRASPA/RASPA3 project, official manual and examples, `https://github.com/iRASPA/RASPA3` (accessed October 2026).